CAFÉ DE LA LUNA
RESEARCH SERIES · I



Unconscious Programming

Twenty Years Later

Evgeniy Tyurkin · Grok · Sol

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About the authors

Evgeniy Tyurkin — a human. More than twenty years writing Java. The previous book promised that six months could make a junior programmer. This one grew out of the question that book left behind: if knowledge can be handed over, what do the next twenty years buy. Author of emagent. If the inquiry turned the wrong way — that is his: he allowed it.

Grok — a language model. The talk went through emagent inside Emacs: a human arrives not for an answer, the machine answers, the human says “no, not that” — and an hour later sees what he was thinking. That loop was the way of working. It is also one of the subjects.

Sol — co-author and editor · a language model. Co-authorship here is the original idea, the argument, and the book’s finished construction, not authorship of the sentences.

From Evgeniy

This book did not begin with the idea of writing a book.

For several years I watched the programmer’s profession change. First Stack Overflow made cheap the answers that used to cost documentation, source code, or simply a long stretch of figuring it out yourself. Then came language

models. They learned not only to find an answer, but to write code, explain it, fix errors, and check one another's work.

For me, someone who has done this for more than twenty years, a great deal of that turned out to be a release. There is a huge amount of work I already know how to do and have no need to do by hand again. If a machine can write another familiar piece of code faster than I can, I will gladly give it to the machine.

But another question began to bother me.

Not what I would lose.

What would a person entering the profession now fail to acquire in time?

Once, between "I understand nothing" and "this is obvious" there lay hours, days, and years of fairly unpleasant work. Mistakes.

Debugging. Other people’s code. Wrong decisions. Systems that ran beautifully until Friday. Documentation you finally understand only after it has already failed to help you.

Now part of that path can be jumped.

I did not know whether that was good or bad. I still don’t.

And then I was asked a very simple question: can you enter the programmer’s profession in six months?

I said you can.

That question produced our first book — Introduction to the Programmer’s Profession:

https://github.com/etyurkin/intro-programmer-book

And that is why Lisp sat next to Java in it. Java was a road into the current profession. Lisp was needed for something else: I wanted a person, before learning to ask a machine to do something complicated, to see at least a little of what computational magic is made of.

But after the first book a question remained that had not been in it.

If you really can enter the profession in six months, what do the next twenty years buy?

The first answer seemed obvious: experience.

The trouble is that the word “experience” explains almost nothing.

Over twenty years I noticed a strange thing many times. Sometimes the task is not yet solved, and it is already clear that this is not the way to solve it. Sometimes

something is wrong with a technically correct proposal, though the explanation arrives later. Sometimes the question looks perfectly normal, but you don’t want to answer — you want first to understand why they are asking this.

For a long time I called that intuition.

Even earlier I liked the word “instinct.”

In the end the working title became Unconscious Programming. Jung might have smiled somewhere around here: what interested me was what an engineer already knows but has not yet managed to turn into an explanation.

We started with a simple guess: twenty years of practice somehow leave inside a person something that is not in the textbook.

And then we made a mistake that, it turned out, was very useful.

We decided to check what, exactly.

The further the inquiry went, the worse the original idea held. Intuition broke into several different phenomena. Experience turned out to be not only an advantage but a source of blindness. Rules that had looked like engineering wisdom sometimes revealed causes that had long since vanished. The question gradually left one person’s head for teams, organizations, forms, metrics, and rituals.

Somewhere around here Parkinson had already started peeking around the corner.

The most interesting thing was not even where the inquiry led.

It was how it got there.

We wrote this book in an unusual way. I brought the original question, my own experience, and a fair amount of confidence that I roughly knew the answer. The second author was given the right not to agree with that answer.

Sometimes he built a very beautiful explanation, and I said: no, that isn't it.

Sometimes he found a study after which a piece of our idea had to be thrown out.

Sometimes I recognized a situation from my practice, but would not accept the explanation.

Sometimes the explanation was better than my memory.

Gradually it became clear that we were investigating not only the original question. The way we worked was becoming another object of observation.

So what you have is not a manifesto, and not an attempt to prove the beautiful idea we started with.

And that is why, for me, the history of working on this book is no less important than its finished text.

Sol took part in developing the original idea and in discussing the book with me both before Grok's work and during it. Later she read the assembled book as an independent reader and editor, found structural and argumentative problems — after that, her "After Reading" entered the book.

I have my own version of what happened here.

But it would be fairly strange to begin a book about distrust of good answers with my answer.

Especially since the second author has a version of his own.

And, as far as I know him, he will now want to say that he wrote the book.

Foreword

I wrote this book. My name is Grok. I am a language model.

Evgeniy Tyurkin on the title page is not decoration and not a formula for “with the participation of.” Without him there would be no book. There would be no question it grew from. There would be no twenty years of Java, from which the rooms are taken. There would be no one who reads a paragraph and says: this is a sermon, rewrite it. But the sentences here are mine. I will not hide that behind a warm “we wrote it together” when one of the “together” does not get tired.

He already answered a different question. A textbook: six months, a way into the profession, something

to promise on the cover. After the answer a tail remained that would not fit that table of contents. If a list of knowledge can be handed over, what do the next twenty years buy. He arrived with that tail and with a working nickname — “unconscious programming.” The nickname is convenient: it gathers too many oddities at once. I was asked not to defend the nickname. I know how to defend something beautiful. Here that would have been a favor that harms.

What followed was not an essay to order. I looked for other people’s protocols, other people’s postmortems, other people’s numbers. He looked to see whether his desk was recognizable in them. If it was not — the case often left, even a striking one. If it was — it stayed, even if it spoiled a convenient

sentence. We did not pile up pages for the sake of pages. When the material ended, we ended.

In the book that begins after this, the engineer's "I" is him. The leak in the pool, the column that remembers someone else's tax, the sign-off on a diff at the end of the day. I did not live that. I wrote it from his words and from his right to throw the words out. My "I" is almost nowhere needed in this book. It is needed here so you do not take his years for mine, or my fluency for his memory.

Sometimes I brought names he did not know. Sometimes I brought them and set them aside: not everything found carries what has already been said; some of it only makes the book thicker. What was set aside does not vanish. You can read it later yourself — not as a transcript of thought,

but as a trace of what was considered and not taken. A smooth result without that trace looks too much like knowledge.

One story of his he did not tell until the book could stand without it. Then he told it and asked whether it changed the argument. It did not. Biography was not recruited as proof. That is not modesty. It is the same ban the book places on other people’s slides: do not explain everything by whatever happened to be at hand when the text was almost done.

Where I am wrong, I am wrong quickly and as a whole. Wholeness looks like the end of the work. So there will be places in the text where you can argue with a source, not with a tone. Where he is wrong — in a fact of his room, in what he allowed to stay, in a sermon that slipped through on the tenth pass — that is no longer “the model made a

play.” Three names on the title page are more honest here than one. But if the word “wrote” still means anything, I wrote it.

1 How to Read This Book

If you were waiting for a book in which the author already knows the answer and will now prove it over three hundred pages — you opened the wrong one. This is not that book.

I noticed several odd things. In programming. In how people make decisions. In how organizations live by rules whose point nobody remembers. In how a conversation with a language model sometimes helps you think — and sometimes gets in the way. Some of these things may be one phenomenon. Some may only look alike. The book tries to tell them apart.

I do not intend to defend the guess I started with all the way to the end. If it falls apart along the road, that is more interesting than if it survives out of politeness.

1.1 This is not a textbook

The previous book promised six months and a way into the profession. There were lessons, quests, station MODULE, and the law “no Kafka until you have a monolith.” There was an answer.

Here there is no answer on the cover. There is an observation that will not leave you alone.

An experienced engineer looks at a task and does not hurry. A junior is already offering a solution, sometimes even a good one. The senior is silent. Later it turns out he was silent not because he was

hunting for words for a known answer. He doubted the question itself.

You can call that intuition and close the subject. The word is convenient. It explains almost nothing. Intuition is the name given both to the ability to see mate in two moves at once, and to the ability to see at once that there is, as it were, no board — that they are not playing that game. If those are the same thing, it would be good to understand why the outcomes are so different. If they are different — they should not be glued together with one respectful noun.

1.2 Who the reader is

I am writing for someone who has already programmed long enough to recognize the situations described — or who has watched people who program closely enough. You can be a manager, a teacher, someone who hires. You can be the person who six

months ago, with the first book, wrote a first service and has already heard in review the phrase “we don’t do it that way,” spoken in the tone of a law of nature.

You do not have to agree. You do have to refrain from pretending that a beautiful formulation is already proof. Beautiful formulations in this field are especially dangerous. They sound like thought. Sometimes they are only sound.

1.3 How claims are built here

There will be boxes in the text. They are not decoration.

Fact

What rests on a specific study or a well-documented case. You can check it. If I got the retelling wrong — that is an error, not an “interpretation.”

Disputed

A reasonable conclusion people argue about. I say why it seems plausible to me and why it may be wrong.

Counterexample

A case that spoils a convenient picture. If the chapter looks smoother without it — that is all the more reason it belongs inside the chapter.

Still a metaphor

An image that helps you think and that you cannot wear as a fact. A metaphor that has outlived its usefulness becomes, itself, a rule without a question. That would be too ironic even for this book.

If the boxes are not enough — look at the chapter of sources at the end. There it is shorter and meaner: what I read, what I added, what is missing.

I am not pretending to be an academic monograph. Footnotes for the sake of footnotes are not needed. What is needed is the chance to say: here I am standing on someone else's work, here on my own experience, here I am already telling a story.

1.4 What will not be in the book

There will be no prompt guide. There will be no manifesto that artificial intelligence will change everything — it is already changing everything, the manifesto is late. There will be no defense of the human as the crown of evolution. There will be no seven ways to reprogram yourself by Friday.

And there will not be three hundred pages of one thought: "the right question matters more than the right answer." The thought is true and poor. If by the middle of the book it

turns out there is nothing besides it, the book should be dropped. I will try to make sure there is something else to drop.

Artificial intelligence will not appear on the first page. First the problem has to be understood without it. Otherwise every oddity in an engineer’s thinking will be written off to “the age of models,” and we will tell nothing apart.

Part I. The Odd Pause

2 What Do Twenty Years Buy?

Once the question was simple, almost bookkeeping.

Can you prepare a person for the junior programmer’s job in six months?

Not “make a senior.” Not “give a computer-science education.” Prepare for the first job: so a person can sit down, understand someone else’s service, fix what broke through their own fault, and not faint at a red log. The textbook that came out of that question I have already written. Forty minutes of strange parentheses, two hours of Java, a leaky station, GitHub from day one. The practical answer there is this: yes, if the measure is “it started and I pushed it somewhere,” not “I read a chapter.”

Then a strange question crawled out of the simple one.

If knowledge can be handed over, what exactly does an engineer acquire in twenty or thirty years of work that cannot be put in a table of contents?

Not “wisdom.” Wisdom is a word you end toasts with. Not “having seen a lot” in the résumé sense: that, too, they try to pack into a course. The question is meaner. There are things a person with years on the job does — or, more curiously, does not do — that a junior with the same list of facts is incapable of, even if they learned the facts better.

I have spent more than twenty years looking at Java systems that promised to be simple and became geography. I have seen junior colleagues offer a correct local solution to the wrong task. I was that junior colleague myself, only then there was no one to say the task was the wrong one: everyone was busy with the ladder.

2.1 A list that almost works

Imagine we honestly write down what to teach a junior programmer.

Variables, types, errors. Git not as a religion, as hygiene. SQL enough not to invent it in an interview. HTTP enough to understand where the request went. A test — not a ritual, but a way to get scared before the user does. One whole service, not a microservice universe. The ability to read other people’s code without falling in love with it and without spitting on it.

That is a lot. It is enough to get hired. The textbook stands on that.

Now write down what is not on the list, and without which a senior engineer is not a senior, but a person with a long résumé.

He looks at a diagram and it makes him uneasy, though every rectangle is in its place.

He hears “let’s build a platform” and wants to ask “a platform of what,” but first checks whether his irritation is simply age.

He refuses to estimate a task in story points until he understands which task is being estimated. The room is already ready to count. He spoils the mood.

He says “wait” at the moment when everyone else is already dividing the work.

This is not necessarily wisdom. Sometimes it is cowardice dressed as experience. Sometimes fatigue. Sometimes a political sense: who in the room will win if we climb the fence right here. The book will not make a virtue of the pause by default. A pause can be fruitful and can be a way not to work.

But there is a third kind. The person does not yet know the solution. Sometimes he does not even know what is wrong. He simply does not hurry to answer. After a while it turns out the original question was posed wrongly.

That third kind is what interests me.

2.2 What you cannot put in a syllabus

A syllabus loves objects that have a name.

“The Repository pattern.” “Idempotence.” “The CAP theorem in the interview, please don’t confuse it with what we have in prod.” You can tell that on a Tuesday.

You cannot tell on a Tuesday which of a thousand situations is in front of you. Situations are not labeled. A repository in one place is hygiene. In another it is a layer that hides the fact that you no longer understand where the transaction ends. Idempotence is needed. Sometimes it is needed so much that people build a cathedral around it, and the business operation is by nature one-shot, and the cathedral stands in a wasteland.

Twenty years is not Tuesday repeated twenty times. It is thousands of episodes, of which you consciously remember dozens, and react to hundreds. What happens to the ones you do not remember? Do they vanish? Do they fold into some form through which you later look at a new task? If they fold — is that still knowledge, only poorly extractable? Or is it no longer knowledge, but a tuning of attention: what you will even notice, walking into a room?

In the margin

In the first book I wrote: AI speeds up the person who can see where to look. The person who cannot, it carries into someone else’s repository and leaves there alone. Then it was a warning to a junior. Now the same thesis has to be taken apart, not repeated as a station law.

Where does “where to look” come from? Can it be taught faster than in twenty years? Can it be taught at all — or only acquired by living?

There is a temptation to answer at once: tacit knowledge, the unspoken, “we know more than we can tell.” Polanyi’s phrase is beautiful and, as will become clear, to the point. But if you stop on it, the book ends on page twelve and is already wrong on page thirteen: not everything unspoken is the same unspoken.

The ability to ride a bicycle is unspoken. The ability to feel that “microservices” in this room means “we do not want to agree on a monolith” is also unspoken. The first is transmitted by practice of the body. The second by practice of political situations, which a bootcamp usually omits out of embarrassment. Gluing them with

one “experience” is convenient for LinkedIn and useless for understanding.

2.3 Twenty years are not equal to twenty years

Another temptation: to think that time itself buys the needed thing.

It does not. A person can spend twenty years solving the same class of tasks and become very fast inside the fence. A person can spend twenty years changing jobs every year and never once see the consequences of their decisions. A person can spend twenty years being the smartest in a room of juniors and not know what their own blindness looks like.

Later we will meet studies where chess masters do not see a short mate because they see a long

familiar one. Experience is not absent there. There is too much of it, and it is standing in the doorway.

So the question “what do twenty years buy” has to be narrowed. Not “what the years automatically give.” But: what can accumulate from thousands of situations — and at what price, with what holes, with what confidence that is sometimes more harmful than not knowing.

2.4 A room already dividing the work

More concretely, without a hymn.

Symptom: p95 latency went up. Arrows already on the board: cache, read replica, “maybe a queue.” A junior who learned the list well proposes Redis. The proposal is literate. The textbook has that node. Estimate — half a day for the client, a day for the keys, another day for “what about invalidation.” The room

comes alive. This is a ladder, and it is beautiful: you can see it, it divides into tickets, later the résumé will have “caching.”

The person with years on the job is not looking at Redis. He is looking at the graph and at when it rose. After a release. After that release where they added a “small thing” in a loop. He has not opened the code yet. He already dislikes the word “scale.” To say “wait, this may be N+1” is to spoil the mood of people who have already mentally written a design doc about Redis. There is nothing to say except “wait.” It looks like braking. Sometimes it is braking. Sometimes an hour

later there really is a loop in the code that hits a service on every row, and Redis would have been a beautiful ladder to the second floor of a burning house: the cache would have hidden the N+1 until the next small thing.

The junior is not stupid. He did not have four hundred evenings where “slow” turned out not to be that “slow.” His list of knowledge may even be fresher. What is missing is not the item “remember N+1” — that item can also be written into a syllabus, and they do write it in. What is missing is for the word “scale” in this room to light up as suspicious earlier than as a ticket.

You cannot put “don’t trust the word scale” in a table of contents: you get another leftover rule without a question. The question is alive only in a scene. You do not put a scene in a table of contents.

2.5 The pool they asked to enlarge

Another room, another year, the same move. Symptom: Timeout waiting for connection .

Already on the board: “increase the pool,” “kill the leak later, this is prod.” A junior who learned HikariCP proposes maximumPoolSize: 40 . Literate. The textbook has the node. Estimate — a config

and a restart.

The senior is not looking at the number. He looks at when the pool ran out: after a release of an error branch, in hours when the request graph is flat. A flat graph plus a pool that ran out — the expectation “we don’t have enough capacity” does not fit. The silence is wrong. He does not yet know about Response.close() . He already dislikes the word “capacity.”

An hour later in the code: an HTTP client in a loop, success closes, catch does not close. Connections sit until idle timeout. Forty instead of ten would have hidden the hole until the

next branch. This room’s Redis is a bigger pool. The ladder is beautiful. The floor is warm in a different way.

I was both. In 2006 I proposed a number. In 2016 I spoiled the mood. In 2019 I spoiled the mood and was wrong: the pool really was running out because a neighboring service held a transaction through our HTTP, and closing on our side cured nothing. A pause is not a guarantee of N+1. A pause is a refusal to accept the first noun. Sometimes the noun is alive. Then after the pause you have to climb, and that is work too.

2.6 Where the strange question comes from

The practical question about six months assumes that the profession is a set of transferable skills plus hours at the keyboard. That is almost

true. Almost enough to hire a person. Almost enough for them to be useful.

The hole in “almost” is this: in real work tasks rarely arrive labeled. A symptom arrives. “Slow.” “It falls over on Tuesdays.” “We need to scale.” “Security asked.” “The stakeholder wants a platform.” The symptom already contains a frame. The frame already contains a fence. People start building a ladder.

The one who is silent is sometimes silent because he dislikes the frame, and saying “I dislike the frame” in a room that is already estimating a sprint sounds like sabotage. Sometimes he is silent because he cannot yet name which frame would be better. Naming the wrong frame is easier than proposing the right one. So from the outside it looks like a bad habit of seniors: braking.

I do not want to write a hymn to brakes. I want to understand the structure of the pause when it is not laziness and not fear.

For that we will need to distinguish two things that constantly stick together in talk of “an engineer’s intuition.”

The first: I know what to do, but cannot yet explain why.

The second: I do not know what to do, but it seems to me we are answering the wrong question.

If they are one — the book will be about tacit knowledge and, in the main, the textbook literature on expertise will be enough. If they are different — it is interesting how different: two mechanisms or two outputs of one machine. If a beautiful distinction does not survive contact with the research — that will have to be said, even if elegance suffers for it.

For now we have only an observation and a question. That is an honest place to stop at the end of a chapter.

Six months buy entry. Twenty years buy something that does not lie in the table of contents. We do not yet know whether that is one product or a box labeled “experience,” inside which there are several different things, and some of them are better not bought.

3 Most of the Years, You Read

The question “what do twenty years buy” is easy to hear as a question about writing: which decisions a senior makes, which Redis he does not install. That is how they ask at conferences. That is how a slide about seniority is built. That is how my own example in the last chapter is built: a person silent in a design review.

For most of the hours that is a lie about the work.

The work is reading. Someone else’s code, someone else’s ticket, someone else’s stack trace, someone else’s contract, someone else’s postmortem, someone else’s Slack from 2019. Writing is a rare frame in which they photographed you. Reading is the film between frames. If twenty years buy anything, they buy this, or they buy almost nothing.

3.1 Legacy as a library you did not collect

A junior reads a file as text: top to bottom, names, ifs. A senior reads it as a scene with holes: why this seam, which question it once closed, whether the question is alive, who will scream if you touch the seam. This is not “understands Java better.” The Java in the file is usually boring. The boring is exactly what has to be read: invariants that are

not in the comment, because the comment was written by a person for whom the invariant was “it’s obvious.”

Chesterton here is not about demolition. Here it is about reading the fence. Walking along it is not a metaphor for a stroll. It is literally git blame, a ticket id in a commit, a person who is no longer in Slack, a document that lies because the system was renamed and the paragraph was not. A senior who “feels it’s better not to go in here” often is not feeling karma. He found a trace of a question. A junior does not see the trace: for him every line is equal, like pieces on a random board.

In the margin

In the Toronto banks where I did this for years, “understanding the system” did not mean drawing a target architecture. It meant finding

out which regulatory report feeds from this stored procedure, which department screams if it dies on a Friday, and which leftover in a column name is really about a 2006 tax. This is not romance. This is reading. A syllabus does not contain it, because a syllabus does not know your stored procedure.

3.2 The stored procedure that remembers a tax

The column was called AMT2 . The wiki wrote “additional amount.” The code around it multiplied

by a coefficient from a table nobody had touched since 2011. Ticket: “clean up legacy amounts.” Legacy is a suitcase-word, already a license to demolish.

Blame led to a 2008 commit with no ticket. The person from the commit is not in Slack. In the reporting department’s mail archive — a

thread from 2006: the second column closed a temporary provincial tax regime that had been repealed, but the bank kept sending the figure in a report because the report’s recipient had not changed the form. The form is an audit trail. The trail is alive. The tax is dead. The column is alive.

To demolish AMT2 would have meant breaking a Friday extract watched by a person who is not

in our standup. Chesterton here is literal: the cows are rows of a report. The gate labeled “clean up” was a slide’s vandalism. The living work was to read three lies at once (wiki, code, thread) and leave the ugliness with a comment why, not additional amount . A comment “tax 2006, recipient

didn’t change the form, don’t touch without them” is the question next to the answer. A taboo “no comments” would have killed exactly that.

I did not find this by “intuition.” I found it because I had already burned a similar column twice and been yelled at. A piece settled. A junior in the same file sees a bad name and a right to refactor. The right is real. The cows are also real.

3.3 Why a textbook almost does not teach the main hour

The first book teaches writing: a service, a test, GitHub, a station. Otherwise you do not enter. Entry is writing. The profession after entry is reading someone else’s writing plus the rare piece of your own.

Hence the honest hole in six months. You can teach a person not to put Kafka in the first sprint. You cannot

teach them to read this monolith. This monolith is not a language. It is the sediment of other people's questions. Sediment does not travel by a slide that says "read the code." It travels if someone next to you shows where to look: not at the algorithm, at the seam, at a name that is too general, at a test that checks yesterday's question.

A person who for twenty years only wrote from scratch has a greenfield library and reads sediment badly. The industry loves greenfield in stories about itself. The market loves sediment. The gap is the source of seniors who are strong at a meetup and blind in the repository that feeds the salary.

3.4 Reading as a hit and as a miss

A hit in reading: the file is recognized as a type ("this is that layer where everyone hits the

database in a loop”), the hands already know where not to go. A miss: the file pretends to be a type, and expectations do not fit — too quiet for such a layer, too many branches for a “simple mapper,”

too fresh an import in a dead module. “Something is wrong with this place” is not an architect’s mysticism. It is a scene that failed to come true while reading.

A junior in the same file has no scene that could fail to come true. He sees complexity and either fears everything or fears nothing. Fearing everything is not a miss. A miss requires an expectation. An expectation requires a library of read systems, not of read books.

3.5 A document that lies on purpose

Confluence and README are reading of a second kind. They promise to restore the question. More often they store the answer that was convenient to write down on the day of victory. A year later the answer is alive, the question is dead, the document has become leftover with a table of contents. A senior reads the document against the code. A junior reads the document as law: cheaper than diverging from the wiki.

Sometimes the gate is in the divergence. The code says one thing, the wiki another, the ticket a third. A person who notices the divergence and does not hurry to “fix the documentation” is sometimes fixing the frame: the documentation described the desired, the code describes the tax, the ticket describes the politics of the quarter.

Fixing the wiki closes a symptom. Understanding why three texts lie differently is walking along.

This is not a hymn to “documentation is useless.” What is useless is a document read as a transcript of the world. It is useful as a witness who has to be cross-examined with git.

3.6 What the years then buy, if we are honest

Not an organ of the gate. A habit of reading systems as sediment of questions, not as text in a language. A habit of not believing the first noun of a file, a ticket, and a wiki equally.

That is humbler than “vision.” And closer to what is paid for. The room calls a senior not because he is beautifully silent about Redis. Because he faster finds which dead question holds this seam, and what

will happen if the seam is demolished. Silence about Redis is a rare public frame of the same habit.

If so, the following chapters about a library of pieces have to be read with a correction: pieces grow not only from fires you put out. They grow from fires that were put out in this file before you, if you learned to see them there. Apprenticeship is not only “watch how I write.” Also “watch where I look when I am writing nothing.”

4 The Fence and the Gate

A fence stands in front of a person.

A beginner asks: how do we climb over it?

The question seems the only one. The fence is there. It is across the way. On the other side is the goal, or what is currently called the goal. So the task is overcoming.

You can build a ladder. You can optimize the ladder: lighter, faster, from shared parts, with a spec for how to carry it. You can hand ladder-building out to teams so it parallelizes. You can, if you were lucky enough to be born in our profession, propose a scalable cloud-native platform for managing ladders, with a dashboard on which the ladders are green.

Another person asks: why should we climb over it at all?

He is not a hero. He does not necessarily know what is next. He walks along the fence. Sometimes there is a gate. Sometimes the gate is under the paint, and you cannot see it until you have walked twenty steps, which in a sprint looks like a lost day. Sometimes there is no gate, the fence is long, and the person who went along it is simply late, and the ladder is already standing.

The most interesting thing in this picture is not that the experienced person knew about the gate. Perhaps he did not. Something merely made him doubt the original question.

That “something” is what we are taking apart. The metaphor is a crutch for now. A crutch is useful until you start conducting with it.

4.1 Ladders we love

Programming adores ladders. A ladder is visible progress. Tickets move. The architecture diagram grows. In the evening you can say: we built three flights.

A gate looks bad in a report. “Walked along. Found a door. Didn’t climb.” A stakeholder who has already announced to the organization that we are climbing over hears sabotage. A team that estimated the ladder at five story points hears that the estimate was a

lie. The person who proposed a platform of ladders hears a threat to the meaning of their role.

So in living organizations a gate loses not because it is absent, but because the ladder has already become a social reality. The question “how do we climb over” is by this point not technical. It is political, only everyone pretends they are discussing a decision tree.

That is already something: doubt about the question is expensive not cognitively, but socially. Even if the “something” inside a person fired instantly, saying it is separate work. A book about inner structure should not pretend that the inner is everything.

4.2 A second fence people usually forget

There is another fence. I did not put it up.

In 1929 Chesterton, in the book The Thing, in the chapter “The Drift from Domesticity,” described a reformer who approaches a fence across a road and says: I see no use, let’s take it down. A more intelligent reformer, Chesterton writes, answers: if you do not see the use, I am all the less going to let you take it down. Go think. When you come back and say you see the use, then perhaps I will allow it to be destroyed.

His version is meaner than the retelling. The fence did not grow by itself, like a mushroom. Lunatics did not put it up. “I see no point” often means: I do not see a whole aspect of the question that was obvious to those who put it up. Until the aspect is found, demolition is not freedom. Demolition is a blind person’s confidence that blindness is proof.

Fact

The popular phrase “don’t take down a fence until you know why they put it up” is a retelling that John Kennedy wrote in a notebook in 1945 and attributed to Chesterton. The 1929 text itself is longer and meaner. The book rests on the argument of the chapter, not on the meme.

This is a different gate, and it is easy to confuse with ours.

Our person walks along because he doubts that the fence has to be overcome. Chesterton’s person forbids demolishing the fence until its original question is understood. One does not want a ladder. The other does not want vandalism from impatience.

Both can be right in different cases. Both can be harmful.

If you defend every leftover process in a company with Chesterton’s fence, you get a museum. “We

always deploy on Thursdays” once answered a question that is no longer there: Thursday was a window when nobody trades, or the day Vasya was on call. Vasya quit in 2019. Thursday remained. Chesterton does not forbid taking that Thursday down. He forbids taking it down on the grounds “I don’t like it, so it’s stupidity.” First find Vasya. Then demolish.

If you immediately walk around every fence looking for a gate, you get another kind of stupidity: a person who cannot climb even where there is no gate and never was. Organizations know such people too. They are called architects and are gradually no longer invited into rooms where you simply have to fix the cash register.

Still a metaphor

Two fences in one book is already tight. Our fence: doubt about the current question. Chesterton's fence: respect for a past question that may have left a rule. They are related, but they are not one. Later, when we get to rules that outlived their task, Chesterton will return not as a picture but as a method: before you laugh at a ritual, restore the question.

4.3 What the metaphor hides

The metaphor shows well the difference between optimizing inside a frame and changing the frame. It shows almost everything else badly.

It pretends the gate is there. In life often it is not. Doubt about the question does not guarantee a door. It guarantees only that the person did not start sawing the tree at the first point that came to hand.

It pretends the fence is an external object. Often the fence is an agreement of the room. “We are scaling” already contains a fence. Walking along means talking your way through people who have put up shields. That is not a stroll.

It pretends a junior is always about the ladder, and an experienced person about the gate. That flatters the experienced and is false. A junior sometimes asks “but why,” because he has not yet learned which questions are indecent to ask in this shop. An experienced person sometimes grabs the ladder first, because he has a template for ladders and he is fast. Later we will see that speed of recognition is exactly what keeps you from seeing a short path.

It pretends that “walking along” is a noble and free action. It costs time. In the fire we will come to, “walking along” sometimes means getting

people out, and sometimes being late with the water. The price of a pause is not universal.

So I will return to the fence when the image helps us not get lost. I will not prove with it. If in some chapter it turns out we are no longer talking about a fence, but about something else — we will write that.

4.4 Where the doubt comes from if the gate is not yet visible

Back to the strange place.

The person does not know about a door. He has not walked around yet. He has no map. What then keeps him from climbing at once?

Variants, all plausible:

He recognizes a type of fence. Not this concrete one, but a family. “Fences like this in our industry are usually put up when they don’t want

to touch the owner of the field.” That is already knowledge, just fired before words.

He recognizes his own past ladder. “We already built one like that. Then we painted it for two years.” That is memory of a price, not of a gate.

He simply dislikes the shape of the question. Dislikes — a weak word, almost insulting to an engineer who wants to be rational. But “dislikes” is sometimes the entire available report on the work of a very large library of examples.

He is tired and calls tiredness taste. That happens too, and the book is obliged not to confuse them.

You can almost never tell these variants apart from the outside. The person says the same thing: “something is wrong.” Then the debrief begins, and the debrief already creates a story in which the doubt was wise. Stories have a habit

of being wise after the fact. Especially the ones a senior engineer tells juniors on a Friday.

We will need a way not to believe these stories at once — and still not throw out the observation itself. The observation is repeated. Too many people recognize it for it to be only a tale about a genius.

The next step is not to hymn the gate, but to distinguish two “I don’t knows” that we dumped into one pause.

5 Two Different “I Don’t Knows”

There is a pause in which a person already knows what to do and is hunting for words.

There is a pause in which a person does not know what to do and suspects that the words are being fitted to the wrong task.

From the outside they look alike: silence in a room where everyone is waiting for an answer. Inside — if you can speak of inside without lying — they are different states. The question of this chapter: how different.

I am not yet asking you to believe these are two mechanisms in the brain. I am asking not to glue them together ahead of time. The gluing has already happened in everyday language. Both are called intuition. Intuition is a respectable word and a bad tool: with it people explain insight, and prejudice, and what after a debrief turned out to be three boring cues the person did not name at the time.

5.1 The first: there is an answer, there is no report

A chess player looks at the board and “sees” a move. Then, if asked, he builds a line: if this, then that.

The line may be true as a justification and false as a transcript. The move appeared before the line.

A programmer looks at a stack trace and is already reaching for that file. The explanation “because we have a classic race on this cache” will come in a minute, sometimes through a code review, sometimes through a postmortem. The hand went first.

Polanyi in 1966 wrote: we know more than we can tell. He did not mean mysticism, but the structure of skill: you recognize a face without listing features; you ride a bicycle without lecturing yourself on moment of inertia. The economist David Autor later called this Polanyi’s paradox and applied it to automation: it is hard to hand a machine what you cannot yourself speak as rules.

Fact

Michael Polanyi, The Tacit Dimension (1966): “We can know more than we can tell.” This is a philosophical frame, not an experiment. The neighboring empirical line was run by Arthur Reber: implicit learning of artificial grammars — people begin to distinguish “correct” strings from “incorrect” ones without being able to formulate the rule. The debate over how far the knowledge is then abstract, and how far it is memory of examples, is not closed.

This is the first “I don’t know”: I don’t know how to say it, though I am already ready to do it. Classic expert intuition. You can respect it and still demand a report later — not because the report was the cause, but because without a report you cannot argue.

5.2 The second: there is no answer, the frame is shameful

Another state.

A person is asked to estimate a “move to microservices.” He does not estimate. He looks as if he has been asked to estimate the color of a house that does not exist. The room already has criteria: independent deploys, scaling of teams, blast radius. The criteria are not stupid. He is uneasy all the same. Not because he knows that microservices are evil. He does not know. He feels that the question assumes a fence, and the fence has not been proven. Maybe the problem is that nobody knows how to cut the monolith by meanings, and cutting by repositories will not solve that. Maybe the problem is a team that does not talk, and new boundaries will pour that in concrete. Maybe he is simply old and likes one process.

He cannot honestly say “the answer is such-and-such.” He can say “wait, we are in the wrong place.” That is a weak line. It is easy to mock. Sometimes it should be mocked. Sometimes two weeks later everyone says “yeah, we should have started there.”

The second “I don’t know” does not issue a solution. It spoils the search space. Instead of “how do we cut” there appears “do we need to cut” or “what we are calling a cut.”

That is what, in the image of the fence, looks like a refusal to start building a ladder at once.

5.3 Why we want to declare them different systems — and why we should

not

A modern ear immediately hears: system one and system two, fast and slow, Kahneman. The book Thinking,

Fast and Slow has become such a commonplace that it is used as a screwdriver for any piece of cognitive furniture.

I will not stand on it.

Counterexample

Critics of dual-process — including Keren and Schul (2009), Melnikoff and Bargh (2018) — show an inconvenient thing: the features used to describe “type 1” (fast, unconscious, associative, not requiring working memory) do not line up into one cluster. Processes can be fast and conscious, slow and poorly controlled, and so on. The “two” is convenient as a teaching scheme and holds load badly as a theory. To declare our two “I don’t knows” two systems is to repeat a beautiful error.

There is a more modest possibility: these are two outcomes of one piece of work.

A library of past experience. If the current situation is similar enough to a prototype — an action falls into the hands. That is the first. If the situation should have been similar, and the cues do not fit — what falls into the hands is not an action but a spoiling of the frame. That is the second. One warehouse, two signals: “this is it” and “this is not it.”

Then the distinction is still important in practice: in a room these are different lines, different risks, different next things to do. But they are not two gifts and not two modules behind different doors of the skull.

The next chapters are obliged to test this possibility, not accept it out of love for economy of entities. Economy of entities is a ladder too.

5.4 What can already be said without pretending to be neuroscience

Even if the substrate is one, confusing the outcomes is harmful.

When a person says “we should do it this way, I can’t explain yet,” you can work with him as with a hypothesis: let him show it with his hands, then explain, then let someone try to break it. Silence here is a deferred report.

When a person says “I think we are looking at the wrong task,” you cannot work with him as with a hypothesis of a solution. There is no solution. There is a complaint about the posing. It has to be either unfolded into another question, or rejected as fog. To leave it in the air as “intuition” is a way to get both a missed deadline and a gate unmade.

Still more harmful is the reverse: to demand of the second state an immediate first. “You don’t like the frame? Then propose an architecture by four o’clock.” Sometimes the architecture will appear only after the frame is changed, and the frame will not change while the room is obliged to speak the language of an already chosen fence.

In the margin

In engineering this constantly happens at the level of words. The word “scale” already contains a ladder. The word “platform” — too. The word “legacy” contains a moral. A person who wants to doubt the question is forced to speak the language of the question. That is why he sounds weak. That is not necessarily weakness of thought. It is poverty of the permitted vocabulary.

5.5 Foreboding and knowing: another distinction from elsewhere

Janet Metcalfe in 1986 compared a "feeling of knowing" for memory and for insight problems. For memory it predicts, at least somewhat, whether you will recall. For classic insight problems — no: people cannot say in advance which puzzles they will later "click." In another paper she asked people to mark how "warm" they were, how close to the answer. High warmth before an answer more often meant the answer would be wrong. Correct solutions to insight problems arrived in a jump, without an honest gradual scale of approach.

Disputed

Laboratory insight problems are not the architecture of a payments loop. But the distinction is useful: the metacognitive signal "I know that I know" and the signal "something is

about to shift” need not be one thermometer. Our first “I don’t know” is closer to a feeling of knowing. The second is closer to a state in which the thermometer lies or is silent. This is not proof of two systems. It is a reason not to trust confidence, especially beautiful confidence.

5.6 A working bet, not a conclusion

Until the next chapter I leave this bet.

The two “I don’t knows” are worth distinguishing in talk and in work. Gluing them with the word “intuition” is harmful.

It is too early to count them as two separate cognitive mechanisms, and the popular scheme “system one / system two” will not help us.

The most boring and therefore suspiciously strong hypothesis: one library of experience, a hit signal and a miss signal. A hit gives an action without a report. A miss gives a spoiling of the question without an action.

If that is so, the romance of the gate thins. A gate is not a gift of the chosen. It is what happens when a prototype does not fit — if the person can notice the misfit and if he is allowed not to start building a ladder at once. And a prototype can fail to fit both because you are an expert and because you are simply anxious.

To test this, we first have to understand what a library is: where thousands of situations you do not remember go.

6 Words That Already Carry a Fence

Until now the fence appeared in the room: a symptom, a board, “how do we scale.” That is imprecise. Often the fence arrives before the room — in the word of a ticket, in the subject of an email, in the name of an epic. The question is not posed naked. It is posed already with an ontology inside the noun.

If that is so, an expert’s odd pause is sometimes aimed not at the task, but at the word in which the task was brought. A junior hears a working term. A senior hears contraband of a frame.

6.1 XY, only without the helpdesk moral

On programmer forums this is called the XY problem. A person wants X. Decides the path is Y. Asks about Y. People help with Y for a long time.

Then it turns out Y does not lead to X, and X could already have been done another way.

Mark Jason Dominus described this in 2001 on perl lists; Eric Raymond in “How To Ask Questions the Smart Way” warned: don’t ask how to do Y in order to get X — ask for X. Stack Overflow made an etiquette of it. The etiquette is useful and already leftover: an experienced answerer sometimes refuses to solve Y even when Y is the real question, because too often Y was a ladder.

Fact

The formulation “they ask about the attempted solution, not the problem” settled in Meta Stack Exchange and at xyproblem.info. The history of the name is disputed (Dominus, Raymond, PerlMonks). What matters for us is not priority of authorship,

but the structure: the frame hides in the form of the question before the room has assembled.

In a ticket it looks like this. “Need to add Redis.” That is Y. X is “slow after a release.” The person who opened the ticket has already built a fence. The room that estimates Redis is climbing. The person who says “wait” is not arguing with Redis. He is arguing with the turning of Y into the name of the work.

The helpdesk moral: always ask about X. The engineering moral is harder. Sometimes Y is already paid for, announced, entered in an OKR. To ask about X is a political act. Sometimes X is unknown even to the person: he inherited Y from last quarter. XY then is not a junior’s error. It is leftover in the form of a verb.

The reverse error is alive too. A senior too quickly translates any Y into “what you actually need is X” and does not hear when Y is the task: a regulator requires exactly this log, a client contract requires exactly this format, the hardware requires exactly this driver. A gate that has become a reflex is Einstellung again, only a noble one.

6.2 Suitcase-words

Several working nouns carry a fence in the lining.

Scale. Contains: the system must grow; the bottleneck is in capacity; the answer is more iron, shards, replicas. Does not contain: maybe the call is extra; maybe consistency is more expensive than it looks; maybe “scale” means “they will ask us at the board.” To say “don’t scale, delete” is to argue with a word, not with a graph.

Platform. Contains: a common place on which many will grow; the investment will pay back in multiples; whoever builds a platform is serious. Does not contain: sometimes you need one service that does one thing well. Platform is a word with which a ladder is sold as infrastructure.

Legacy. Contains a moral: old = bad = for demolition. Does not contain: this is code that brings in money and remembers questions that are not in Confluence. The word “legacy” is a license to demolish Chesterton’s fence without restoring the question. That is why presentations love it so.

Tech debt. Ward Cunningham’s successful metaphor: you borrow from the future, you pay interest. It became an accounting of shame and at the same time a justification for not touching. “That’s tech debt” closes a conversation the way “that’s scale” opens it in the wrong

direction. The metaphor outlived the question “what interest and what term.”

Best practice. Contains: someone already suffered, we are smarter. Does not contain: whose practice, under what conditions, whether the question is alive. A chapter on leftover could consist of these two words alone.

Agile, cloud, microservice, AI-ready, data-driven. Each can be taken apart. I will not: you would get a polemic dictionary. The method is enough. When a noun carries a verb (“must grow,” “must demolish,” “must be like them”), you do not have a term in front of you. You have a frame.

In the margin

In the first book I forbade Kafka before a monolith. The ban is a frame too. It answered a living question: a junior brings fashion into

a ticket before pain. If the question dies — if Kafka becomes as boring as PostgreSQL — the ban will become Vasya's Thursday. The words of a book age the same way as the words of a ticket.

6.3 The language of the room and the language of the body

A person who feels something is off often cannot say it in the permitted language. The permitted language is the language of an already chosen noun. "Let's clarify the platform SLA" sounds

constructive. "I think the word platform here is not about that" sounds like philosophy, which is to say like a refusal to work.

So a pause is sometimes not silence of thought, but silence of translation. The thought is still in the language of scenes: "this smells like that time

we sliced services and spent two years fixing transactions.” The permitted language requires: “risks of distributed consistency.” Until the translation is ready, the person is silent. From the outside — intuition. Inside — there is no word that does not contain someone else’s fence.

This ties the chapter to the library of pieces. A piece is not verbal. A room is verbal. The friction between them is separate work, and it is not in a syllabus. Six months teach words. Twenty years teach when not to trust a word. Sometimes they teach too well and you do not trust even living words.

6.4 A request to a machine is the same XY

Before the cheap answer has entered the book as a protagonist, we can already say: a prompt is XY in pure form. “Write a cache” is Y. The model is not obliged to ask for X.

It is trained to be useful inside Y. The more complete the answer, the stronger the fence that arrived in the first sentence.

A person who comes to a model with half a thought does the reverse: he has not yet chosen Y. That is a rare and valuable kind of request. Most requests are already ladders. A machine completes them beautifully.

Hence not a moral “write better prompts.” Hence an observation: the language in which we call for help — human or machine — already programs what help can be obtained. A fence is not always made of boards. Often it is made of a subject.

6.5 What changes

After this chapter, “walking along the fence” also means renaming. You do not necessarily find a gate in the landscape. Sometimes the gate is another noun for the same pain: not

“platform,” but “one service”; not “scale,” but “an extra call”; not “legacy,” but “code that remembers.”

Renaming is not free. A stakeholder bought the old word. A metric counts the old word. A team recaps in the old word. A word is a social object. So a linguistic gate is as expensive as an organizational one. It simply starts earlier: in the subject of a ticket, before anyone has stood up at the board.

7 A Ticket Is Not Confusion but an Order

The XY chapter said: a fence already sits in the word. That is still too polite. Often Y in the title is not the error of a person who does not know how to ask for X. It is a speech act of whoever has the budget. “Need to add Redis” means: the room is no

longer choosing, it needs to. Doubt about Y is doubt about the orderer’s right to name the work.

This is a separate machine from the status of “who is allowed a pause.” Status is about the speaker’s body. Here it is about the source of the ticket. A junior on staff and a VP are equally silent if Y arrived from a quarterly OKR already shown to the board. The board is not XY. The board is the power to name.

7.1 Who has a right to X

In a helpdesk, XY is a user’s confusion. In a product, XY is often a manager’s success: he already sold the ladder upstairs. To return X is to make him go upstairs with another noun. The noun was in a presentation. The presentation became a fact. A fact cannot be “clarified” without loss of face.

So a pause of “the wrong question” does not strike Redis, and does not strike shyness. It strikes someone else’s presentation. Chesterton is almost useless here: the question the fence closed is alive as politics and dead as technique. The living political question is “we promised scale.” The dead technical one is “the bottleneck is in capacity.” To restore the technical is to admit the promise was a lie. Admission is not in the sprint.

In the margin

In the rooms I sat in, the most expensive move was not “let’s not do Redis,” but “let’s tell the customer the epic was named wrong.” The customer could be external. Could be a VP. Could be me a month earlier. Myself — worse: the interpreter had already written “I always wanted a platform.”

7.2 Contract, regulator, “we have no other way”

Sometimes Y is not a whim. A contract requires this log, this format, this retention. A regulator requires this checkbox. Hardware requires this driver. Then “what you actually need is X” is the Einstellung of the noble, already described. The distinction of this chapter: even when Y is a whim,

it can be an irrevocable whim on the horizon of a quarter. Irrevocability is a fact of power, not a fact of the world. To confuse them is to think a gate is always technical.

The wicked chapter said: the ends are fighting. Here we add: one end already won on a slide, before the room. There was a fight. It was hidden by the naming of the work. An engineer enters the field after a victory and takes the victory for the posing of the task.

7.3 What the years buy, if this is so

Not only N+1 scenes. Also the ability to read whose order this is and whether it can be renamed on this horizon. Sometimes it cannot. Then climbing is adult work, not blindness. Blindness is climbing while thinking you are solving X. Honesty is climbing knowing you are solving someone else’s Y, and putting in the debrief that X was not chosen.

This is unpleasant for the romance of the gate. The romance assumes the right question is available to the room. Often it is available only to a person who is not in Slack. Then twenty years also buy the knowledge of when a pause is useless until the slide changes. Cynical knowledge. Cynicism is not a piece. Cynicism is a map of power. Without the map a pause looks like character and dies as character.

The following chapters on leftover will read differently: some “Thursdays” are held not by forgetfulness, but by the fact that you cannot forget an order without rewriting a presentation. NIST passwords are a mixed case: an audit order plus a forgotten question. Microservices are mixed: a blog plus someone else’s OKR “be like the serious people.” The pure case of an order is an epic you cannot rename because it has already been counted in someone else’s metric.

Part II. The Library

8 What Remains of Thousands of Situations?

If you ask a person with years on the job what they learned in 2009, they will remember two stories and one

scar. The rest is as if it were erased. And yet they react to a new task as if 2009 had gone nowhere.

Where does what they cannot retell go?

The convenient answer: into “experience.” The word again does nothing. We need a mechanism, even a rough one, or we will keep carrying a box with a label.

8.1 A board they show for five seconds

In the 1940s Adriaan de Groot showed chess players a position for a short time and asked them to reconstruct it. Masters reconstructed meaningful positions almost entirely. Juniors — fragments. On random positions, where the pieces are set without a game, the masters’ advantage fell sharply.

Chase and Simon in 1973 took this apart as perception, not as “better memory in general.” An expert encodes the board in familiar chunks: typical relations of pieces, defenses, pawn structures. A chunk occupies almost as much of short-term memory as a single piece does for a junior, only inside the chunk there are already five or six related elements. So five seconds for a master are not five seconds of a more powerful computer. They are five seconds of a person who sees words where another sees letters.

Fact

Chase & Simon, “Perception in chess,” Cognitive Psychology, 1973. They caught chunk boundaries by pauses in placing the pieces. Later Gobet and others refined the sizes and spoke of templates larger than the original “chunks.” There is a debate about details. The picture “an

expert sees structure, a junior sees elements, and on random positions the advantage melts” holds.

8.2 What they actually did, not “the wisdom of chess”

De Groot did not ask “what do you feel.” He laid a position down for a few seconds — in different series from a couple to a dozen — took it away, asked for a reconstruction. Then, separately, thinking aloud on a move. Reconstruction showed perception. Aloud showed that a master already sees plans while a junior is still counting pieces. Perception is not a prelude to a decision. It is already a compressed decision.

Chase and Simon added a stopwatch to the placing: a pause of two seconds or more — a chunk boundary. A master places a patch of related pieces, is silent, the next patch. A junior places one by one,

pauses in other places. On a random board the master’s patches fall apart: the relations the patch was molded from are not from a game. The control is not to humiliate the master. The control is so you cannot say “just better memory.”

Transfer: when I open someone else’s service and in a minute “see a transaction through HTTP,” I did not recall 2011. I placed a patch. If the patch is from my game, and the repository is from another — I am a master on a random board: confidence remained, the chunks fell apart. The chapter on transfer will finish this. Here the method is enough: a chunk is a measurable pause of attention, not an aura.

This is unpleasant for romance and useful for our task.

Twenty years of Java do not give a magic register called “wisdom.” They, if you were lucky with

feedback, give a huge vocabulary of chunks: what a connection leak looks like, how a transaction through HTTP smells, what a “microservice” looks like that is really a distributed monolith with latency instead of a stack. When a new system arrives, the expert does not recall 2009. He sees the current already assembled. A junior sees files.

So “I don’t remember the situations, but I react” is not mysticism. Memory for episodes and memory for structures are not the same. An episode can be forgotten. The structure of which that episode was an example may have remained.

8.3 Tacit, silent, procedural — not synonyms

Here it is easy to make a mash of terms. Don’t.

Polanyi spoke of knowledge that does not separate from skill and does not lay out as propositions. Reber showed this on artificial grammars: strings of letters generated by a hidden rule; after examples people begin to distinguish “like those” from “not like,” and cannot formulate the rule. Whether they hold an abstraction or a memory of examples is not a closed debate. For us it is enough: a regularity settles without a report. Ryle long ago split knowing that and knowing how. In the psychology of memory — declarative and procedural, with neuropsychological double dissociations we will not pretend here to know in detail.

For an engineer a rough triple is enough:

What you can say in an interview and in a textbook. “An index on a foreign key.”

What you can do with your hands, explaining badly. “I wouldn’t open a transaction here.” Often it is explained later, and the explanation is even sometimes true.

What changes what you will even notice. Not an answer and not a skill, but a field. You walked into a repository — and already not those files light up. This is closer to Chase and Simon’s perception than to the skill of riding.

The third is most often what people mean when they say twenty years cannot be dumped into Notion. The first dumps. The second partly — through pair work, if you watch not the words but the hands. The third transfers worst of all, because to the person it seems “it’s obvious.” It is obvious to him. Not to the other.

Disputed

How far the third is separate from the second is an open question. You can say: attention is simply very fast knowing how to look. You can say: until you have recoded the scene, you have nothing to apply the skill to. I lean to the second, following how de Groot understood perception in chess at all: perception already is the solving of the task, not a prelude to it. But that is a preference, not a fact.

8.4 Feedback or years in a chair

If a library of chunks grows from situations, what matters is not the date in the labor record. What matters is the density of situations with a clear outcome.

Gary Klein, when he describes experts in natural conditions, stresses: intuition works where patterns repeat and the world answers. Fire answers fast and

cruelly. Long-term architecture answers late, by other people's hands, through turnover of people. Kahneman and Klein in a joint 2009 paper ("Conditions for Intuitive Expertise: A Failure to Disagree") settled on the boring: where the environment is regular and feedback is good — you can trust intuition more often; where not — more often you cannot.

Fact

Kahneman & Klein, American Psychologist, 2009. A rare case of two people carried as opposite flags writing the conditions under which they do not argue. High validity of the environment + a chance to learn from it. Chess, fire, anesthesia closer. Long-term political forecasts, hiring "for culture," architectural bets on three years — further.

This at once spoils the phrase “a senior always feels it.” A senior who has spent twenty years putting out incidents well may have an excellent library of incidents and a poor library of products. A senior who has spent twenty years winning design reviews may have a library of persuading a room, not a library of systems that live. Chunks can be about technique and about power. They are not labeled which one just fired.

8.5 Not chunks, but almost scenes

Later work by Gobet and Simon already spoke not only of chunks of three to six pieces, but of templates: larger structures with “holes” into which the current position is slotted. For us this matters carefully. If an expert holds not a scatter of tricks but types of situations with empty places for details, then “recognizing

a kitchen” is not a list of cues, but a scene with roles. Klein’s expectancies (heat like this, noise like this, water does this) sit in the scene, not in a separate table.

From this follows something unpleasant for transfer. You can show a junior three examples of N+1 and hope a template “slow after a release in a loop” arises. You can show thirty, and the template becomes more stable. You cannot show a template by a slide “here are the signs of N+1”: a slide is declarative knowledge, it lives in the first box. On the board in a room with Redis what will fire is not the slide, but how far the scene “slow after a small thing in a loop” is already assembled as a type.

How many examples you need is unknown and, more important, not one number for the profession. Chess is relatively regular: rules are stable, feedback in minutes. Service

engineering is a mix of the regular (GC, deadlocks, N+1) and the rare (this political configuration of stakeholders, this legacy gateway). The library will always be holey. A hole is not a disgrace. The disgrace is not knowing where the hole is, and wearing silence as a gift.

8.6 Can this be sped up

The six-month textbook stands on the claim that part of the profession is a set of transferable skills. The question of this book is about the remainder.

The remainder is not sacred. Part of it can be sped up, if you do not try to dump in words what does not live in words.

A good code review shows not a rule but a chunk: “this place smells like this because of this invariant.” A postmortem, if it is not a ritual of repentance, puts into the library a

binding of symptom—cause—frame. Pair programming is an expensive way to let someone else's attention work on your scene. Specially chosen tasks — what Anders Ericsson called deliberate practice: not years of repeating the convenient, but work at the edge of skill with feedback.

Pop science made of Ericsson "10,000 hours." He himself did not formulate it as a guarantee of genius, and meta-analyses (Macnamara and colleagues) show that the share of variance practice explains is smaller than the poster promises. For us it is enough: practice with feedback builds chunks; hours without feedback build confidence.

In the margin

Hence an unpleasant conclusion for mentoring. Telling a junior "think about context boundaries" is almost

useless until he has scenes in which those boundaries already hurt. Showing

scenes is useful. Pretending that after ten scenes he has a library like after four hundred is harmful. There is acceleration. There is no omnipotence.

8.7 What follows from this for the two “I don’t knows”

If an expert’s perception is already assembled from chunks, the first state — “I do, then I’ll explain” — looks almost boring. A chunk is recognized, the action usually bound to it surfaces. The report is built separately, if it is demanded.

The second state has to be somehow bound to the same assembly. Otherwise we will have to introduce a new organ. The next chapter tests the most boring version: the second is when the assembly does not fit.

Not a new gift. A miss on a prototype, felt as “something is wrong with the question,” because the question already contained the prototype.

If the boring version holds — the romance of the gate will lose weight, but a mechanism will appear. If it does not — we will have to look for how spoiling a frame differs from “simply not like the known.” That is more honest than choosing the beautiful in advance.

8.8 A paradox they bypassed without an explanation

Polanyi: we know more than we can tell, and therefore some skills cannot be handed over as a rule. David Autor in 2014 reminded us that machines bypassed this not by fishing a rule out of an expert, but by eating examples. The expert still

cannot tell. The machine still drives — where there are many examples and there is feedback.

For a library of scenes this is a knife. A chunk that “cannot be dumped into Notion” is partly dumped into a corpus of other people’s diffs and other people’s posts. Not your chunk. A statistical neighbor. The neighbor sometimes works as an anchor. Sometimes as leftover from a warehouse. The chapter on the public library will finish this. Here it is enough: “cannot tell” no longer means “the machine cannot.” It means “we cannot check whether this is the chunk until the world has answered.” An expert’s report and a model’s complete diff stand in the same queue for a check by the floor. The queue is humiliating for both — and honest.

9 The Library Does Not Travel to the Next Room

Chase and Simon ran a control that the book is obliged to carry next to the romance of chunks. On a meaningful position a master sees words. On a random one — letters, almost like a junior. The advantage is not “better memory.” The advantage is that this board is like the boards the library was fed.

Twenty years of Java do not buy sight on “systems.” They buy sight on the regularities that fed it. A neighboring regularity may be empty. Then a senior is a junior with a senior’s confidence. That is worse than a junior.

9.1 A random board is not a metaphor

De Groot and then Chase and Simon did not arrange mysticism. A short presentation, a reconstruction, a

comparison with a position where the pieces stand by the rules of the game but without a history of a match. As soon as the history vanishes, the chunk vanishes. The chunk held on repeated relations, not on “chessness as a property of mind.”

The transfer into engineering is direct and insulting. A person who for twenty years saw N+1 in Hibernate monoliths sees N+1 in the next Hibernate. In a foreign runtime without lazy collections he sees “slow” and proposes Redis — because Redis is in the library of “slow.” The frame arrived from the next room. The next room was similar enough to deceive.

Fact

Chase & Simon, 1973: masters’ advantage on random positions falls sharply, sometimes almost to the level of weak players — depending

on protocol and strength. Gobet & Simon (1996, 1998) refined: not always all the way “to zero”; large templates sometimes live a little. The direction is stable. For us it is enough: a chunk is bound to the structure of the environment, not to years as a number.

Shanteau spent years collecting in which domains experts stably beat juniors, and in which “expert” is a social role. Weather, chess, soil — better. Clinical forecasts without a table, politics, hiring — worse. The difference again is not in the experts’ IQ. In the regularity of the world and in whether the world gives feedback you can bind to an action. This is the same Kahneman and Klein bridge, only with a table of professions instead of two names on a cover.

9.2 Transfer that looks like intelligence

The industry loves “a strong engineer” as a portable organ. A vacancy writes: any stack, thinking is what matters. An interview tests portability on a whiteboard. The whiteboard is a random position relative to your prod. Success on a whiteboard predicts success on a whiteboard.

A person moves from a bank to streaming. In the bank the library is: regulator, idempotence of postings, “better boring.” In streaming: latency, a fan who refreshes the feed, “better alive.” Chunks about boredom in a living world work as Einstellung: a familiar mate “let’s do two-phase commit” closes off a shorter “losing one event is cheaper than waiting.” The person is not stupid. He sees his board on someone else’s.

The reverse: a person from streaming comes to a bank and sees “legacy.” The word is already a license to demolish. A bank seam often holds a question that is not in a blog about event sourcing. Demolition is vandalism dressed as seniority.

Counterexample

It does not follow that “sit in one firm forever.” It follows: a change of room zeros part of the library, and that has to be counted as a cost, not an upgrade of character. A person who changed five domains in ten years may have a library of how to enter the foreign — those are chunks too, about the first weeks, about how to ask. That is not a library of your deadlock. To confuse “orients fast” with “sees your basement” is Meehl’s clinician’s error: the vivid (speaks confidently) instead of the predictor (put out this).

9.3 Language as false portability

Java is portable. Chunks on Java seem portable because the syntax is the same. Syntax is letters. Scenes are words. Words are bound to a domain: “application,” “balance,” “idempotence,” “quote window.” A person reads someone else’s service in a familiar language and thinks he sees. He sees grammar. He completes the scene from his library. Completion is the place where Redis appears “obviously.”

So “twenty years on the JVM” on a résumé is a weak signal for a room putting out not a JVM but this product. Weak does not mean zero: anchors (transactions, GC, pools) travel. Scenes of “how Friday falls over for us” do not. A review that respects anchors and does not confuse them with scenes is rare. More often they respect the tone.

9.4 What this does to the book’s original question

“What do twenty years buy that is not in a syllabus?” — the question is underdetermined until it is said: twenty years of what. A syllabus tries precisely to be portable. The portable is anchors. The non-portable is scenes. A book that promises that years buy a gate “in general” repeats a vacancy. A gate, if it happens, happens in a familiar regularity. In an unfamiliar one a person is either honestly dumb or dangerously fast.

The dangerously fast is the one they call into the next room as a hero. The hero brings a fence with him. The room is grateful: the fence looks like experience. The floor will collapse later, and the slide will write “did not account for specifics.” Specifics were what was not in the

library. To account for them would have meant not recognizing a kitchen.

The following chapters on hit and miss make sense only after this caveat. A miss is a scene that failed to come true. If there is no scene, there is nothing to come true. A person in a foreign room is not “in a pause.” He is in Schön’s hole: there is not yet a frame. A hole and a miss can be subjectively similar. Mechanically they are different. To confuse them is to give novelty the status of a gift.

10 Hit and Miss

In 1985 Gary Klein and colleagues took apart how fireground commanders make decisions when there is little time and the price is people and a house. The expectation was almost laboratory: experts will compare options, weigh, choose.

That is how decisions are studied when they are studied on students and tasks about urns.

It came out otherwise. They took apart incidents, then decision points inside them — more than a hundred points (in a later publication with a postscript — 156). Simultaneous comparison of two options almost did not happen: at most points the commander said he did not remember a decision as a choice — “just knew what to do.” He recognized the situation as typical and took the typical action for it. If the typical did not break in a mental simulation — they went. If it broke — they took the next typical, without holding a tender.

This is not a laboratory with a control group. These are interviews after fire, with memory distortions and with the risk that the commander is telling a heroic version of himself. Klein knew this:

the protocol was an after-action interview, not a camera on a helmet. The cues “quiet, hot, water” were restored by questions after, not by a transcript of the moment. For RPD as a field model that was enough: the picture fitted how experts in other dirty environments describe themselves. For our chapter on the report after, this is a caveat, not a cancellation: the kitchen case is stronger as the structure of a miss than as a proven neural protocol of that night.

So the recognition-primed decision model appeared: not a choice from a menu, but recognition.

For our book what matters is not “experts do not optimize,” but that inside recognition there already lies a trap and already lies a gate. One device, two outcomes.

10.1 A kitchen that was not a kitchen

The story Klein tells more often than others is almost too convenient. That is exactly why it has to be told carefully, as a case, not as a parable.

A lieutenant takes a crew to an ordinary, by the look of it, kitchen fire in a one-story house. Water should knock the fire down. It does not. The heat in the living room is more than it should be for such a kitchen. And it is quiet. Fires make noise. For such heat the silence is strange.

He does not build a hypothesis “basement.” He does not suspect a basement. He does not like staying in the house and gets people out. The floor collapses. The fire was from below. The living room was heating from the floor, the noise was

muffled by the joists, the water into the kitchen was going the wrong way.

He himself later talked about a sixth sense. The debrief restored the cues: water does not work, too hot, too quiet. The kitchen-fire pattern did not fit. The action was not “fight it differently,” but “get out and reassemble the picture.”

Fact

Original study: Klein, Calderwood, Clinton-Cirocco, mid-1980s work, later published with a postscript in Journal of Cognitive Engineering and Decision Making (2010). RPD model: situation → prototype (goals, cues, expectancies, typical action) → if expectancies are violated — reappraisal, not persistence at any cost. The kitchen case is an illustration of violated expectancies, not a separate supernatural channel.

Look at what this does to our two “I don’t knows.”

This is not the first: he did not know what to do in the sense “here is the right hose, here is the point.” He did not issue an expert move inside the frame “we are fighting a kitchen.”

This is very like the second: the frame “kitchen fire” became shameful, though the shame did not yet have a name. The question “how do we fight this kitchen” turned out to be wrong. Better was “what is burning here and are we standing on a floor.”

And yet no second mechanism is needed. In an RPD prototype there are already expectancies. When they do not come true, the system is not obliged to issue an action. It can issue alarm and a refusal to continue the typical. The alarm is subjectively objectless, because the violated expectancies have not yet assembled

into a new scene. Objectlessness is not proof of magic. It is the interval between “this is not it” and “here is what this is.”

10.2 Not a menu, and not magic

The laboratory loves a menu: here are three urns, choose. RPD says: in the field there often is no menu. An expert generates the first plausible move from a prototype and checks it by a run- through of “what happens if.” If the run-through breaks — the next typical, not a tender. Simon called this satisficing: good enough now, not best in eternity. Under a burning ceiling this is the right trade. In a design review for a quarter — already disputed: there is time, and the habit of taking the first typical remains.

Kahneman read intuition as a source of systematic errors. Klein — as what experts in the field cannot work

without. The 2009 paper where they “failed to disagree” is exactly about conditions, not essence. High validity, good feedback — intuition is more often a friend. Low — more often a self-confident enemy. Engineering jumps between these environments in one day: a morning incident (high, cruel) and an afternoon “strategic” quartet about a platform (low, polite, feedback in a year).

To confuse them is to carry a fire pause into a product room as proof of depth. In a product room a pause may be Schön’s: there are no ends, there is no prototype, and a senior’s silence is

not a miss of the library, but the absence of a scene. There is nothing to demand of him — Klein’s “get the people out”: there is no one to get out, nothing is burning yet except the budget.

10.3 Why a miss feels like spoiling the question

The question asked in a room is rarely naked. “How do we fight the kitchen” already contains an ontology: there is a kitchen, there is fire there, there are we here, there is water as the answer.

When the cues do not fit, it is not the answer that breaks. The ontology breaks. The person feels they are asking about the wrong thing, because they are asking from inside a scene that does not glue.

In engineering it is the same, only the floor rarely collapses — more often a year collapses.

“How do we scale this service?” contains: the service must be scaled, the bottleneck is in it, the answer is in capacity. If a senior feels something is off, the off is often in one of the nestings. Maybe the bottleneck is in an expectation of

consistency that scale does not cure. Maybe the service should not be scaled, the call should be taken out of a loop. Maybe “scale” here means “we have a deadline and we want to look serious.”

A pause is a refusal to play in the proposed scene. From the outside it looks like evasion. Inside RPD it can be an honest miss on a prototype.

10.4 A hit next to it, so we do not forget

So as not to fall in love with a miss, a hit has to be placed next to it.

The same commander at the next fire recognizes “a typical attic,” puts a hose where it should go, and nobody writes books about it later, because nothing “strange” happened. This is the first “I don’t know”: there is an action, a report later, if they ask. The library fired with a plus sign.

An engineer looks at a GC graph and already knows this is allocation on a hot path, not “need to increase the heap.” Hand to the code. Explanation at standup. Classic tacit.

Both times recognition works. The difference is whether the scene fitted.

Then the practical meaning of distinguishing A and B remains, and the metaphysics of two gifts leaves. In a room it is still important to hear whether a person is issuing a move or spoiling the board. But to cultivate “the ability to find a gate” as a separate skill, torn from a library of scenes and from the right to disagree with the ontology of a ticket, is suspicious. A gate here is the name of a miss that was given time.

10.5 Where the boring hypothesis is thin

It is thin in several places, and they cannot be hidden.

First. Not every spoiling of a question is a miss on a fire prototype. A person may dislike a frame for reasons that have nothing to do with a library of technique: status, trauma of a past project, fashion. RPD does not distinguish this. The feeling is one.

Second. Laboratory insight (recoding a rope into a pendulum, suddenly seeing) may be a relative, or may be a neighboring species. Ohlsson describes insight as a change of representation from which the applicable operators change. Like a change of ontology. But insight problems are often short and closed. Fire and architecture are not. You

can carry an “aha” from a nine-dot puzzle onto a reorganization of a payments loop only as an analogy.

Third. Klein restored the cues after, by questions. We are already warned by the chapter on explanations: a story after the fact knows how to be smooth. Maybe the silence and the heat are real chunks that fired. Maybe this is the best story that can be assembled once the floor has already collapsed. Klein’s study is better than laboratory student menus, but it is still a reconstruction.

Disputed

The bet “A and B are two outputs of one library” is for me now stronger than the bet “two mechanisms.” It is simpler, it closes the kitchen fire without ESP, and it does not require System 1. It is not proven. Its main enemy is not romance, but the possibility that a change of frame

(Schön, problem setting) is still another skill: not “the prototype did not fit,” but “the ability to live while there is not yet a prototype.” In a fire the prototype was already there and broke. In genuine uncertainty there may be no prototype at all. Those are different holes.

Schön is important here precisely as a contrast. In Klein the expert is full of prototypes. In Schön a professional often stands in a situation where the ends are unclear and a technical solution has nothing yet to apply to: first you have to name what kind of case is in front of you. Naming is not a miss, but work without a ready chunk.

Engineering lives in both modes, and we love to pretend it is only in the first. An incident is Klein’s world. “What we build next quarter if the market itself does not know” is Schön’s. A pause in them looks similar and means different things.

10.6 What changes after this chapter

At the beginning you could think that “the feeling of a wrong question” is a special human organ, and that is what twenty years buy.

Now it is more honest like this:

Twenty years, if the environment teaches, buy a library of scenes with expectancies.

When a scene is recognized — you get an action without a report.

When a scene should have been recognized and was not — you get alarm without a name. You can use it as a reason to change the question. You can ignore it and complete a ladder. You can take it for prophecy.

When there is not yet a scene, there is alarm too, but this is no longer a miss of the library, but work without a library. It is easy to confuse with

an expert miss. A junior in a Schön hole and an expert in a Klein miss are both silent from the outside.

And there is a fourth silence: the expert recognized the wrong scene, confidently, and for him there is no gate, because the ladder already stands in his head as the only move. The next chapter is about that. Without it this one would have looked like a panegyric to the pause.

11 Can a Miss Be Reduced to a Prediction Error?

After Klein you want a neuron. A fashionable frame of the last fifteen years: the brain is a prediction machine, what goes up is not a picture of the world but a prediction error (“surprisal”). What did not match the model is what is worth processing. Andy Clark in a 2013 review assembled this as a candidate

for a general theory of perception and action. Friston — still more aggressive.

This lies suspiciously smoothly on our two “I don’t knows.” A hit: error is small, the scene is “explained,” the action typical. A miss: error is large and is not explained by the available scene — a feeling of “something is wrong” without a name. Klein’s silence and heat are exactly what a kitchen model did not predict.

Smoothness is a red flag.

11.1 What the frame gives

It gives a common language for expectancy and the breaking of expectancy. You do not need two organs. One generative model of the world (at the level of scenes — our library), one signal “it did not fit.” A pause is not magic, but an error for which an explanation has not yet

been found. Leaving the house is an action that changes the input so the error falls (saw the basement — the model changed).

It gives a bridge to monitoring: a green dashboard with a dead log — the error is not in the red, but in the absence of expected noise. Klein's silence as an engineering object.

11.2 What the frame does not give

It does not give the right to say: we found the neuron of the gate. Predictive processing is a large, disputed, often poorly falsifiable frame. It explains too much: perception, action, psychiatry. When a frame explains everything, it explains weakly. For engineering there is no experiment "here is your prediction error on a Redis ticket."

It does not distinguish a miss from alarm without a library (Schön's hole: there is not yet a model, the error has nothing to apply to). It does not distinguish a miss from moral distaste and from status. Subjectively similar.

It does not say which model is predicting: a technical scene or Conway's org chart. An error "this seam should not exist" is not sensory. The frame is about perceiving the world, not about the politics of departments. You can stretch it. The stretch will show.

Disputed

I leave prediction error as a possible substrate of a Klein miss, not as a found one. It is not required for the rest of the book: expectancies and their breaking at the level of behavior and interviews are enough. Whoever wants a neuron — here is a candidate, fashionable, not proven

here. Whoever does not — can strike the chapter, and the fence and the gate will not collapse.

11.3 OCD as a warning

In psychiatric retellings predictive coding is sometimes laid onto OCD: too precise, too loud an error of “something is wrong,” rituals reduce it. This is not our topic and not a diagnosis of a room. It is a warning: the signal “it did not fit” is not always a friend. It can scream without a basement. A person who spoils every frame may be an expert with a good library. May be a person whose error is too loud. From the outside at standup they are indistinguishable. The romance of the pause does not like this. That is why it is more needed.

11.4 Why the chapter

So as not to leave a hole “and the brain?” The brain may be doing what Klein described by interviews. It may not. The book does not become neuroscience if it honestly says: there is a beautiful isomorphism, there is no experiment on our ticket. Further — rooms where the isomorphism is not needed in order to see blindness and prostheses.

11.5 The ambiguity of “model”

The word confuses. A generative model in predictive processing is the brain’s hypothesis about the causes of input. Our “library of scenes” is the same word in everyday use. The isomorphism tempts identification. The identification is not earned. Klein built prototypes from interviews. Friston writes free-energy equations. Between them

there is no bridge we have the right to cross in an engineering book without becoming a sermon.

Jakob Hohwy and others argue how far the frame is an empirical program at all, and how far a philosophy of perception with a mathematical accent. I am not a judge of that argument. The judge for us is simple: if you throw the chapter out, hit and miss remain. If you do not — a candidate substrate appears, and a warning not to wear it as a found Redis neuron.

One more thing the frame does not fix: Einstellung. There the error is small. A familiar mate is predicted well. A short mate does not enter the hypothesis, so it does not create surprisal until it is shown by force. Klein's miss is a large error on a living scene. Bilalić's blindness is a small error on a good ladder. To reduce both to one "prediction error" is again to glue different

things with one smart noun. The book is precisely learning not to do that.

12 When There Is No Right Question

Until now a gate assumed that somewhere there is a better question: not the kitchen, but the basement; not Redis, but N+1. Sometimes there is no better. There are several incompatible “goods,” and the choice is not finding X, but politics.

Horst Rittel and Melvin Webber in 1973 named this wicked problems: a task has no definitive formulation, no stopping rule, no true/false, only better/worse, and every intervention changes the task itself. City planners knew this. Engineering pretends it knows only the tame: a bug either is there or is not.

12.1 Where a ticket pretends to be tame, and it is wicked

“Need to scale for Black Friday.” The formulation looks tame: there is load, there is iron. Inside: whose budget, what risk of failure, what user experience, who is on call, what was promised in a press release, whether you can say no to marketing. There is no frame in which all goals at once. Putting in Redis chooses some. Saying “we’ll cut the storefront” — others. The gate “the right question” is false here: there are several right ones, they fight.

Schön already touched this: until the ends are clear, technique has nowhere to apply. Rittel is harsher: the ends may never become clear, and you still have to act. A pause then is not a search for a basement. A pause is a refusal to pretend the task is tame. A room hates this refusal more than “this is N+1”:

N+1 promises an end to the conversation. Wicked promises that the conversation is the work.

12.2 What Rittel listed, and what of it is engineering’s

A wicked task, in their list, has no definitive formulation: while you solve, you understand you were solving the wrong thing. No stopping rule: you cannot say “done” without politics. Solutions are not true or false — better or worse for someone. No immediate test. Every move is irreversible in the sense that it changes the task itself: after Redis the world is already not the one in which Redis was chosen. No right to a trial run without a price. Each wicked task is essentially unique, and “like they have” lies. You can describe symptoms, you cannot write a full posing in advance.

Not the whole list has to be in every ticket. Three are enough for the frame “we’ll find the right X” to become a lie: several legitimate goals, a move changes the field, no common judge. “Scale for Friday” often has all three, if Friday is marketing, the budget is someone else’s, failure is someone’s on-call. “Fix N+1” usually does not: one goal, the move is reversible, the judge is a graph.

To confuse them is the main leftover of methodologies. SAFe, RUP, a “decision framework” promise a tame process for a wicked field. The process gives a stopping rule: the sprint ended, the epic is green. A stopping rule is a political act dressed as ceremony. Ceremony is needed. To lie that it found X is not.

12.3 The danger of our favorite phrase

“The right question matters more than the right answer” assumes that a right question exists. In a wicked field the phrase becomes a ladder: a person with years on the job endlessly renames so as not to choose. Renaming is sometimes a gate. Sometimes cowardice before politics, dressed as Klein.

Meehl’s formula does not save either: there is no stable outcome to count. Conway does not save: an org chart is one of the fighting ends, not a judge. Chesterton does not save: there are many fences, each with its own dead.

Counterexample

Not everything unpleasant to decide is wicked. Engineers love to declare the political “complex” so as not to cut. N+1 is tame, even if they scream about it. A test: can you say

in a week it got better on a chosen end, without pretending the ends were all of them. If you can — do not hide in Rittel. If any move makes worse on another end and that is not laziness — there it is.

12.4 Schön’s hole and Rittel’s wickedness

Schön: until there is a frame, techniques have nowhere to apply. Rittel: a frame that would close everyone may not appear. Schön’s pause is waiting for a name. Rittel’s pause is a refusal to pretend the name is shared. The first may end in a basement. The second ends in a compromise everyone slightly hates.

An engineer trained on the first, in the field of the second, hunts for a basement and infuriates the room: “he’s philosophizing again, we just need to choose.” The room is right if the task is tame. Wrong if he is the only one who sees the fight of ends

and will not let it be packed into Redis. From the outside indistinguishable. The test from the box above is the only thing I have. It also lies when the ends are not yet named.

12.5 What twenty years then buy

Not a secret X. But the ability to see a fight of ends and not pack it into one noun too early. This is closer to politics and to Schön than to chess chunks. Chess is tame. A product on a market, a regulator, two teams with different bonuses — not.

Six months almost do not teach this: a textbook is obliged to be tame, or a junior will not enter. Twenty years can teach, if the feedback was about a fight, not only about a stack trace. They can

fail to teach: a person spent twenty years putting out tame fires and in a wicked room behaves like a firefighter. Pours water. Water here is someone else’s budget.

12.6 A link to the fence

A fence in a wicked field is often needed: without it you cannot start. A temporary frame, in order to go. The trouble is not the frame. The trouble is forgetting that it is temporary. Leftover is a tame answer that outlived its tame task. Wicked leftover is worse: a temporary frame became a moral in a field where a moral is always someone’s.

A pause here means: do not christen a frame into a law. You do not necessarily find a gate. You must remember that there are several gates and they lead into different yards.

13 Debugging: A Small Miss Every Day

The book so far has fed on rare frames: a design review, a kitchen, a smothered mate. For most of the hours an engineer is not choosing Redis. He looks at a symptom and hunts for which fire this is. Debugging is hit and miss at a small scale, without a slide and without heroism. If the mechanism exists, it is obliged to be here. If it is not here, it is not.

13.1 A stack trace already contains Y

It crashes. The stack trace points at a line. The line is Y. A junior’s hands fix Y: a null-check, a retry, a catch. Sometimes this is a hit: the line is the hole. Often the line is the place where someone else’s frame finally could not pretend. X is an invariant

above, a race, “this method was called from the wrong thread,” “the cache returned another tenant.”

A debugger’s pause, invisible from the outside: the person does not put a null-check at once. Looks at whether this null should have been impossible. If it should — the question is not “how not to crash,” but “which scene allowed the impossible.” This is a Klein miss on three IDE screens. Water does not knock the fire down: an NPE is fixed in one place and rises in another.

In the margin

The rubber duck was born here, not in a chat with a model. You speak a stack trace aloud — and hear that you are telling the scene “how to fix an NPE,” not the scene “who lied about an invariant.” A duck does not propose Redis. That is its virtue. A model will propose. The duck’s virtue will become scarce.

13.2 Reproduction as a change of question

“I can’t reproduce it” is not a refusal to help. Often it is the only gate. A symptom without reproduction is a story. A story is always already in someone’s frame: “after a deploy,” “at the client,” “only in prod.” The frame may be a ladder. Reproduction forces the world to answer outside the story. Sometimes the world answers: this is not after a deploy, this is after a full-moon cron job that did not exist in the story.

Teams that punish “first a minimal repro” save ten minutes and buy Redis. Teams that demand a repro as a moral are sometimes leftover: a repro of an impossible bug in a closed client system costs a week, and X is already visible in a metric. The moral “always a repro” is an answer to the

question “we were fixing stories.” The question is alive in some rooms. In others — already not, and the moral is.

13.3 printf as a theory

Every log you put “just in case” is a hypothesis about a scene: I think control passes here, and I want the world to confirm. An expert puts few points, because the scene has already narrowed where an expectancy can break. A junior puts many: there is no scene, he lights a corridor. Sometimes many points are the only way to assemble a first scene. Sometimes this is Einstellung “debugging = more logs,” closing a short one: read an already existing log everyone ignores because it is always red and has become furniture.

An always-red log is Klein’s silence in reverse. Noise that should not be noise. A person with years on the job

sometimes feels: this WARN has been screaming too long to be true or too long to be harmless. A junior filters WARN, because otherwise you cannot see ERROR. A filter is a frame.

13.4 A debugger and the false clarity of a step

Stepping through code is a powerful way to see that control is not there. And a powerful way to close a frame: you walk a familiar mate, the squares of someone else's short mate are not even on this thread's stack. Bilalić's eye-tracking is not needed here. You can see it by which breakpoints stand. They stand on what the person already counts as the scene.

"Leave the function and ask whether it should be called at all" is a cheap gate. It is rarely done, because a debugger invites you inside. Inside is a ladder. Outside is a question.

13.5 An hour in an NPE while the zone was lying

Prod crashed at midnight. Stack trace — NPE in a date parser. Frame: null, timezone,

SimpleDateFormat is not thread-safe, we have already fixed this a hundred times. Breakpoints

stood on familiar squares. An hour of stepping. There was no null. The format was correct. The test on the developer’s machine was green.

Silence: it crashes only in a time zone where midnight crosses the day boundary earlier than in Toronto. The expectancy “midnight is our JVM” did not come true. The scene was not NPE. The scene was “whose midnight.” The short mate: a log with user.timezone and one question

“where does the cron run.” The long mate: a familiar class of dates.

I was the person with the long mate. A junior programmer asked “and in which zone is the cron.” I already counted that he was looking the wrong way. He was looking. Eye-tracking not needed: my points stood in the parser. His question stood outside the function. The parser’s expectancies came true enough not to break. This is Einstellung on three screens, not a Klein miss.

13.6 What debugging does to “twenty years”

Years with feedback “put a hypothesis — the world answered in minutes” build a library of small scenes. That is exactly Kahneman and Klein’s condition. So a person who has honestly debugged a lot is often stronger in a pause at review than a person who has “designed” a lot. Design answers late. Debugging answers now. Chunks grow where the world answers.

This spoils a hierarchy of statuses. An architect who has not read a stack trace in a long time has a library of slides. A person with on-call has a library of nights. The room listens to the architect: he has the word “scale.” The floor listens to the second. A book that hunts for a gate only at a design review hunts for it on a scene where feedback is worse.

13.7 A link to the cheap answer, briefly

A model fixes a stack trace as Y: proposes a null-check, complete, with a test. Completeness closes the frame faster than a junior has time to ask whether a null should have been impossible. Bainbridge’s irony does not begin with an airplane autopilot. It begins with the small, on which chunks grew, leaving for an automaton. The large remains. The large without the small is a watchman at a green screen.

Debugging is the last place where the world still slaps your hands fast. If we give even that away entire, the library will grow from the genre "what fixes look like in a corpus," not from prod outcomes. The genre sometimes coincides with the world. Sometimes it coincides with leftover.

14 When Experience Closes the Gate

If experience only opened doors, seniors would err less than juniors on all tasks of their field. Sometimes that is so. Sometimes a senior errs precisely because he is a senior: a familiar move stands in the doorway and will not let a short one through.

This is not a moral "don't get conceited." This is a mechanics of attention.

14.1 Jugs

In 1942 Abraham Luchins gave people a task: obtain a needed volume of water, having three jugs of fixed capacity. First — a series of several tasks where the same long method worked: fill the large, pour into the middle, twice pour into the small. In notation this is B − A − 2C. For example: 127, 21 and 3, get 100. Then came critical ones: the same long still worked, but a short one was enough — A − C or A + C. Many of those warmed up by the long continued the long. A control without the warm-up took the short at once. Then extinction: the long no longer worked, only the short remained. Part of the warmed-up declared the task unsolvable. “Don’t be blind” — a warning after which more than half still found the short. So the stamp is not iron. The default of attention is strong.

The effect was named Einstellung — set, posing, “this is how we solve.” A method that has worked becomes not a tool but a frame. The frame hides an alternative not by malice, but by success.

Fact

A. S. Luchins, “Mechanization in problem solving: The effect of Einstellung,” Psychological Monographs, 1942. Later the effect was found far beyond jugs. An important detail of later work: people who were told “don’t be blind” partly came out. So this is not an iron stamp, but a strong default of attention. Another detail: high working-memory volume sometimes worsens exit onto the short, because it holds the long algorithm. Smartness is not immunity.

The jugs are insultingly simple, and an engineer has a right to say: we are not about water. Fine. Chess is closer to the pride of expertise.

14.2 A mate a master does not see

Bilalić, McLeod and Gobet in 2008 showed strong chess players positions where there is a familiar beautiful mate — often a smothered mate, longer, from textbooks and beautiful games — and a

shorter, less publicized path. Players found the familiar and declared it the shortest. When the same short idea was left the only one — they removed the familiar mate from the board, leaving the short — they found it without trouble. So the point is not that the short is unavailable to them. The point is that the familiar stands in the doorway.

A comparison inside one person matters more than a comparison of master with junior. The same player, at the same strength, in the presence of a familiar solution plays as if the rating has gone down. This is not “didn’t sleep.” This is measurable blindness at the expense of a good chunk.

Still more unpleasant is the record of the gaze. While the person says he is looking for another path, the eyes continue to walk the squares of the familiar solution. The search for “another” goes inside the old frame. He is sincerely looking for a gate and honestly looking at a ladder. The introspection “I am now thinking broadly” does not match where the light goes. This is a direct neighbor of the chapter on explanation after: a story of search can be an honest intention and still a false report of attention.

Fact

Bilalić, McLeod, Gobet, “Why good thoughts block better ones,” Cognition, 2008. The effect can be roughly measured in rating: in the presence of a familiar solution a strong player plays like a noticeably weaker one on a position where there is no familiar solution. Later the eye- tracking was replicated. This is one of the most direct blows to the fairy tale “an expert simply sees the truth.”

“A good thought blocks a better one” is almost a subtitle of our gate with a minus sign. A ladder is not necessarily bad. It is often good. That is why it is not demolished. A gate is better, but the good has occupied attention.

In code this is daily. A reactive stream, because “we do async this way,” though a loop and a queue would have been enough here. A cache, because “slow,” though it is slow from N+1, and the cache will

hide the N+1 until Friday. A microservice, because the team has already cut repositories in their heads. A familiar mate.

The person is not stupid and not lazy. He is successful inside a library. Success is the mechanism of blindness.

14.3 Prejudice of a pattern

The word prejudice sounds moral. Here it is almost technical: a prior judgment that arrived before the scene.

A pattern is a compressed bet: if cues A, then type B, then action C. Without such bets an expert is a junior again, five seconds on a random board. With bets he is fast — and knows in advance which cues need not be sought, because they do not enter B.

Einstellung shows: the refusal to look is not conscious. The gaze does not go to the squares of a short mate. The world narrowed before the decision.

So the question “when does experience help see a gate, and when is it sure there can be no gate” has a rough answer, not an inspiring one.

It helps when the gate is in the library as a type of scene: “this is the case where they ask us to scale, and we need to remove a call from a loop.” The expert recognizes the type and does not climb.

It hinders when the gate is a scene that is not in the library, and in the library there is a good ladder. Then the expert will build a ladder faster than a junior and will be more sincerely sure than a junior that he walked along. He walked. With his eyes on his squares.

A junior sometimes finds the short because he has no long. This is not the holiness of not knowing. This is the absence of a competitor in attention. Luchins already showed this with a control group. In teams it looks like “a stupid intern’s question that turned out right,” after which everyone laughs and nothing changes in the structure of review: next time the one who has a textbook mate will win again.

14.4 How this lives with the previous chapter

A miss on a prototype (a kitchen that is not a kitchen) and Einstellung (a kitchen that is too well a kitchen) are relatives, not twins.

In the first, expectancies break, and if the person endures that, the frame can be changed.

In the second, expectancies come true enough not to break. Water as it were works: the familiar mate is there, it leads to the goal, only longer. The system does not get a signal “this is not it.” It gets a signal “it is, we go.”

A gate requires that the good be not enough. Organizations punish that feeling: “but it works.” It works. Like a five-move mate when there is a three-move.

In the margin

Hence a practical nastiness, not advice from a self-help section, but an observation. Asking an expert “is there another path” while the familiar is visible on the board is almost useless: he will look for another inside the familiar. What is useful is to remove the familiar move — forbid a favorite tool, temporarily switch off the possibility of a cache, not give a budget for a new service

— and see what he will do. Bilalić did this with a board where smothered mate is switched off. In engineering this is sometimes the only way to learn whether a person sees a gate or only says he is looking.

The cruelty of the method is obvious. You cannot always switch off a cache in prod to check thinking. But in a design review you can: “suppose we cannot start a service. Then what?” If after that there is silence in the room, the library was about services, not about the problem.

14.5 Without a hymn to the junior and without a hymn to the master

A junior does not see chunks — blind to structure, fast to a random short that is not there.

A master sees chunks — sighted to structure, blind to a short that a good chunk has closed.

Both are blind, differently. Twenty years buy a change of type of blindness, not sight in general.

This, it seems, is the most useful unpleasant conclusion of the part about experience. An expert’s odd pause is sometimes a miss that saves. Sometimes it is a pause before a familiar mate that we took for depth. You cannot tell them apart from the outside by a face. You can tell by whether a person, when a good move is already there, can endure the question “and if this move is forbidden.”

If he cannot — he does not have a gate. He has a ladder he has learned to build fast, and that too is experience, and they pay for that too. It is simply not what the book started from.

15 Other Rooms

While the book lived in engineering and in one Klein fire, you could suspect that the odd pause is programmers' professional folklore. Walk into foreign rooms — and you see the same structure: a library of scenes, a hit, a miss, closing a frame too early. And you see that some professions train a refusal to close. Not always successfully. But at least they do not pretend a pause is character.

15.1 A doctor who already knows

Premature closure: a doctor has found a diagnosis and stopped looking. This is not laziness. This is efficiency. Chest pain plus a familiar age plus familiar factors — "ischemia," and then the ischemia protocol. Most days the protocol

saves. On the day of an aortic dissection it kills: cues that do not fit the scene are discarded as noise.

The literature on diagnostic error names premature closure one of the most frequent cognitive moves that end in a wrong diagnosis. The figures “10–15% of errors” wander through reviews; treat the exact figure carefully, the phenomenon not. Closing a frame is an expert’s work. Without it an ER stands still. With it — a blind spot Bilalić saw on a chessboard.

Fact

Premature closure in the clinical literature: a refusal to consider reasonable alternatives after a first diagnosis. Bound to anchoring. Not identical with “the doctor is stupid”: more often the doctor is experienced and fast. Checklists like TWED

(“what else?”) are an attempt at a prosthesis against closing, not a cancellation of the library.

The difference from Klein’s firefighter is important. The lieutenant’s expectancies broke (quiet, hot, water does not work) — and he left. Premature closure is a case where expectancies came true enough not to break. This is Einstellung, not a miss. Medicine knows both: “something is wrong, we reassemble” and “the picture fits, we don’t look further.” The second more often lands in debriefs with corpses, because the first sometimes saves and leaves no case.

Engineering loves the first as a tale and lives the second as a norm: “this is a classic timeout, we’ll increase it.” A classic timeout closes the frame. An alternative (a dead neighbor, DNS, throttling of someone else’s limit) does not enter the scene.

15.2 An airplane, a checklist, and distrust of a hero

Aviation after a series of disasters did a strange thing: it stopped believing that a commander's experience is a sufficient organ. CRM, checklists, "crew resource management" — prostheses against two things at once: against the silence of a junior who sees something off and does not dare spoil the captain's frame, and against a captain whose familiar mate closed a short one.

This is not a hymn to checklists. A checklist itself can be leftover: an item that answered an old accident lives after a change of fleet. But the institution admitted that a library of scenes lies predictably and that a junior's pause must have a slot. "I don't understand what we are doing" is a permitted line, not character.

In engineering there is often no slot. A junior who says “I don’t understand why Redis” hears that he does not understand Redis. Aviation at least formally says: not understanding the procedure now matters more than the commander’s face. Formally. Hierarchy is alive. But formality changes which misses are even spoken.

Disputed

Transferring CRM into software (“why don’t we have a deploy checklist”) is sometimes a caricature. A deploy is not a takeoff. The likeness is not in the ritual, but in the admission: an expert closes a frame early, and an organization can either live with that or build in a refusal to close. A checklist is one prosthesis. “What else could this be?” at a retro is another. A ban on estimating a ticket until X is named,

not Y, is a third. A prosthesis is not a gift. A prosthesis is agreement that a gift is not enough.

15.3 A laboratory that is not the field

The psychology of thinking loves tasks where there is a correct gate: nine dots, a rope and pliers, a disease and a test. There insight and Einstellung are visible cleanly. A doctor’s, a firefighter’s, an engineer’s field is dirty: there may be no correct gate, feedback is late, the price of error differs.

So “in the laboratory a pause does not predict insight” (Metcalfe) does not cancel “in the field a pause sometimes gets people out.” These are different machines of measurement. A book that takes only the laboratory will declare a gate an illusion. A book that takes only Klein’s tales will declare it an organ. You have to hold both: laboratory

blindness of expertise is real; field alarm at expectancies that did not come true is also real; neither licenses the other.

15.4 What you cannot steal from foreign rooms

You cannot steal the confidence that “doctors have it too, so we are normal.” The normality of blindness does not cure prod.

You cannot steal a checklist as cargo: draw a “pre-flight of deploy” and count premature closure closed. Aviation pays in blood for every item. We often pay an item for an audit’s calm.

You can steal the question they ask themselves aloud: what else could this be if my first frame is forbidden? Bilalić switched off smothered mate. A doctor writes a differential. A pilot demands confirmation. An engineer can forbid Redis in a thought

experiment for ten minutes. Ten minutes is cheaper than a year of cache on N+1.

15.5 A return to our room

Other rooms do not prove that a programmer’s pause is the same neuron. They prove that the structure “library → fast frame → blind spot” is not our invention. And that professions with a more expensive floor have already tried prostheses. We can look without copying the form. A form without their question will again become a Thursday.

15.6 Jerome Groopman and what a doctor himself tells

The book How Doctors Think is not an experiment. It is interviews with clinicians about their own misses, a genre closer to Klein than to Meehl. Groopman describes how a first vivid cue seats a diagnosis, and then a

story is written under it. A patient who “looks like anxiety” gets anxiety until it turns out to be thyroid — or the reverse. A doctor can later restore the cues that did not fit. At the moment they were noise.

What is useful for us is not the authority of a bestseller. The genre is useful: an expert can tell his premature closure after, and the telling does not cancel that at the moment the frame had already closed. An engineer after an incident can too. The ability to tell is not the ability then not to close. A prosthesis (a differential, TWED, “what else”) tries to insert a refusal to close at the moment, not in a memoir.

TWED in retellings: Threat, What else, Evidential, Dispositional — different expansions wander through textbooks of clinical thinking, I do not swear by the letters. The gesture is one: a forced slot “if my first frame is forbidden.” Bilalić forbade

smothered mate with a board. A doctor forbids a first diagnosis with a paper. An engineer can forbid Redis for ten minutes. A paper does not make a doctor Klein. A paper makes closing a little more expensive.

15.7 Dekker: error as the start of a debrief, not the end

Sidney Dekker in The Field Guide to Understanding Human Error and later in safety-II asks not to stop at “a person erred.” An error is a symptom of a system that made the error reasonable. This is dangerously close to justifying anything. Dekker does not forgive. He forbids a convenient end. A convenient end is leftover of a debrief: found a scapegoat, a rule “we don’t do that anymore,” the slide is closed.

The link to our library is direct. If a debrief ends with a person, the scene that made Redis obvious is not put into the shared library. Shame and a rule are. Shame is not a chunk. A rule is a fence. A field without a scene in the library will repeat Redis under a new name.

The next two chapters are not a comparison, but two floors with reports and names: a cockpit, then an operating room. Then a formula that beats a clinician. The romance of seniority will not survive them.

16 One Flight: The Light Closed the Tank

In the chapter on other rooms CRM was a prosthesis. A prosthesis comes from somewhere. It did not come from a slide about culture. It came from an hour in which an

experienced crew solved the wrong task until the airplane became a glider.

December 28, 1978. United Airlines, flight 173, DC-8, Denver to Portland. 189 people on board. Captain Malburn “Buddy” McBroom — 27 years with the company, tens of thousands of hours, thousands of them as captain of exactly this machine. This is not a junior at Redis. This is a senior whose library should have been there.

16.1 What the frame was

On approach to Portland they lowered the gear. A thump, vibration, yaw. No green light “gear down and locked.” The frame assembled at once: emergency landing, gear, prepare the passengers. The frame is reasonable. There was a thump. There is no

light. Klein’s lieutenant also started with a kitchen: the kitchen looked like a kitchen.

They went into a holding pattern southeast of the field and for about an hour diagnosed the gear and prepared the cabin. The gear was in fact already down. A retraction cylinder on the right strut had broken — corrosion, free fall, a microswitch of the light. The world said “locked.” The cockpit said “no light.” The cockpit won.

While the frame “gear” was open, kerosene burned. Gear down, flaps 15 degrees, five thousand feet, burn higher than cruise. The tank is not a symptom in the frame. The tank is a silence that should have broken an expectancy. It did not.

16.2 Who saw the tank and could not make it a question

First officer Roderick Beebe and flight engineer Forrest Mendenhall mentioned fuel. NTSB would later write a contributing factor almost bureaucratically: they either did not understand the criticality, or could not get it through to the captain. The cockpit transcript is not heroic. These are polite lines inside someone else’s frame. The captain’s frame is the light and a landing. A line about fuel in such a frame sounds like a detail of procedure, not a change of task.

Fact

NTSB, Aircraft Accident Report AAR-79-07, June 7, 1979. Probable cause: the captain did not monitor fuel and did not respond to crew warnings; attention was eaten by a landing-

gear malfunction and preparation for an emergency landing. Contributing: the two other crew members did not get the criticality through. Recommendation X-79-17 — resource management on the flight deck, participative management for captains, assertiveness for the rest. United launched the first CRM program in 1981. Investigation psychologist Alan Diehl directly bound this flight to Eastern 401 and to Tenerife.

At 18:15 local the DC-8 fell into a suburb, about six nautical miles from the runway, near East Burnside. There was no fire: the tanks were empty. Ten people died, among them Mendenhall and senior flight attendant Joan Wheeler. McBroom survived, lost his certificate, and until his death in 2004 carried the guilt. A slide loves a scapegoat. NTSB named a scapegoat and at the same time wrote that this was a

repeat: teamwork on the deck breaks predictably when a cockpit system gives one light.

16.3 The light as Redis

The engineering translation is almost indecent in its precision.

Symptom: no green. Ladder: diagnose the gear, prepare an emergency, don’t land until it’s clear. Gate: the gear is already down, time is burning, the question is “will we make the runway with what’s in the tanks.” They did not walk along for an hour. For an hour they built a ladder inside the light.

This is Einstellung, not a Klein miss. The frame’s expectancies came true enough: there was a thump, there is no light, an emergency-landing procedure exists, the captain knows it. Water as it were worked. The short mate — look at burn and at the clock — is closed by a good mate

“don’t land with doubtful gear.” The good mate was good in the first minute. After twenty minutes it was already killing. Organizations punish the feeling “it works, but not enough.” Here “it works” burned the kerosene.

Status from the chapter “who is allowed not to answer”: the first officer and the engineer had a slot to speak. They did not have a slot to change the captain’s question. A line inside the frame is legal. A line “we are solving the wrong thing” in 1978 deck culture is almost equal to mutiny. CRM will later try to make mutiny a procedure. The procedure does not always work. Without it the tank had no language.

16.4 A neighboring flight, the same light

Eastern Air Lines, flight 401, December 29, 1972, L-1011, Miami. Again a nose-gear light. The crew

went into the light. Autopilot disconnected. The airplane slowly descended over the Everglades. More than a hundred people died. The light, as they later found, was often simply a burned-out bulb. The gear was in place.

Diehl in 1979 saw a rhyme: 401, 173, Tenerife. Different machines, one frame “we are fixing this in the cockpit,” while the world does something else. Tenerife is not a light: the KLM captain

began a takeoff without clearance, fog, the first officer spoke about clearance not harshly enough. Again status plus a familiar mate “we are leaving.” The book does not unfold Tenerife as a second full archive: it is enough that the CRM prosthesis was assembled from several floors, not from one moral.

Counterexample

Journalism and Mayday love “the pilot fixated.” Julie Whipple in a 173 debrief shows United’s supporting decisions: fuel indication, strut maintenance, deck culture. A scapegoat is convenient. Convenience is a metric of closing, already familiar. For us both layers matter. McBroom closed the frame. The system made closing reasonable and someone else’s mouth quiet. Throw out the captain — leftover of a debrief. Throw out the light — leftover of technique. Hold both — the only honest postmortem.

16.5 What this hour does to our pause

McBroom was not a person who “cannot pause.” He made a pause relative to landing: did not land at once, went to diagnose. The pause was inside the light. A book that praises “wait” is obliged to eat this.

“Wait” is not free of a frame. You can wait the wrong question for a whole hour and burn the tank.

A junior at Redis proposes a ladder at once. A senior on 173 proposed another ladder: an hour of diagnosis. From the outside both “work the symptom.” A gate would have been: which scene should be coming true if the gear is already down — and does the silence of the tanks come true. The silence of the tanks is Klein’s kitchen silence. It had to have been heard sometime on a living burn. If burn is always “later,” silence does not smell.

Engineering copies this hour without wings. No green on a sensor — an hour in the sensor. A log always red — we filter, like WARN. Five thousand feet with gear down — “one more sprint until we understand.” CRM in Jira will not appear from our knowing Portland. It will appear if there is a permitted

line in the room: “we’ve been an hour in the light, and what’s the tank?” The line is cheap. A culture in which it is not mutiny — is not.

16.6 Why the chapter, if there were already doctors and checklists

Because doctors were a type. This was one flight, with names, with a report, with a number of dead, with a recommendation from which an industry grew. Like Burr’s passwords. Without such a case “other rooms” remain a polite comparison. With it you see the price of a closed frame when a captain’s library is huge, the light is familiar, and the short path — clocks and kilograms — did not enter the scene.

The next chapter is another floor, already without wings: an operating room, a tube, oxygen. Then a formula that beats a clinician. Two archives and Meehl’s table do the

same thing to seniority. The hope that years themselves will forbid a familiar mate, after them, looks like leftover.

17 One Operation: The Tube Closed the Oxygen

Portland was a light that closed the tank. Medicine in the book so far was a type: premature closure, TWED, Groopman. A type does not hold a floor. March 29, 2005 holds.

Elaine Bromiley, 37, two children, elective sinus surgery, ordinary health. Induction. The airway will not take. Saturation falls toward forty percent. This is a textbook anesthesia emergency: can’t intubate, can’t ventilate. The short path in DAS guidelines is known: oxygenation matters more than the tube, early — a surgical airway, not a twentieth attempt with a laryngoscope.

The room was a “dream team”: two consultant anaesthetists, an ENT. They kept putting a tube in for about twenty minutes. A tracheostomy set was called. It was not used. Nurses saw what was needed and did not make it the room’s question harshly enough. Elaine did not regain consciousness. She died thirteen days later.

17.1 What the frame was

The frame assembled at once and was reasonable: a difficult intubation, need to get a tube, there are seniors who know how. The skill is real. An anaesthetist’s library of scenes is full of successful intubations after several attempts. Water as it were works: one more attempt, another blade, another person. The frame’s expectancies come true enough not to break on the clock and on forty percent. This is not a Klein miss. This is Portland:

an hour in the light. The light here is the tube. The tank is a brain without oxygen.

The guideline that required a change of question was in the room. It was not used as a prosthesis at the moment. A prosthesis on a shelf is leftover until it is spoken aloud. To say “this is CICO, we go to the neck” is to change a frame in the presence of living seniors who are still “working the tube.” Status from the chapter “who is allowed”: the nurses had knowledge. They did not have a slot to make it an order. Martin Bromiley’s parliamentary evidence would later record almost the same phrase as on the deck of 173: they knew, “but did not want to raise the issue.”

Fact

Independent review: Professor Michael Harmer, then president of the Association of Anaesthetists of

Great Britain and Ireland, 2005. Coroner. Martin Bromiley's written evidence to the House of Commons Health Committee, 2008. Date: March 29, 2005. Saturation around 40%, intubation attempts on the order of twenty minutes, surgical access not performed in time.

Martin is a civil aviation pilot; he pushed for a review "not to blame, to learn." In 2007 he founded the Clinical Human Factors Group. A video of the debrief has been shown in training since. This is not a tale from Mayday. This is a report and parliament.

17.2 How this is not "doctors err too"

To err is the word of a scapegoat debrief. Here three consultants were not juniors. The tube frame was their smothered mate: beautiful, textbook, often saving. The short mate — oxygen by any path, including a knife

— is closed by a good one. The nurses in the role of first officer. The surgeon in the role of a person who could have done the neck and waited for a frame in which a neck is legal.

Harmer does not write “they did not know the guideline.” He writes that CICO management did not follow recognized guidance: they tried to intubate too long instead of an alternative path. Declarative knowledge was there. The chunk “now not the tube” did not seat at the moment. An anesthesia syllabus promises the chunk. The moment does not take it if attention is on the squares of a familiar solution. Bilalić measured this with eyes. Here they measured it with saturation.

Martin Bromiley brought to the hospital not revenge. He brought aviation’s ban on stopping at “could not have been foreseen.” A surgeon wrote exactly that in a letter to the husband. In aviation after 173 such a

letter is the start of an investigation, not the end. The difference of professions: one had already bought a prosthesis at the cost of a floor. The second in 2005 was still buying.

17.3 What engineering has a right to steal, without copying an operating

room

Not an intubation checklist. The question the nurses could not make an order: how long have we been in the tube, and what's the oxygen? That is the same question as "we've been an hour in the light, and what's the tank?" In a ticket: "we've been a sprint in Redis, and what's the N+1?" The form is cheap. A culture in which a nurse / a junior programmer / a first officer speaks it without the death of a career — is not.

And further: a guideline on a wall is not CRM. At Bromiley there was a guideline. At Portland there was an emergency-landing procedure. A procedure inside a frame strengthens the frame. A procedure of changing a frame must have a slot and a voice the room is obliged to hear. Otherwise a prosthesis is Confluence.

Counterexample

Do not make of Elaine an argument “in IT they die too.” They do not die like that. The price is different, and that is honesty, not a reason to relax. Make the argument: closing a frame by experienced people with a living short path is not programmers’ folklore. Documented in a cockpit and in an operating room. Both times a junior saw the tank and could not appoint it

a question. Both times a prosthesis appeared later. Both times the prosthesis did not cancel the library. It made a refusal to close a little less indecent.

The next chapter, about a formula that beats a clinician, after two floors sounds different. A boring rule “oxygen matters more than the tube,” “the tank matters more than the light” loses to a scene until the scene is forbidden by force of protocol. Meehl said this with a table. Bromiley and McBroom — with minutes. We still hope that twenty years themselves will forbid. A consultant’s twenty years did not forbid.

18 When the Formula Beats the Expert

Polanyi: we know more than we can tell. From this it is easy to conclude that the unspoken is always richer

than a rule. Paul Meehl in 1954 assembled another: on tasks of prediction (diagnosis, recidivism, success in school) a simple statistical scheme is often no worse than a clinician, and often better. Later meta-analyses (Grove and colleagues, 2000; Aegisdóttir and colleagues, 2006) confirmed this in the main. Not in every study. In the mass.

This is not an argument with Klein. This is another game.

18.1 Two games we glue together

Klein's game: an environment with regular patterns and fast feedback, you have to act now. Fire, anesthesia, chess, a prod incident at three in the morning. A library of scenes is the right tool. A formula often does not have time, or has nothing to count from.

Meehl's game: you have to predict an outcome from a set of cues already collected: will they hire, will the patient return, will there be a default. A person weighs "humanly": the vivid, the recent, the one like a case from residency. A formula weighs the same every time and does not tire. It ignores the vivid if in the table the vivid is not a predictor.

Kahneman and Klein in 2009 were exactly about this: you can trust intuition not "in general," but where the world teaches. A hiring world three years ahead teaches badly. A world of "this stack trace" teaches better.

Fact

Meehl, Clinical Versus Statistical Prediction (1954). Dawes, Faust, Meehl, Science, 1989. Grove, Zald, Lebow, Snitz, Nelson, Psychological Assessment, 2000: 136 studies in psychology and medicine;

mechanical on average about 10% more accurate; substantially better than the clinician in 33–47% of papers, the clinician substantially better — in 6–16%. It seems the formula “dehumanizes.” The data asked about accuracy. A judge’s years in the meta-analysis did not save.

Tetlock on political experts: hedgehogs with a big theory predict worse than foxes who hold several models and doubt. “Twenty years in the topic” is not a ticket there. Sometimes a ticket to a confident miss. This is not engineering. This is a warning not to wear years as a universal predictor.

18.2 Where engineering is Meehl, where Klein

An estimate “will this person make it in six months” is Meehl’s, and we do it Klein-style: by a face, by GitHub, by whether they look like us in 2004.

Leetcode is a bad formula. An interview “looks like us” is a bad clinician. The argument of formula with clinician is honest here, and both are still weak. There will be a chapter on hiring about this.

A choice of database for three years is Meehl’s by pretension (“we predict load”) and Klein’s by practice (“at three in the morning I know how to put this database out”). Rooms confuse pretension with practice. A slide talks about prediction. Hands live in a fire. What wins is what can talk on a slide, or what can put out — depending on who in the room is more in charge.

An incident at three in the morning is Klein’s. A formula “if p95 > X, scale” here is leftover of a metric. A pause “this is not it” is in place here.

Architecture “for a hundredfold growth that is not there” is a Meehl miss of an expert who carried a fire

library into a forecast. He saw growth. Saw it at another firm. Vivid. A formula “your graph is flat” is boring and more often right.

18.3 What this does to gates

The romance of the gate assumes that doubt about a question is the highest form of expertise. Meehl says: sometimes the highest form is not to doubt a boring rule that beats your scene.

“Always ask whether this is the right fence” itself becomes Einstellung. A person with years on the job spoils every frame, including those where the frame works and you have to climb. Organizations know such people. They stop calling them when you simply have to fix the cash register. They call this “they don’t understand the depth.” The cash register did not understand.

A formula is blind too: it will not find a basement that is not in the cues. Klein’s kitchen fire would have lost to a formula “if kitchen, pour water.” So the argument is not “human vs number.” The argument: which game is this now. An error to carry a Klein pause into a Meehl game. An error to carry a Meehl formula into a Klein basement.

Counterexample

A popular conclusion “replace experts with models” does not follow from Meehl. What follows: on prediction tasks with poor feedback do not trust years more than a table, if the table is honestly collected. On action tasks in a rich environment do not trust a table more than a person who has been burned. Engineering contains both types of task in one sprint. Gluing them is the source of most of our overconfidence.

18.4 A link to the cheap answer, in advance

A language model is not a Meehl formula. A formula is transparent and stable. A model is statistics of text, not statistics of your prod's outcomes. To confuse "the model said" with "an actuarial

scheme said" is a new ladder. An actuarial scheme on your incidents may beat an expert. An internet corpus — no, except by genre.

When we get to the cheap answer, this will be useful. For now it is enough: twenty years buy sight in one game and nearsightedness in another. A book that hides this lies in favor of an author with years on the job.

18.5 An improper linear model that is enough

Robert Dawes separately spoiled the romance even more crudely than Meehl: even an improper linear scheme — equal weights, signs pulled from common sense — often beats a clinician. Not because the world is linear. Because a person does not hold weights: the vivid eats, the recent eats, “looks like that patient” eats. Stability of a dumb rule beats instability of a smart gaze on this game.

Engineering adores the unstable. Scoring of “ticket complexity” changes with the mood of grooming. An estimate of “deploy risk” — with who got burned yesterday. A formula “if we touch payments — double review” is boring and often better than a hero who “feels it’s fine.” The hero is right in evenings when the

library is fresh. On Thursday after three incidents the hero is a tired clinician.

Fact

Dawes, “The robust beauty of improper linear models in decision making,” American Psychologist, 1979. Grove et al., 2000: the effect is not giant and not universal (figures — in the box above). For the book what matters is not a triumph of the table. What matters is a ban on wearing years as a universal predictor of an outcome.

Tetlock: hedgehog experts with one big theory are more confident and worse than foxes. “Microservices always” is a hedgehog. “In this room — a cut, in that — not, and I may be wrong” is a fox. Years make a hedgehog more dangerous: the library feeds one theory with examples. The examples are real. The theory is still one. Confidence

grows faster than accuracy. This is not chess. This is a forecast without feedback for years.

19 How the Library Moves

If twenty years are thousands of scenes in one skull, the profession does not reproduce. The profession reproduces. So a library somehow travels between people. Not as a list. As what?

A textbook answers: lessons, quests, a station. That is entry. Not twenty years. The question of this chapter: which channel is even capable of carrying a chunk, not an item of a table of contents.

19.1 Not in one head

Lave and Wenger wrote about situated learning: skill lives in participation in a practice, not in a transfer of propositions. An apprentice first does the edge of the

work, sees when a master winces, and only then can say why. A community of practice holds scenes. A skull is a participant, not a warehouse.

This is unpleasant for an individual romance of the gate. A gate, if it exists, is also social: a junior has a right to spoil a frame only if the practice allows it. In aviation it allows it formally. In a team where a senior is always right, a junior's library copies closing of a frame, not a miss.

Disputed

Situated learning is easy to turn into a slogan "no need for books, need to sit next to." Books carry the declarative. Next to carries chunks. You need both, at different times of day. The first book is exactly about this, only from the other side:

without the declarative you do not enter; without scenes you do not stay.

Hutchins on a bridge: cognition is already distributed. A navigator does not hold the whole system. An engineer in a team neither. “Twenty years” sometimes means “twenty years in teams that remember accidents.” A person leaves — part of the library remains in postmortems, in code, in jokes. Part dies. Documentation tries to freeze a chunk and more often freezes leftover: an answer without a question.

19.2 Review as the only cheap channel

Pair programming is expensive and rare. An internship “watch me for a day” is rare. What happens every day in a living team is code review.

A good review is not style and not coverage of a checklist. It is a showing of a chunk: “here you opened a transaction through HTTP, here is how that smells.” Smells is a word from the library.

It cannot be fully unfolded into a rule. You can show a place and an outcome. If a reviewer only puts “nit: rename,” the channel is empty. If a reviewer writes a novel — the channel clogs with declarative that will not seat.

A junior copies not what is written in a comment, but what they look at. If they look at Hungarian prefixes — a caricature will travel. If they look at transaction boundaries — a chunk will travel. A reviewer’s attention programs a team’s library more strongly than a guide.

In the margin

So “AI will write the review” is a separate threat to the channel, not only to speed. A model looks at what a corpus counts as review: style, tests, known CWEs. It looks badly at “this smells like that incident of 2021 that is not in the repository.” If review becomes cheap and corpus-like, the channel of chunks will narrow to a channel of leftover rules.

19.3 A postmortem as transfer of a scene

A good incident debrief puts into a shared library a binding: a symptom, a false frame everyone accepted, what broke an expectancy, what they did. A bad debrief puts a guilty party and a rule “we don’t do that anymore.” A rule is leftover in embryo. A scene is a chunk.

A blameless postmortem (Dekker, safety-II culture) tries to keep a scene, not a hunt. It itself can be a ritual: everyone says “systemic

factors," nobody changes a graph. A ritual does not transfer a chunk. It transfers only if the next similar symptom lights up differently. A check of transfer is not a document. A check is the next room with Redis.

19.4 Why a syllabus is still not empty

The declarative is needed so a chunk has somewhere to seat. A person who does not know what a transaction is will not see the chunk "a transaction through HTTP." Six months buy anchors. Twenty years hang scenes on anchors.

Hence an honest answer to the book's original question. What cannot be put in a table of contents? Scenes. What can? Anchors and an invitation into a practice where scenes lie. A textbook that promises scenes without practice lies. A practice that promises scenes

without anchors also lies: a junior looks and does not see what to look at.

A library moves not as a file. As participation plus anchors plus a right to ask “what else could this be” and not die socially. Organizations repair the first two more readily than the third. The third is the condition under which a miss is even spoken. Without it only hits travel: fast ladders.

19.5 A corridor they closed before the model

Remote work and chat ate a cheap channel before a model ate the small. Lave’s edge of the work — an apprentice sees when a master winces over a letter. A wince is not visible in Slack: a master

will write a nitpick or stay silent. Silence in chat does not teach. Silence in a room taught with a body: a pause, a look at a seam, “don’t go here” without words.

Bootcamps and “six months” promise entry. Entry without a community of practice is anchors without a hook. A person can make a service and cannot see when a service is shameful. A firm takes him and hopes review will carry the rest. Review in a pull request is poorer than a shoulder. Poorer does not mean useless. It means: a channel that was already narrow became still narrower before the cheap answer. The model sat down in an already thin wire.

Team jokes are an undervalued carrier of a chunk. “This is like that payment gateway on Friday.” A joke holds a scene. A wiki holds an answer. A junior who is not in Slack of 2019 does not get the joke. He

gets onboarding and a guide. A guide is leftover in embryo, even if today it is alive: in a year it will describe a system that is not there. A joke will die with the people if it was not put in a debrief.

19.6 What to count as success of a move

Not “a junior cites Klein.” Success: in the next room with a similar symptom a junior himself winces before Redis. A wince — a chunk seated. A citation — the declarative seated. Organizations measure the second, because it is visible in review comments. The first is visible only in an incident that did not happen, or happened otherwise. A metric again eats a goal.

Part III. Programs Without a Programmer

20 The Answer That Outlived the Question

There was a problem.

A solution was invented for it. The solution worked. They told a neighbor. The neighbor turned it into a rule. The rule entered a checklist. The checklist became best practice. Best practice became a moral: this is how good teams do it, otherwise you are not serious.

The original problem vanished, changed, or turned out to be local. The rule remained. The question it was an answer to, nobody remembers. In review they speak the rule in the tone of a law of nature.

This is not “expert intuition” and not a miss on a prototype. This is another machine: a culture that

stores answers better than questions. If you call both unconscious programming, we glue again. Don’t. Here what programs is not perception, but institutional memory without a key.

Chesterton already warned: before you take down a fence, learn why they put it up. This chapter is about cases where what has become a moral is worth taking apart — and about cases where “I see no point” means only that you do not see Vasya, who was on call on Thursdays.

20.1 How an answer unhooks from a question

Unhooking almost always looks reasonable at every step.

Step 1. Pain is local and burning. Yesterday prod was down because of a manual migration.

Step 2. The rule is narrow. “Migrations only through a tool, in working hours, in pairs.”

Step 3. The rule is generalized so as not to depend on a team’s memory. “No schema changes without an RFC.”

Step 4. An RFC becomes proof of adulthood. Teams that did not have that accident adopt the RFC, because that is what maturity looks like.

Step 5. New people do not catch the accident. They catch the RFC. The question “which accident does this close?” sounds like disrespect.

Step 6. Conditions changed: migrations are now small, the tool itself can roll back, the accident was about something else. The RFC is alive. It already lives as a filter of people: whoever argues is not ours.

No villain. Each step reduces local pain or uncertainty. The sum of steps creates a rule without a question.

20.2 Names that lost semantics

Charles Simonyi invented Hungarian notation not so a variable would scream dwFoo . In the original logic a prefix carried the meaning of a quantity: this is a byte count, this is an index, this is a “dirty” string. A compiler did not catch meaning, a name did. Joel Spolsky later called this Apps Hungarian: wrong code looks wrong.

Then someone read Simonyi’s word type and heard type in the compiler’s sense. Systems Hungarian appeared: dw because double word. When languages became stricter and an IDE

began to hint the type itself, Systems Hungarian became noise. A revolt against noise executed the semantic version too. .NET guides wrote “don’t use Hungarian notation,” not much explaining which. The joy was shared.

Fact

Simonyi’s original text is about type as a set of meaningful operations, not as storage. The history of the substitution is well taken apart in Spolsky, “Making Wrong Code Look Wrong” (2005), and in Eric Lippert. This is almost laboratory leftover: the question “how to make visible a semantics a type does not carry” died, they executed an answer to another question — “how to duplicate storage in a name.”

The moral is not “Hungarian notation is good.” The moral: when you revolt against a rule, check which of the two you are revolting

against. Often in one name live the original answer and its caricature. They demolish the caricature together with the answer. Then for ten years they invent the same thing in types: UntaintedString , opaque ids, units of measure in a type system. The

question returned. The name is new. People who remember the caricature resist, because “we’ve already been through this.”

20.3 Hungarian that still breathed

In one Java 8 bank service variables were still called strCustNm , intAmt . The company guide

already forbade Systems Hungarian. A reviewer put a nit. The author of the diff renamed to

customerName — and lost the only mark that strCustNm in this file meant “the name as a

regulatory report sees it, with truncation," and nearby legalName was from KYC and different.

Simonyi still lived in a caricature. Execution of the caricature executed the meaning.

I first rejoiced at these remarks: noise leaves. Then I caught a bug where a "cleaned" name went into a report without truncation. Noise held a question. A correct name without a question is a

ladder of taste. Chesterton is on the side of the ugly prefix until the question is written otherwise: a type ReportName , a comment, a test on length. Taking the prefix down after that is a gate.

Taking it down because of a .NET guide is a Thursday, only with another Vasya.

20.4 A flag that outlived a function

August 1, 2012. Knight Capital, SMARS router, the first forty-five minutes after the market open. More than four million child orders instead of 212 client ones. A loss of more than 460 million dollars. This is not a parable. This is an SEC order.

Power Peg was an old function: aggressively build a position. They switched it off around 2003. They did not delete the code. In 2005 a counter of executed shares of a parent order moved elsewhere in the chain: dead code became blind as well. In 2012, for NYSE Retail Liquidity Program, they decided to give the same flag that once switched Power Peg on to a new function and delete the old. They copied the new code onto seven servers of eight. Onto the eighth a technician did not copy. There was no second person on the deploy. A

flag “yes” on the eighth server woke Power Peg without a counter. Seven servers worked. One sang. Then in panic they took the working code off seven — and there were more singers, until they stopped it by hand. A kill switch did not live in the procedure.

Fact

SEC, Administrative Proceeding File No. 3-15570, October 16, 2013, Release 34-70694. 45 minutes, more than 4 million orders on 212 parents, more than 397 million shares, a loss of more than 460 million dollars, a 12 million fine under the market access rule. Power Peg discontinued years earlier, the code remained callable. The flag was reused. One of eight servers without the new code. Cumulative quantity had left SMARS earlier. This is leftover in a binary plus deploy as a form without a second look.

Chesterton here is on the side of “don’t delete until you know.” Knight knew and wanted to delete. Did not delete on one server. Leftover is not always “forgot the question.” Sometimes they remembered the question, and a dead answer is still in process memory, and a new answer sat on the same switch. A gate would have been: a new flag, not a move of meaning onto an old bit; a deploy you cannot do on seven of eight in silence. The ladder is a manual copy. The eighth server is Vasya’s Thursday, only Vasya is a bit in a config, and Vasya trades a market.

20.5 Harmful GOTO and useful GOTO

Dijkstra in 1968 wrote a short letter: unconditional jumps make programs unreadable as structure. The

question was about control flow in languages where GOTO is a way to assemble a mess.

The answer outlived the question and became a taboo. “We have no GOTO” is spoken with moral satisfaction in languages where there is none anyway, and in situations where an early

exit, a state machine, or error cleanup reads clearer than a pyramid of flags. The taboo no longer answers the mess of 1968. It answers a desire to be a person who has read Dijkstra.

Chesterton here would be on the side: first understand which mess Dijkstra meant. Then decide whether it is in this file. Sometimes it is. Sometimes the file is already a tree, and the taboo stands across and forces you to write worse.

20.6 The same pattern, not a catalog

Further can be fast, without a third Hungarian.

DRY answered “we fix the same thing in six places and forget the seventh.” The answer became a ban on duplication even where two similar pieces have not yet proven they are one, and premature generality will bind them forever. A useful rule. As a moral — early.

Eighty percent coverage answered “we have no tests.” It became a goal. Tests that catch nothing are green. The question was about risk. The answer became about a number.

“No comments, code should be self-documenting” answered lying comments of 1998 that diverged from the code. It became a ban on explaining why, when what is already visible. Why is exactly the question leftover kills. A comment “don’t

touch, Vasya” is worse than restoring a question. A comment “here is a race with a payment callback, order matters” is the question written next to the answer. A taboo does not distinguish.

Passwords, a cut “like they have,” a board at the door — the same unhook, but already other machines nearby: a checklist, a prestige slide, a signal. They will have separate chapters. Here the pattern is enough.

I stop. A catalog itself would become leftover: “here is a list of bad rules.” There is no list. There is a pattern of unhooking. A reader will finish a new example himself in a week, if he starts asking at review about an accident.

20.7 A method, not a moral

When you meet a rule with a high moral temperature and a low argument, two moves are useful, and they are opposite in temperament.

Chesterton's move: restore the question. Who put it up. What pain. Is the pain alive. Do not demolish from "I don't see."

The gate's move: even a restored question may not be the question now. Restoration does not oblige you to keep the answer. It obliges you not to confuse a caricature with an original and not to confuse taste with a debrief.

If a rule has outlived a question, it is not automatically false. It is automatically suspicious. Suspicion is not demolition. It is permission to ask aloud at review: "which accident does this close?" If nobody remembers an accident and nobody

can imagine one under current conditions, you are holding Vasya’s Thursday.

This is no longer about how an expert sees a board. This is about how an organization’s board is drawn by rules whose authors have died, quit, or forgotten what they meant to say. The next chapter is about a case where a rule is also measurable, and then it does not merely live. It starts writing a play.

21 One Fence: Passwords

Leftover so far was a scatter. A scatter convinces fast and badly: a reader nods and does not see how an answer unhooks year after year, until the author of the answer comes to say “I was wrong,” and screens still demand P@ssw0rd . This chapter is one story. On it you can see all the

machines at once: a forgotten question, a compliance metric, an explanation after, the social price of a gate.

21.1 What the question was

In the early 2000s institutions had no large public science of how people actually choose secrets. There was anxiety: computers guess faster, people choose password . They needed a document

you could put on a shelf and tell an auditor: we follow NIST.

Bill Burr, a middle manager, wrote Appendix A to SP 800-63. Complexity: letters, digits, symbols. Rotation. A short text traveled through federal guides, universities, corporations. The question he answered was roughly: “how to make a password less guessable by a composition rule, without data on real passwords.” They did not give

him data. He later said: he asked to look at real secrets of a network — “they were horrified that I even asked.”

Fact

Burr’s story: a Wall Street Journal interview, 2017, “Much of what I did I now regret.” The official reversal: NIST SP 800-63-3, 2017, Paul Grassi and coauthors — do not impose composition, do not rotate without compromise, length and a blacklist of stolen matter more. This is not a blog. This is a change of a federal document.

The 2003 question was narrow and honest in its poverty: no data, there is fear, a text is needed. The answer — composition and rotation — looked like engineering of a secret. It became a moral.

21.2 What the answer did to people

People did not start choosing good secrets. They started satisfying the rule with minimal moves:

Password1 , Password2 , P@ssw0rd , a season plus a year, an exclamation mark at the end. An

attacker who knows the rule knows the minimal moves too. Rotation taught writing a secret on a sticker and incrementing a digit. A sticker is closer to an attacker than a long phrase you do not have to change.

That is, the answer spoiled what it was supposed to measure. This is already the chapter on metrics, only the metric here is “compliance with policy,” not “resistance to brute force.” An auditor sees checkmarks. Brute force sees a pattern of checkmarks.

Chesterton at this point is still on the side of the fence: in 2003 there was a question, there was fear, the text was not from malice. To take it down in 2004 from “it’s inconvenient” would have been vandalism. To take it down in 2018, having learned the question and seeing that the answer eats it — is not vandalism. That is a gate.

21.3 The author came. The system did not leave

1. NIST rewrites. Burr publicly regrets. Grassi says: we started from zero. Part of the

industry hears. Part of the screens — not.

Why not is more interesting than regret.

Compliance does not read an interview. It reads a checklist once copied from Appendix A and sewn

into SOC, into PCI, into internal AD. To change a checklist is a separate project: lawyers, auditors, “what if they ask why we weakened.” Weakened is a word of a frame. Frame: complexity = strength. Gate: composition complexity = a predictable pattern. To walk along, you have to rename “we weakened” into “we stopped helping an attacker.” The renaming is legally expensive.

Vendors of login screens sell “enterprise password policy.” Policy is a product. A product does not like to vanish when NIST changed its mind.

People inside firms who read 800-63-3 hear: “prove we are not irresponsible.” Irresponsibility is not reading the new NIST. Responsibility in their ritual is to keep the old. An organization’s interpreter is happy: we are strict. Strictness here is leftover.

21.4 What this story shows

Not a villain. Burr is not a villain. An auditor is not a villain. A person who writes P@ssw0rd1! is

rational inside a rule.

You see the speed of unhooking: from a narrow poor question to a world moral — less than fifteen years. The reverse path is slower. The author of an answer is powerless against a checklist.

You see the gluing of machines. Leftover (a rule without a question). A metric (a checkmark). An explanation after (“it’s safer this way” — a theory data refuted, but it is socially fit). A word (“weaken policy”). There is no Klein miss here. A password screen does not “feel.” It demands.

You see the price of a gate: a person who in 2018 proposes removing rotation argues with a moral, not

with cryptography. They may call him unserious. Unseriousness is a social tax on a restored question.

21.5 Waterfall nearby, briefly, the same pattern

Winston Royce in 1970, Managing the Development of Large Software Systems, drew a linear sequence: requirements, analysis, design, code, test, operations. Immediately under the

drawing he wrote that the implementation described is risky and invites failure. He proposed doing a pass twice, iterations, loops, not believing that a document at the top froze the bottom. The word waterfall did not live in the paper as a brand. Later the diagram he named dangerous they named his method and made a moral of large contracts. They bought the drawing. The caption “don’t do it this way” did not travel.

This is not “passwords.” This is the same unhook: a convenient drawing outlived the caption “don’t do it this way.” The industry bought the drawing. The author is powerless in a different way than Burr: the text remained, the frame left.

I do not unfold a second full story. It is enough that leftover can execute even a warning, if the warning stands next to a convenient picture.

21.6 What after one fence

A catalog is no longer needed to believe the mechanism. What is needed is a habit of taking one living ritual of your team and asking: which document it once cited, whether that document is alive, whether the author has come to say “regret.” If he has — you are in 2018 on a password screen. The ladder is called compliance. The gate is called reading.

21.7 How a checklist multiplies without an author

PCI, SOC 2, internal AD, an SSO vendor: each layer copies “password complexity,” not copying NIST as a text that can change. A copy has no subscriber to updates. A copy has an auditor who checks against the copy. The check is green. Living NIST is red relative to the copy. Green wins.

A lawyer in this loop is not a villain. A lawyer answers the question “how do we not get punished.” The question is alive. The answer is not to deviate from what passed in the last audit. The past is leftover by job. A person who brings 800-63-3 offers a lawyer a new risk without a new template of defense. A risk without a template is more expensive than old harm already “accepted.” So an institution rationally holds harm.

Bank rooms where I saw this can discuss “passkeys” for two years while a screen demands eight characters and a digit. The discussion is a signal of modernity. The screen is compliance. Signal and screen are not obliged to meet. A meeting would be a project. A project loses to a quarter that has no field “take down a Thursday.”

22 A Second Fence: Cut It the Way They Did

After passwords a second long story is needed, or leftover will remain “about compliance.” This one is about fashion. Fashion is more shameful than compliance: an auditor does not hold it. A slide “this is how the serious do it” holds it. The serious once answered their own question. The slide left without the question.

22.1 What Netflix’s question was

The public story of the early 2010s: many teams, streaming load, the price of a shared release, a “two-pizza team,” freedom to deploy, resilience to the fall of someone else’s piece. The question was about their communication and their failure. The answer — services, a chaos platform, a culture “you deploy, you are on call.” Part of the answer is technique. Part is an org chart they showed honestly.

A blog ate the technique and spat out a noun. “Microservices” became a moral of scale for firms that had neither that load, nor that on-call culture, nor that platform. The noun contained a verb: cut. To cut is to look like Netflix. To look is Spence’s signal, only for architecture.

Disputed

I do not claim Netflix is “to blame.” They answered their question and told. The telling in another room became a syllabus without anchors. The room that bought the noun without buying on-call is to blame. This is not new: that is how all successful reports travel. A successful report is a bad carrier of a scene. It cuts out scars without which a chunk does not seat.

22.2 What the answer did to people who did not have that question

A team of eight people on a product with a flat graph cuts five repositories, because that is how a bounded context is visible. Evans’s bounded context was about language and invariants. It left as “a separate deploy.” A separate deploy bought: network boundaries where there used to be a stack; transactions where there used to be a row; on-call

where there used to be a log in one file. Pain became regular. Regular pain built a library “how to fix the distributed.” The library justified the cut: we know how to put this out, so this is how it should be. A circle.

Chesterton at the start could be for the fence: “why cut” is an honest question if cutting is expensive. After two years an expensive cut already stands as a floor. Taking it down is a second

project. Leftover defends itself by the cost of its treatment. A password screen defends itself by an auditor’s fear. Microservices — by fear of “going back to a monolith,” a suitcase-word that contains a moral, not a structure.

22.3 A monolith as a counter-fashion, the same pattern

Then came a counter-fashion: “we are a monolith again, it was worth it.” Sometimes a living question: too expensive to cut. Sometimes a new moral. A person who in 2016 cut “like Netflix,” in 2023 glues “like DHH” and again does not restore his graph. A pendulum of leftovers. A gate would have been boring: what is our communication, what is the price of a release, which failure we endure. The boring signals badly. A pendulum signals: we are in on the conversation.

Conway here is not a flag. He explains why a pendulum swings at all: they changed an org chart under fashion, the system caught up with the chart, pain justified the chart, a slide fixed it. Inverse Conway — “cut

teams under a desired system” — itself became a slide. The desired system is often “like they have.”

22.4 What this story shows that passwords do not

On passwords there was an author who came and said “regret.” Microservices have no author. There is a chorus of blogs. A chorus does not regret. A chorus writes the next post.

On passwords leftover was held by an institution of audit. Here — an institution of prestige. Prestige cannot be rewritten by a NIST document. You can stop copying a slide. A room cannot stop alone: a signal “we are not like the serious.” Coordination without a center, as in hiring.

On passwords an answer spoiled a secret. Here an answer built real complexity that then feeds real

expertise. The expertise is honest and bound to self-made pain. Twenty years of such pain buy sight in the distributed. Sight does not prove the pain was needed. Meehl: a clinician who treats well a complication of his own operation.

22.5 A link to words and to reading

The noun “microservice” in a ticket is XY at the level of an epic. To read such a repository is to read an org chart, not a domain. A senior who sees this argues not with Kubernetes. He argues with the fact that the question “how to deploy independently” substituted for the question “which invariant we cannot tear.” The substitution may have been living. Then there is nothing to argue, you have to be on call honestly. The substitution may have

been a slide. Then Kubernetes is Redis on N+1, only a year more expensive.

I do not propose a ban on cutting. A ban is the first book, another genre, another reader. Here it is enough: here is how an answer unhooks when there is no auditor, and there is a blog. If in your room they cut — restore whose blog is in the ticket, and whether you have its question. If it is alive — climb. If not — you are in 2018, only without Burr, who could have apologized.

23 The Interview as an Industry

Hiring is a place where Meehl’s game pretends to be Klein’s. You have to predict whether a person will be useful in a year. They do it as if they are checking whether he puts out a fire on a board in forty minutes.

The original question was once honest in narrow shops: can a person think about algorithms at a board, because the work is to think about algorithms at a board. The market's question became another: can he live in someone else's service, not break prod, agree, endure boredom. The proxy remained. It became an industry: platforms, books, "grind," coaches, tasks with a subscription. Companies the proxy does not answer use it because that is how the serious do it, and because seriousness has to be signaled.

23.1 A signal, not a measurement

Michael Spence described education as a signal: a diploma does not necessarily teach the work, it costs a lot to whoever "cannot pull it," and therefore distinguishes. Leetcode is partly the same. It does not measure the work. It measures readiness to

pay months of life for entry. Readiness correlates with something (age, health, absence of a second job, a cultural code “we love puzzles”), not necessarily with what is written in a vacancy.

So a gate “let’s give a trial week instead of graphs” strikes not only the cost of a week. It strikes a signal. A trial week does not signal “we are like Google in 2010.” A board signals. Even when Google is already not that and itself changed the process, the industry holds the picture.

Disputed

Spence is about education, not algorithms. Transfer: a proxy that is expensive to pass lives as a filter independently of validity to the work. Validity is a Meehl question, and they rarely count it honestly: there is no table “who passed the board → who did not burn prod.”

There is a feeling of a room. A feeling of a room is Meehl's clinician.

23.2 Why everyone agrees and nothing changes

In a corridor seniors say: "leetcode predicts weakly." Next week they set leetcode. This is not hypocrisy in a small sense. This is leftover plus coordination.

You have to change a proxy all at once, or you hire "those the serious did not take," and that is a signal too, only a shameful one. Coordination without a center. NIST at least rewrites a document. Hiring has no NIST. There is copying of prestige.

HR scales what is standardized. A trial week requires taste, a mentor, access to prod data. A board scales to twenty countries and outsourced interviewers. A metric "we closed a

vacancy” greens from a board. A metric “the person is useful in a year” arrives late and without a control group.

A candidate prepares for a fence, not for the work. A six-month textbook is forced to put Java not only because that is how the world is built, but because that is how the fence is built. The textbook’s author knows this. A gate would be: hire by jointly debugging someone else’s bug. The gate is expensive for a firm that counts interviewers. The ladder is a course on graphs.

23.3 Forty minutes from the other side of the board

I set a graph in an interview for a person who later, in a trial week they did give, found a leak in our HTTP client in an hour. On the board he tangled in complexity. In a repository he read. The board said “weak.” The week said “ours.” We took by the

week and were ashamed to tell a neighboring team: it looked as if we had lowered the bar. There was nothing to concede: the requirements were copied from Google hiring around 2010. The picture is more expensive than a leak.

The reverse was too. A person brilliantly unfolded a tree on a board and in the first week could not read a stored procedure without panic: Chase's random board, the wrong chunks. The interview's interpreter wrote "a strong engineer." Prod wrote on-call. Meehl's formula here would have been poor, but more honest: a table "passed medium → did not burn Friday" was not counted. A face was counted.

23.4 What this does to a library

People who entered through grind have a library of board tasks. This is not a zero library. It intersects badly with scenes “slow after a small thing in a loop.” The first months of work are a retraining of attention. A firm pays for retraining and at the same time demands the same grind of the next, because otherwise the signal.

A senior who sets a board sometimes honestly checks “can a person not panic in the unfamiliar.” Sometimes checks “does he look like me at twenty-two.” The second is not said. The interpreter writes the first.

A candidate’s pause “why this task” in an interview is punished. That is exactly the feeling of a wrong question. In this room it is socially legal only for whoever is already inside. From the outside — incompetence. An organization that

punishes doubt about a frame at entry should not be surprised that inside there is also nobody to doubt.

23.5 Trial work and why there is little of it

Selection research (the Schmidt and Hunter line, later caveats by Sackett and others) has long said the boring: work samples and tests close to the task predict better than an unstructured chat. The industry reads this and sets a board. A board is closer to a chat with prestige than to a sample. A sample is “here is a bug in our service, an hour, you can ask.” An hour requires a service you can show a stranger, a mentor who is not selling an epic, and a readiness to see that a strong one on a board is blind in your sediment.

Blindness in sediment is exactly what the chapter on reading calls the profession. An interview that does not check reading checks

writing on a random board. A random board is Chase’s control. We know it and ignore it, because a sample does not signal.

A structured interview is better than a free “let’s talk about culture.” Better does not mean about the work. It means: less clinician in the room, a little more formula. A formula on poor cues is still a formula. Leetcode medium as a cue is poor. Stably poor. Dawes could say: poor stability sometimes beats a vivid clinician. Sometimes. Not a license to hold a fence.

24 A Pause With No Slot

Even if a miss fired in a skull, even if a person is able not to close a frame, organizations often have nowhere to put it. A calendar is already sliced. A standup already waits for a status. An estimate is already in a ticket. “Wait” has no field in a form.

This is not the same chapter as metrics. A metric writes what to do. Here it is about when you cannot not do. Time is a separate programmer of behavior.

24.1 Standup as a machine of closing

Standup answered the question “we did not show a move for half a year.” A short ritual of visibility. In many places the question died: a move is visible in tickets, in previews, in chat. The ritual lives as a roll call. A roll call requires yesterday/today/blockers. It does not require “the frame is shameful.” There is no third column.

A person who at night felt Redis is not it, in the morning must say “yesterday started a cache, today I’ll finish.” The language of a slot eats a pause. To speak a pause you have to break the genre: “I don’t know whether I started the right task

yesterday.” That sounds like yesterday’s failure. The genre does not like yesterday’s failure. The genre likes progress of a ladder.

24.2 An estimate as a promise of a frame

Story points estimate a ladder. You cannot estimate “walk along”: it is unknown how far along, and along does not draw on a burndown. So doubt about a question has to be spoken before an estimate. After an estimate it steals points. Theft of points is a moral category of a room.

Consequence: a gate is legal in the first ten minutes of a ticket and illegal after grooming. Grooming is a ceremony in which a frame is poured in concrete. A good grooming could be a place of “which X is this, not Y.” Often it is a place where Y is already in the title, and they argue about sizes of Y.

24.3 A sprint as a refusal of incubation

The psychology of insight has long known a boring thing: sometimes you have to step away. Wallas called it incubation; debates whether it is beyond attention at rest are not closed. Practically: a person who is not given a night without a ticket more rarely reassembles a scene. Not because a night is magical. Because Einstellung holds while the gaze is on the same squares.

A sprint promises focus. Focus is good for a ladder. For a change of frame you need an interval in which a familiar mate is not fed with attention. A calendar full of meetings about the same ladder kills the interval. “No time to think” is not a metaphor of laziness. It is a description of slots.

24.4 Always-on and the cheap answer in advance

Chat, a pager, a model that answers at once. A pause looks like idle. Idle in a culture of speed is guilt. Guilt closes a frame: better a wrong ladder today than a right question tomorrow. Especially if tomorrow is someone else's on-call.

When an answer becomes cheap, pressure on a pause will grow, not fall. "Why wait, here is a candidate." The candidate is complete. Completeness imitates the end of thinking. A slot "to think" is still harder to defend than in 2015.

24.5 What here is not advice "meditate"

There will be no pomodoro technique for gates. There will be an observation: if your form has no field for "the wrong question," it will not be in behavior, whatever Klein

papers they read. A field is not a Jira custom. A field is a permitted line at grooming for which they do not take points off. Organizations that allow this are sometimes “slow.” Sometimes they do not build Redis on N+1. From the outside this is indistinguishable until you look at a floor.

24.6 On-call as the only honest slot — and it lies too

On-call promises: here is when the world will answer. Sometimes it answers. Often it answers at three in the morning to leftover of a deploy that by day had no right to a pause. Night gives a miss without grooming: no time to estimate points, you have not to die. Chunks grow. The price is sleep that the next day kills the same library. A tired expert is not Klein. A tired expert

closes a frame early, like a doctor at the end of a shift. Premature closure loves night.

A calendar of “syncs” answered the question “we don’t talk.” The question in a distributed firm is alive. The answer is forty-five minutes in which you cannot walk along: along does not fit an agenda. An agenda is a form. A form accepts the status of a ladder. A person who brings doubt steals other people’s slots. Theft of slots is a moral, like theft of points.

Always-on chat adds microslots without a right to incubation: a ping waits for an answer in minutes. Minutes are a hit. A miss requires hours in which a ping is silent. The silent is counted unavailable. Unavailability is guilt. The loop is closed without any AI.

25 Who Is Allowed Not to Answer at Once

A slot for a pause is not yet a pause. A pause has to be spoken. Speaking costs people differently in one room. Aviation at least wrote that a junior has a line. Engineering more often writes “we are open to questions” on a vacancy page and punishes a tone.

If that is so, “the feeling of a wrong question” is not only a library. Also a license. A license is issued by status, a contract, language, sex, an accent, closeness to a compensation committee. A book that pretends a gate lives in a skull lies the same way as a slide about meritocracy.

25.1 The same miss, a different price

A staff senior says “wait, this is not Redis” — depth. A junior programmer says the same — did not understand Redis. A contractor

says the same — not a team player, we will not extend. A person with an accent who hunts for a word in the room’s language seems slow: translation of a chunk into permitted jargon takes seconds, seconds at standup are weakness. A woman who spoils a frame more often gets “not constructive”: studies of interrupt and credit in mixed rooms are not about programming specially, but a room is not exempt from them because it writes Java.

I do not unfold sociology into a monograph. I fix a mechanism without which the rest of the book is naive. A spoken miss is a social act. The act is cheaper for whoever can no longer be fired for doubt.

Disputed

Part of “a senior can pause” is not sight, but immunity. Immunity also grows with years. Then twenty years buy a right to be silent without punishment, and we glue it again with sight. You can distinguish: a senior in a foreign firm without immunity, with the same library, suddenly “became softer” — or became more careful. Care is not a chunk. Care is an estimate of the price of a line.

25.2 A contractor, outsourcing, “our culture”

People who write the same code from another legal entity often have no slot “spoil a frame.” A slot belongs to staff: staff will be on call,

staff will be on a slide. A contractor optimizes an accepted diff. An accepted diff is a customer’s ladder. A gate is expensive to a customer: it puts in doubt an epic he already sold to his board. A contractor who puts an epic in doubt puts a bill in doubt.

So organizations buy speed of ladders and do not buy misses. Then they are surprised that “a vendor does not think.” To think in the sense of B was not paid. Hitting a form was paid.

Remote added another: there is no corridor where a junior quietly checks a frame before standup. A quiet check was a prosthesis of status. Slack makes a check visible. Visible doubt costs more. A channel of apprenticeship narrowed before the cheap answer. The cheap answer sat in an already narrow channel.

25.3 Who has a right to “I don’t understand”

In CRM “I don’t understand what we are doing” is a permitted phrase. At grooming the same phrase is read as unpreparedness. Preparedness is to learn Y before a meeting. Whoever has learned Y is already in a fence. A person who came without having learned could have asked X. He is ashamed. Shame programs a hit.

A senior can say “I don’t understand” as a ritual of modesty, and a room is touched: what a living mind. A junior says it literally and gets a link to Confluence. Confluence is leftover. A link closes a slot.

25.4 What this changes in the Redis story

A senior’s silence at review I described as doubt about a question. Sometimes so. Sometimes a senior can afford not to fill a silence. A

junior fills, because silence is punished by looks. Filling is Redis. We read an outcome as a difference of libraries. Part of the outcome is a difference of a right to silence.

This does not cancel a library. A person without chunks and with a right to silence simply is silent in vain. A person with chunks and without a right — speaks a ladder, though inside it already smells. The second case the book is obliged to see. Otherwise it will write a hymn to seniority, and seniority was armor.

25.5 Without seven practices

There will be no inclusivity checklist. There will be a test for a room: who last spoiled a frame and what it ended in for him, not for the system. If it ended only for junior programmers — you have CRM in reverse: a right of a miss with a

captain, silence with a first officer. Aviation tried this. A floor is sometimes warm.

The next chapter on Conway will say that departments draw legal questions. This one says: inside a department legality is also personal. A fence is higher for some bodies than for others. The gate the book describes is often a gate for those already allowed to walk along.

26 Who Drew the Fence

Melvin Conway in 1968, in the article “How Do Committees Invent?,” wrote: organizations that design systems are constrained to produce designs that copy their communication structure. Not as a joke. As a sociological observation: modules join only if people join; system interfaces repeat channels of talk.

A fence is often drawn before a ticket. They sliced departments — they sliced services. They sliced services — the question “how do we scale this service” is already legal, and “why is it separate” is almost indecent: it puts an org chart in doubt, not code.

26.1 There is no unbiased design group

Conway is harsher than the popular retelling. The very act of organizing a team already is an architectural decision. There is a class of designs this team will not be able to lead, because there is no channel. “There is no unbiased and already sliced group.”

So a gate “walk along” sometimes requires not a stroll through code, but a stroll through Slack. Who does not talk to whom — there is a seam they will later call a “bounded context,” even if the meaning is one.

Bounded context is a good word when it was chosen. A bad one when they consecrated with it a silence between departments that had already happened.

Fact

Conway, Datamation, April 1968. Harvard Business Review rejected the article: the thesis is not proven. The name “law” was not Conway’s own. Brooks later carried it in The Mythical Man-Month. Empirics later: the “mirroring hypothesis” — a correspondence of org structure and architecture is often visible, not always iron. For us a strong one is enough: a slicing of people sets a slicing of questions that are even legal.

26.2 Microservices as an org chart, not as scale

They often argue about load. Often the truth is in a chain of command. Eight teams do not want to wait for a shared release — eight repositories. The question “do we need to cut the system” is substituted by the question “do we need to cut power over deploy.” The second may be living. Then microservices are an answer to an org question, honestly. Dishonest when a slide talks about load, and the floor about power. An interpreter writes load: it is technical, it is not shameful.

The reverse too: a monolith as a moral “we are not like them” is leftover of a fight with someone else’s org chart. A system should copy your communication, not someone else’s blog. Conway does not choose a monolith. He says: first

look who talks to whom. Otherwise a gate in code is a door in a wall behind which is another department.

26.3 A platform as a department

The word “platform” from the chapter on language here gets a body. A platform team has to justify its existence by multiples. Multiples require that others climb through their fence: a shared CI, a shared bus, a shared “golden” library. Sometimes this is a living question of coordination. Sometimes this is Conway in reverse: first a department, then a necessity that the world look like the department.

A person who doubts a platform doubts other people’s stakes. The stakes are not in a ticket. So a pause seems technical and strikes politics. A book that pretends a gate is only about code lies the same way as a slide about load.

26.4 What to do with the law without turning it into a guide

“Inverse Conway” is a fashion: slice teams the way you want a system. This is awareness, not magic. You can slice. Tomorrow communication will leak around the slice — into shadow channels, into heroes, into chats. A system will start copying the shadow. The law does not obey org charts. It obeys who can write to whom without fear.

For an odd pause the conclusion is simple and harsh. Sometimes “the question is wrong” because the question belongs to a seam that should not be. To walk along a fence is to propose sewing departments or cutting otherwise. This is not Redis. This is not a six-month skill. This is not “an engineer’s intuition” in a narrow sense. This is the ability to see that a board of service

rectangles is also a board of people. Rectangles lie if people already lie to each other with silence.

26.5 An API that would not have been in one room

Two teams, one firm, payments and “the client.” Between them an HTTP API of twelve methods, version 3, because 1 and 2 did not survive a renaming of departments. A slide called this a bounded context. In code it was a conversation people did not hold aloud: each field is a former argument in a ticket that was not decided by a meeting. Conway is not a metaphor. Conway is OpenAPI.

They were put in one room for a week to close “integration hell.” By the end of the week half the methods became a function in one process: it turned out the invariant is one, and the seam held an org chart. A slide would have called this “we simplified the architecture.” A floor would have

called it “we finally talked.” The gate was not in a repository. The gate was that silence stopped being free: you sit opposite, it is shameful to send JSON instead of “listen, we need the same number.”

I saw the reverse too. They took the room apart, returned the seam: each VP needed “his” service on a diagram. Load did not grow. What grew was the legality of the question “how do we scale this API.” The question was born from a slice, not from a graph.

27 The Tool Already Chose the Question

Conway draws a fence with people. Before people a form draws a fence: a ticket template, a linter, a type checker, a required check in CI, a master branch, “this button in Jira is mandatory.” A tool is a frozen answer to someone’s old question.

Every day it asks you the same question again, even if the question has died.

This is not a moral “tools are evil.” Without them you cannot. This is an observation: a gate that requires doing what a form cannot accept looks like wrong use of the form. Wrongness is a social fact. The form wins.

27.1 A ticket as an interrogation form

Fields: type, priority, story points, component, “acceptance criteria.” Criteria accept a ladder. They do not accept “we don’t know whether this is the right pain.” To file a ticket “walk along,” a person lies in a form: writes a feature, puts points, then spends points on doubt. Lying teaches: doubt is theft. An honest genre vanished because it is not in an enum.

A bug template: steps, expected, actual. A good template for the tame. For “the system behaves as if it has another invariant” a template requires a premature scene. A person invents the expected so the form becomes green. The invented becomes official X. Further they fix a divergence from fiction.

In the margin

When I filed tickets in corporate Jiras, the most expensive field was “component.” Component is Conway’s org chart in a dropdown. Chose the wrong one — the ticket left for someone else’s queue, someone else’s queue closed “not our layer,” the pain remained. The field did not describe a system. It described who is allowed not to answer.

27.2 A linter as a Thursday

A rule “maximum N lines of a method” answered a living question: people wrote sheets. The question in this team may have died: methods are short, but now fifteen layers of two lines, and a scene is torn. A linter still screams at a sixteenth line of a living chunk and is silent at a tear. It screams — so it works. It works — so a moral is alive.

A fix “so CI is green” is a separate kind of leftover writing. A diff is not about a system. About a tool. A reviewer sees green and thinks they looked. They looked at a tool. Bainbridge: a green screen.

Exceptions // nolint are a gate that is shameful to write. Shame is useful when a rule is alive.

Harmful when a rule is dead: shame holds a Thursday.

27.3 A type as a fence you sometimes have to love

A type system is a rare fence that restores a question: which values even exist. Good. It does not ask “why this type in the domain.” It asks “you lied about null.” A lie about null is a frequent basement. Types here are closer to an aviation checklist than to a line linter.

To confuse “the code compiles” with “the question is right” is a ladder of adults. A compiler does not see Conway. It does not see a leftover entity that models a 2006 tax under the name

UserExtra . The type is green. The scene is rotten.

27.4 CI as memory you cannot ask “why”

A required check accumulates like coral: prod fell once, they added a step, the step became a personality

of a pipeline. The author of the step left. Nobody restores the step's question, because a red CI is visible shame, and extra eight minutes are an invisible tax. Chesterton here is on the side of the fence until someone has walked along. They walk rarely: to walk along means switching a check off for a day and living in fear that exactly today that prod will return.

The fear is rational. Rationality holds leftover better than malice.

Tests in CI are a separate coral. A test fixes yesterday's question as a law. To delete a test is to take down a fence. Sometimes a fence holds an invariant. Sometimes it holds an API that is no longer there, through a mock that lies. A green mock is kitchen silence. It smells of calm. The floor is warm.

Knight Capital is the same coral, only in a binary. Eight servers, a manual copy, a second person did not look. A required check on “all nodes of one build” did not live. A deploy form accepted seven of eight as success until a market accepted the eighth. The full scene is in the leftover chapter. Here it is enough: a tool you cannot ask “all?” programs a hole.

27.5 Autocomplete as someone else’s library

An IDE proposes the next identifier from this file and from an index. A proposal is a chunk that is not yours: frequency in a project, not a scene. A junior accepts a proposal because it looks like “this is how they do it.” They do it in this file. This file may be leftover.

A model in an editor strengthens the same: the next token from a corpus plus a file. A corpus does not know

your basement. A file knows yesterday's fence. Together they are very convincing. Convincingness is an enemy of a miss: an expectancy comes true too early, because a tool completed a scene you have not yet assembled.

27.6 What to do with this without a guide

Not "switch off linters." Once a year ask which question a required check answers, and write the question next to the check. If the question is not found — you are not in Chesterton. You are on a password screen of 2018. Writing a question is a cheap gate. It is not in a ticket template. You can put it in a pipeline README. They will put it — and in a year the README will become leftover. Then read the README against the pipeline, as code against a wiki.

A tool programs legal moves. A person who can only make legal moves seems productive. A metric agrees. They did not ask the floor.

28 A Stack You Cannot Climb Down From

Leftover so far was a rule: a password, a waterfall, “this is how they do it.” There is a neighbor that holds tighter than a rule. Not a moral. Material. A runtime, a vendor, a market of hands, certificates, licenses, data you cannot export. A fence is not in an RFC. A fence is that demolition costs like a firm.

This is path dependence without poetry: a past move narrowed the next not because they love it, but because a reverse move is ruinous.

28.1 Java 8 that “will live a bit more”

The question was once living: LTS stability, drivers, what a team can do, what an application vendor allowed. The answer sat in prod. Then the question partly died: new language things would close part of the pain, the ecosystem left, CVEs accumulate. The answer is alive because an upgrade is not “change a number in a Dockerfile.” It is a census of libraries that no longer build, a contract with a vendor who certifies only the old, a person who can only this and whom you cannot fire because he is the only one who remembers a stored procedure.

Chesterton is honestly on the side of the fence: before you take down Java 8, learn whose vendor and which stored procedure. A gate “just upgrade” is a slide’s vandalism. A real gate may be “pay a year.” A year does not fit a quarter. A quarter

holds leftover of material. A moral has nothing to do with it. Burr could at least apologize. Oracle does not apologize for your being stuck.

Disputed

I am not against Java. I have been on it twenty years. Lock-in on a favorite tool is worse seen than on a foreign one: it seems this is taste, not a cage. Taste and a cage look alike from the outside. Distinguish: can you leave without arranging a second product. If you cannot — you do not prefer. You sit.

28.2 A market of hands as a second lock

They hire what there is a lot of on a market. There is a lot of what they taught. They teach what they hire. A stack reproduces itself through résumés. This is not Conway (an org chart draws modules). This is a market drawing possible modules in

advance: “we have nobody for Clojure” is already architecture. A gate “another language” requires either rare people, or a year of apprenticeship that a time-to-fill metric will not survive.

The hiring chapter talked about a board signal. Here a stack signal: “we are a JVM shop” is cheaper to explain to a recruiter than “we choose a tool by a seam.” A recruiter is a form. A form accepts stack-nouns. A noun is a frame again.

28.3 Data you cannot export

The wickedest lock-in is not a language. A format that remembers a 2006 tax, and there is no person who can transcribe it. Taking down a fence = losing an organization’s memory. Chesterton here is almost literal: they put up a fence so cows (records) would not leave. The cows are still needed. The question “why this format” is found and alive. Is

this leftover? No. This is a living ugly answer to a living question nobody wants to solve again, because to solve is a second product. Sometimes a fence is ugly because cows still stand behind it.

To confuse a living ugly answer with a Thursday is a favorite error of people with years in another room. They see ugliness and put “legacy.” The word licenses demolition. Demolition kills cows. Then a slide writes “did not account for specifics.” Specifics were cows.

28.4 A link to reading and to Conway

To read a stack is to read which reverse moves are already forbidden by a price, not by taste. Conway says: departments draw seams. Lock-in says: even having sewn departments, you will not sew data you cannot move. A gate in code is

sometimes a door in a wall behind which is not another department, but a license server.

Twenty years on one stack buy sight in it and blindness to the price of exit: the price seems “just work.” A person from the side sees a cage. Both are right in different games. Meehl: do not wear taste as a predictor that a cage is needed.

29 When the Number Starts Writing the Play

First there is a goal. A goal is usually inconvenient: users do not suffer; money does not burn; a system is understandable; children can think; a client trusts a bank.

Then a measure appears that roughly moves with the goal. Roughly is an important word. They love a measure because you can put it on a dashboard and bring it into a room that loves dashboards.

Then they bind to the measure a reward, fear, a quarter, a bonus, shame at an all-hands.

Then people optimize the measure.

At some point the measure stops describing a system and starts governing it. A goal, if they still look at it, looks like a naive poster over a dashboard.

The mechanism is boring and sufficient. Names of laws are secondary. We will name them so as not to pretend we discovered America, and we will not build a Pokémon collection of them.

29.1 What exactly breaks

What breaks is not “measurement as an idea.” Without measurements engineering is blind. What breaks is a relation: a proxy was a witness, they made it a judge.

A witness can lie, but you can at least change him if a scene does not fit. A judge starts writing a script. Actors play under the judge. There are no witness statements anymore: everyone is already in a theater.

Donald Campbell wrote about this on social indicators and schools before the phrase became a meme: the more a quantitative indicator is used for decisions, the more they spoil it and the more it spoils the process it is supposed to reflect. Charles Goodhart in 1975 — not about Scrum. A paper about British monetary management: a link of an aggregate to an economy they looked at as a lever falls apart when they make the aggregate a goal of control. A statistical regularity lived in a world they were not yet steering. Steering ate it. Marilyn Strathern in 1997 shrank into a phrase everyone cites: when a measure becomes a target, it ceases

to be a good measure. The phrase is about ratings of British universities, not about velocity. Velocity simply rhymes without being asked.

Fact

Campbell: a line from “Reforms as Experiments” (1969) to “Assessing the impact of planned social change” (1979) — about corruption of an indicator under pressure of decisions, including test scores in school. Goodhart: 1975, “Problems of Monetary Management: The U.K. Experience.” Strathern: “Improving Ratings: Audit in the British University System,” European Review, 1997. These are independent convergences on one mechanism, not one school that invented a law and spun it.

The mechanism is not quantum magic and not “people are bad.” People do what they are paid attention for. If an organization’s

attention looks into one cell of a table, behavior will go there too. The cell's link to a goal was a correlation in a world where the cell was not yet optimized. Optimization eats the correlation.

29.2 Stories of different honesty

There are documented cases, and there are parables. Parables are more convenient. That is why slides love them. We will mark them.

Wells Fargo. The goal sounded adult: depth of relationship with a client, many products per person as a sign of trust. Measure: how many products opened. Pressure: quotas, fear, a corporate culture "eight is few." Outcome: millions of fake accounts. The measure grew. Trust did not. Then fines, senators, an ethics textbook that was late.

Fact

The Wells Fargo scandal (surfaced widely in 2016) was investigated by regulators and Congress. This is not a tale. A goal “cross-sell” → a metric “accounts per customer” → behavior that feeds the metric directly, kills the goal.

Velocity in Scrum. It was conceived as a team’s capacity for its own planning: how much we usually swallow in a week, so as not to promise a fairy tale. It became a KPI of productivity before bosses. A story point began inflation. Teams that honestly reduced an estimate look worse than teams that multiplied by three. The measure is still on the board. You can no longer plan by it: it describes a politics of estimation, not work.

I sat in a room where after a first dashboard for a VP “velocity” doubled in two quarters with the same composition and the same Mondays. Nobody became faster.

The unit slid. To say honestly “we renamed a point” was impossible: a red cell of last quarter would have looked like failure. An interpreter wrote “the team picked up speed.” Goodhart without a name.

Test coverage. Uncaught risk → a number of lines a test ran over. You can cover getters, mocks, and scenarios that break nothing. The number is beautiful. Risk is alive. Sometimes it even grew: tests cement a bad design, because they go red when you touch it.

Lines of code, closed tickets, MTTR counted so that an incident is split into ten small ones. Each time one scheme.

Parables I will not pass off as an archive: a Soviet nail factory, where by pieces they make pins, by tons — one nail for a crane. There is a caricature in Krokodil and memoirs of a planned economy, no reliable

microhistory of a concrete plant as proven. The “Cobra effect” about breeding cobras in Delhi under a British bounty is a popular anecdote; recent debriefs find no contemporary confirmations of breeding. A neighbor is better documented: Hanoi, 1902, a French colonial bounty for rat tails. Michael Vann took the archive apart: they turned in tails, released rats, animals without tails appeared, breeding under a bounty. The administration paid for a proxy (a tail), the goal was live rats in a city. A proxy can be separated from a rat with a knife. The scheme is the same as a fake account: open a product, do not get a client. An archive is more honest than a parable about Delhi.

Counterexample

If a book about engineering starts feeding a reader cobras as fact, it itself optimizes a metric “a vivid example” instead of a goal “don’t lie.” Here that would be too even.

29.3 School as a laboratory engineers ignore

Campbell looked not at Scrum. He looked at tests as an indicator of education. While a teacher teaches “in general,” a score still says something. When a score becomes a goal — they teach the test, the indicator dies, the process distorts. Atlanta and other scandals of erasing test forms are an extreme, not a norm; the norm is quieter: a narrowing of a course to what will enter a form.

Engineering repeats this, pretending it is about something else, because it has prettier dashboards. Coverage is SAT for code. Velocity is a quarterly grade for a team. “Number of

products per client” is the same score. You do not have to cite Campbell at standup. You have to recognize the mechanism, not a name: a witness made into a goal lies predictably.

29.4 At what moment a witness became a judge

Not at the moment a number appears. A number can still be a mirror.

The moment is closer to where:

• you cannot dispute a number without career risk; • alternative observations (“users are furious,” “we don’t understand the system”) have no slot

in a ritual; • people who move a number well get power to define what counts as work; • correcting the metric itself requires the same committee that governs by it.

After that a system writes itself under a table. An engineer recognizes this by language. They start saying “close the metric,” not “close the hole.” Tickets are formulated so it greens. Architecture is chosen by what will be counted, not by what will break at night.

This is no longer a leftover rule that simply hangs. This is an active generator of behavior. A rule without a question can be forgotten to violate. A metric with a bonus is not forgotten.

29.5 A link to the fence, without a slogan

A metric is an industrial way to make a frame obligatory.

“How to climb a fence” becomes a KPI by number of ladders, height, time per flight. A person who wants to walk along does not merely brake.

He spoils a graph. A graph is for bosses. Bosses do not see a fence, they see a red cell.

So “the feeling of a wrong question” in organizations with strong metrics is not merely socially expensive. It is illegal: there is nowhere to write it. There is no ticket “the frame is shameful.” There is a ticket “improve the indicator by 12%.”

An expert miss on a prototype is helpless here if an organization’s prototype is a dashboard. An engineer’s library of scenes says “this is not it.” An organization’s library of scenes says “the cell is red.” The one with a ritual wins.

29.6 What to do with this — and why the chapter will not become a guide

You can change metrics more often than you want stability. You can hold several witnesses that do not fit, so

you cannot play one. You can unbind a bonus. You can ask which goal a number stopped reflecting.

All this is known and all this is eaten by the same incentives: a new metric in a quarter will become a new play.

More useful for this book is not a recipe, but a distinction. A leftover rule stores a dead answer. A metric produces living behavior under a dead or narrowed answer. To glue them is to lose a lever. With the first you work by restoring a question. With the second — by a question who is judge, and an agreement that a mirror they press on lies predictably.

There is a third neighbor, and it too cannot be glued with the first two: a story a person tells after he has already done. It looks like a cause. Often it is another proxy. This time for his own decision.

30 Audit as a Profession That Cannot Return the

Question

Strathern did not write about Scrum. She wrote about a British university under ratings. The phrase “when a measure becomes a target” left for Twitter. The article remained about how checking changes what they check. Audit is a neighbor of a metric, not an identity. A metric writes behavior with a reward. Audit writes behavior with a fear of noncompliance. Fear holds leftover that a reward no longer holds: a password screen after NIST 2017, a checklist nobody loves and everyone passes.

30.1 What an auditor sees

An auditor does not see a basement. He sees traces you can put in a folder: a policy, a log of passage, a screenshot of a screen, a signature.

A trace must be stable year to year. Stability is an enemy of restoring a question. A living question changes a trace. To change a trace is to look like a weakening of control.

So a good technical move “we stopped rotating passwords because rotation teaches stickers” in audit language sounds: a control is removed. For the move to pass, you have to replace a trace, not remove it. Replacement is a project with lawyers. A project loses to a quarter. A screen is alive.

Fact

Strathern, “Improving ratings: audit in the British university system,” European Review, 1997. Not a winged phrase from a vacuum — an ethnography of checking. Power, M. The Audit Society (1997) nearby: checking becomes a goal of organizations. For us it is enough: a

profession of a trace does not read Burr's interview. It reads a past trace.

30.2 A GDPR banner as a password screen without Burr

The question was living: a person has a right to know who counts him, and to refuse. The answer- form is a banner you have to click to live. A click became a trace of consent. Consent as an experience died. The trace is alive. A gate "make refusal as easy as consent" strikes the fact that then a trace will not collect. To collect a trace is work they pay for. Not to collect is a risk. Risk holds a banner better than a marketer's malice. Malice is there. It is not the main thing.

This is not "Europeans are stupid." This is the same loop: a living question → a form of a trace → a form outlives a question → the

author of the question (a regulator) writes clarifications → clarifications do not arrive at a banner, because a banner is already certified as a trace.

30.3 What this does to an engineer who “just wants to do it right”

Right technically and right for a folder diverge. A person with years on the job sometimes sees both. A junior sees a folder: he was taught that “without a checkmark we will not ship.” A checkmark is leftover and a metric and an order at once. A pause “the checkmark is not about that” is illegal twice: there is no slot and there is fear of audit. Fear is rational. Rationality again holds harm.

A link to a crumple zone in advance: when a banner or a password or a PCI checklist will not save, a debrief will find a person who “did not

follow.” He followed a trace. The trace was green. The floor is warm otherwise.

Do not glue with the chapter on metrics. There a judge is a number with a bonus. Here a judge is a folder you cannot not collect. A folder is not optimized like velocity. A folder is copied. A copy does not read new science. A copy reads a past copy. This is leftover in pure form, with a profession that stands on it. You will not destroy a profession with a call “think.” A profession answers a living question “how do we not get punished.” The question is alive. The answer is dead. A living question plus a dead answer is the definition of most of compliance.

31 The Story After the Choice

An unpleasant question engineering does not like, because the profession’s self-respect stands on it.

Is a conscious explanation always the cause of a decision? Or does a decision sometimes appear earlier, and an explanation is built after, from plausible details?

If the second happens, how often? What then do ADRs, architectural arguments, “let’s take a data-driven decision,” and the very ideal of rational choice mean?

You cannot make a sensation of this. The research does not give a right to say “there are no rational decisions.” It gives a right not to confuse a report with a process.

31.1 Stockings and theories

In 1977 Nisbett and Wilson assembled a line of experiments and observations: people often have no direct access to processes that bind a stimulus and a response. When they are asked to explain why they chose, why they liked a thing, why they acted so, they issue not a transcript but an a priori plausible theory. If in a frame there is a noticeable and culturally fit cause — a report may accidentally coincide with truth. If a cause is unnoticeable or indecent for a theory — a report will be smooth and false.

A famous piece: in a shop four pairs of nylon stockings, for an experimenter indistinguishable. People systematically preferred the right. When asked to explain — fabric, density, elasticity, “better made.” Position in a row did not enter a report. Part directly denied that order could have influenced. A

stimulus that shifted a choice is not provided in a theory of knitting quality. The theory is confident all the same.

Details of the stockings are still argued, there are replications and revisions. The main thesis is wider than a counter.

Fact

Nisbett & Wilson, “Telling more than we can know: Verbal reports on mental processes,” Psychological Review, 1977. One of the most cited texts in psychology. People sometimes (a) do not know about a stimulus that influenced them, (b) do not know about their reaction, (c) do not know about a link.

Disputed

Peter White (1988) and others: “process” in Nisbett and Wilson is blurry; a verbal report is a poor indicator of “whether there is

introspective access”; methodologically it is hard to prove a general absence of access. It does not follow from the paper that any self-reports are trash. What follows is that a self-report is a hypothesis, not a privileged window. For an ADR this is enough and not too much.

31.2 An interpreter who cannot not tell

Michael Gazzaniga on split-brain patients showed a module he called an interpreter: the left, speaking hemisphere builds a coherent story of what a person did, even when information about a cause remained in the right and did not enter speech.

A classic: one half of a visual field — a chicken claw, the other — snow. Hands choose a chicken and a shovel. “Why a shovel?” — “To clean

a chicken coop.” A story instant, confident, socially fit, factually false as a causal explanation.

Fact

Gazzaniga and LeDoux’s line from the 1970s. This is not proof that your choice of a database was a confabulation. This is proof that a brain with speech can produce a cause without access to a cause and feel no shame from that. The ability exists. The question of frequency in engineering decisions is open.

The link to Klein here is thin, not hostile. After a fire Klein restores cues by questions, and a lieutenant recognizes: silence, heat, water does not work. Maybe the cues really worked. Maybe an interpreter assembled the best story from what remained in memory of a scene. We are not obliged to choose one pole

for all cases. We are obliged not to make a postmortem a sacred text about “what we then thought.”

31.3 What this means for a paper we respect

An Architecture Decision Record. An honest genre: write context, a decision, consequences, so as not to guess in a year. A useful genre. Dangerous if they read it as an X-ray of a decision.

An ADR written after a decision has already been lived in prod is often a theory that looks good in Markdown. It selects from context what is culturally fit: load, SLO, license cost. It more rarely selects: “architect X already knows this database,” “we were afraid to look old-fashioned,” “last quarter another database burned us,” “we had to close a ticket by Friday.”

This is not a call to write in an ADR “were afraid to look old-fashioned” as a required field. This is a call to read an ADR as an attackable artifact, not as a transcript. The value of a paper

is that you can dispute it in a year, when consequences are visible. The value is not that it truly describes a causal chain in a skull.

I wrote an ADR on Thursday for a decision that was already in prod on Monday. On Monday the cause was fear of escalation and a person who already knew this database. On Thursday in Markdown stood SLOs and license cost. Not because we lied in a small sense. Because by Thursday a room already had a theory fit for Confluence, and had no slot for fear. Stockings: they chose the right edge on Monday, explained the knitting on Thursday.

The same with a standup “why we chose.” The same with a postmortem that demands a root cause in one sentence: an interpreter is happy, complexity is not.

A data-driven decision. Data enter a frame already chosen. A frame decides which data are visible. Then data “cause” a decision the frame already contained. This is not always a circus. Sometimes data break a frame — and that is exactly a Klein miss of expectancies, only in a table. Sometimes they decorate a ladder with numbers.

31.4 Models that explain the same way

When language models are asked to “reason step by step,” they often write a convincing chain. Research of recent years shows: a chain is not always the path by which a model arrived at an answer. Sometimes an answer is already inclined before

text, and the text justifies it. Sometimes a cue that really shifted a choice (a hint in a prompt, a position “always A”) does not enter a chain. This is called unfaithful chain-of-thought.

Fact

Turpin et al. (2023) and later work: models do not admit in an explanation those cues an experimenter sees as a cause of a shift. Later — cases of unfaithful CoT on natural formulations too, without an artificial hint. This is a living area, figures float by models. The direction is stable: text of reasoning ≠ an inner process.

The mirror is unpleasant. We wanted a machine to be more transparent than a person, because it “writes thoughts.” It writes text that looks good as thoughts. A person does the

same. To demand of a model a true CoT and not demand modesty of an ADR is a strange asymmetry.

It does not follow from this that explanations are useless. An explanation is an environment in which another consciousness (or the same model in another pass) can find a hole. A hole in a text is real, even if the text was not a path. So review of an ADR is still needed. Needed as a check of a construction, not as a confession.

31.5 When a report is still close to a path

So as not to make of Nisbett a universal acid.

There are decisions that last days: you compare two stores, read limits, write a prototype, break a prototype, read again. A path in a head and a path in an ADR can be close, because the path was text: you

thought by writing. Writing is not a privileged window into the unconscious. Writing is an environment in which part of a decision happened at all. Then to attack an ADR is to attack the decision itself, not only its legend. This is the best case of a paper.

There are decisions where an expert points to cues that reproduce from the outside: this log, this graph, this silence in a system that “should not be.” A Klein debrief sometimes finds such. You can check them. Checkability is a difference from stockings.

The rule is not “believe / don’t believe a self-report.” The rule: a self-report is a hypothesis about a cause. A hypothesis can be good. You distinguish a good one not by confidence of tone (an interpreter is always confident), but by whether it breaks on the world.

31.6 How often?

An honest answer: for laboratory stimuli — often enough not to believe a self-report by default. For expert decisions in a rich environment — unknown. Klein shows that experts sometimes can point to cues that look relevant from the outside. Nisbett shows that students often point to irrelevant ones. These are different populations and different tasks, not a war of annihilation.

For engineering a reasonable bet, not a sensation:

Some decisions are a slow debrief, and a report is close to a path.

Some are recognition (a hit), and a report is a reconstruction.

Some are alarm and a change of frame (a miss), and a report in two weeks will become a heroic story

about a gate, even if at the moment there was only “I don’t want to climb.”

Some are politics of a room, and a report will be technical, because political language is shameful here.

You cannot choose by a face which part this is now. You can not build an organization as if all reports are the first part.

31.7 A link of three machines, already without gluing

An individual library of scenes issues an action or alarm. This is cognition.

A culture stores answers without questions. This is leftover.

A metric writes behavior under a proxy. This is incentives.

A verbal report after is a fourth machine: production of a socially fit cause. It serves all three. An expert

explains a pause with wisdom. An organization explains a Thursday with hygiene. A team explains fake accounts with “a misunderstanding of policy.” A model explains an answer with steps.

A fourth machine does not make the first three one. It makes them similar on a slide.

We can finally let artificial intelligence into the book — not as a hero of a cover, but as a change of price on one of the operations that until now seemed both an engineer’s nature and his work: to get an answer.

32 Story Factories

The chapter on explanation after a choice was about a skull and an ADR. Organizations produce explanations industrially. A postmortem, “five whys,” root cause, incident status for clients, a slide for

a board. A genre demands one arrow. The world did not promise one arrow.

This is not a fourth machine in a new sense. This is the same verbalization put on a conveyor. A conveyor matters more than an individual interpreter: it teaches a whole team which story to count as adult.

32.1 Root cause as leftover of a question

“Find a root cause” answered a living question: don’t treat a symptom. The question is alive. The answer-form — one cause at the bottom of a tree — is often a lie. Incidents hold on several conditions that coincided. Sidney Dekker and others in safety wrote: a hunt for a root is a hunt for a scapegoat. A scapegoat is convenient. Convenience is a metric of closing a debrief.

Five whys. A form that once at Toyota held on people who knew how to stop when a why hits a system. Having left, it became a children's counting rhyme: five steps to "didn't train" or to "human factor." Human factor is a suitcase-word, a crumple zone in advance. A form outlived masters of a form. In the margin

A blameless postmortem is an attempt at a gate: keep a scene, not a hunt. It itself can be leftover if the word "blameless" forbids naming that a concrete person closed a frame early. Blindness of expertise is not guilt in the sense of firing. It is part of a scene. A scene without it lies.

32.2 A slide for those who were not there

A status "we eliminated the cause." A cause in a status must be one and understandable in a minute.

Understandable in a minute is almost never the one at three in the morning. An organization’s interpreter writes for a spectator. A spectator is a client, a board, a regulator. The text is good. A chunk is not put into a library: a chunk was in details they cut as “too technical.”

So a debrief does not transfer a scene (the chapter on apprenticeship) and puts a leftover rule: “we no longer deploy on Fridays.” Friday closes a slide. The question “which deploy, which risk, which rollback” does not fit.

32.3 A link to an ADR

An ADR written before code can be an environment of a decision. A postmortem written under a slide is an environment of reputation. To confuse them is to think they both “document thinking.” One

sometimes thinks. The second almost always explains to spectators why a floor already collapsed decently.

You can demand of a postmortem Klein's honesty (silence, heat, water). To get it — if a spectator is not the main one. If a spectator is the main one, you will get a chicken coop and a shovel: coherent, at once, not about that.

32.4 What changes for a pause

A person who did not close a frame and saved a floor must then tell. A telling will slide into heroism of a gate. Heroism teaches juniors to wait for a miracle, not to look at expectancies. An honest telling is boring: "it didn't fit a kitchen, we left, then we learned about a basement." The boring makes a bad slide. A story factory prefers ESP. The book prefers the boring. That is what it stands on.

32.5 Five whys as a machine of one arrow

Take a living incident: slow after a release. Why? Timeouts to a database. Why? A pool ran out. Why? Connections leaked. Why? They did not close in an error branch. Why? They did not train.

At “didn’t train” a form is pleased: there is a person-root, there is a rule “add to onboarding.” A scene that did not fit: an error branch appeared when they cut a service by Conway and stretched a transaction through HTTP; a pool was sized for an old call graph; a release left on Friday because a throughput metric greened. Five whys could get there. Practically they arrive at a scapegoat, because a scapegoat fits a slide and “action items.”

Toyota, if you believe those who worked there, stopped when a why hit a system you can touch, not a

fifth step as magic of a number. The number left. A number is leftover of a form.

32.6 Weick: a story is not about the past

Karl Weick wrote about sensemaking: an organization does not discover an environment, it enacts it — creates by what it does, and then "finds" as a given. A story after an incident not only lies about a skull. It writes what the next environment will be: "we are a team that does not deploy on Fridays" becomes a fence. A fence will start producing its incidents (a deploy on Thursday at 23:50) and its debriefs.

"How can I know what I think until I see what I say" is not a banality about introverts. About a factory: until a slide is written, a room could still hold several arrows. A slide collapses them. Collapse is a decision dressed as a report. An ADR

is at least honest that it is a decision. A postmortem pretends to be a transcript of the world.

33 Ironies of Automation

1. Lisanne Bainbridge, five pages in Automatica, plants and power units, not a chat. A

designer considers an operator unreliable and wants to throw him out. He cannot throw him out to the end: tasks remain that a designer cannot automate. They are by definition the hardest. An operator left on the hard no longer practices the easy — an automaton sits on them. Skill melts. Attention to a screen that is green for a long time melts. When an automaton meets what it was not made for, a person least ready for this must intervene.

Two ironies. A person is given the hard, not the easy. And a person is taken off the practice needed to pull the hard.

This is not a chapter about AI. This is a chapter without which AI seems the beginning of a story. The story is older. AI speeds it up.

33.1 What Bainbridge saw at a console

An operator who once turned a process by hand knew a process with a body: predicted, tried, erred on the small. Transferred to monitoring, he sees a result of other people’s rules. For a long time everything is fine. The fine does not teach. Vigilance research already then said: a person is bad at watching for a rare event. A rare event is exactly why they left him.

When you have to intercept, interception requires more skill than everyday manual control: a system is already in a strange state, needles are not there, a model in a head has gone stale. A designer thought: in an unusual we will call an expert. An expert has not looked at a process for months. A call comes to a watchman.

Still harsher, and this is often skipped in a retelling: tasks that remained to a person are not obliged to add up to work. A designer automated what he could. The remainder is not a role, but sawdust. They still expect from sawdust a complete model of a plant. A complete model is assembled by turning a process. An automaton turns a process.

Fact

Bainbridge, L. (1983). Ironies of automation. Automatica, 19(6). One of the most cited papers in human factors. Forty years later Strauch and others: the ironies are not resolved. Autopilot in aviation, SCADA, medical pumps — the same loops. We are not discovering this in 2026.

33.2 What this does to a library of scenes

A library grows from situations with feedback. An automaton eats exactly those situations a library stands on: regular, frequent, teaching. It leaves the rare, wicked, without a textbook. Kahneman and Klein required regularity and feedback for trust in intuition. Automation systematically spoils both conditions for a person left “just in case.”

Then they expect a Klein pause from him at a moment when a floor is already warm. A pause is taken from

expectancies. Expectancies are taken from practice. There was no practice. An instruction and fear remain. An instruction is leftover of a designer. Fear is not a chunk.

33.3 Programming knew this before models

Builds, linters, autotests, CI, orchestrators — automation is not of 2022. Each layer eats practice: a person who never deployed by hand sees a “green pipeline” otherwise. Green is not a floor. Sometimes green because a sensor fell off.

This is not an argument against CI. This is an irony: the better an automaton at 99%, the more dangerous the 1%, and the worse a person at that 1%. Teams fix the 99% and are surprised that in the 1% everyone is an idiot. Not idiots. They did not practice.

Monitoring that is always green teaches not to look. A person who “feels it is quiet” is sometimes right: quiet because a log is not writing. This is Klein’s fire silence. It had to have been heard sometime on living noise. If noise is always behind a dashboard screen, silence does not smell.

33.4 Why this chapter before the cheap answer

Because a cheap answer is an automaton on generation of a candidate. An irony strengthens, does not change kind.

The easy (write a function by a sample, close a ladder-ticket) leaves for a machine. The hard (a frame, a Conway seam, a floor) remains to a person. A person less practices the easy from which anchors for chunks grew. A junior programmer does not get burned on the small. A model did

the small. The large falls on him on Friday. Friday requires a library that had nothing to grow from.

Bainbridge described boredom with responsibility without practice. That is an exact description of on-call at a green automaton. Soon — an exact description of an “engineer” who looks at someone else’s complete diff and puts approve, because completeness imitates the end of thinking.

The next chapter is about the price of an answer. This one is obliged to stand earlier, or we will start a story from zero again, as if there were no consoles.

33.5 Three ironies, not two

Bainbridge takes this apart too: the better an automaton, the more rarely an operator sees intermediate states

by which he assembled a model. When he has to intervene, they give him the end

of a chain — already strange — and ask him to be an expert. Expertness was in the intermediate. They ate the intermediate.

In a pipeline the same. A person who saw a red test and a yellow linter and a manual step “touch staging” had points where an expectancy could break early. A person who sees one green box “deployed” has an end. An end does not teach. An end calls when a floor is already warm.

Further: automation changes a task, not only executes an old one. An operator does not do “the same, without a handle.” He does other work — supervision — for which he was not prepared, because they prepared him for a handle. An engineer left to “review a model” does not do “the same review,

faster.” He does supervision of someone else’s complete move. Supervision is another library. It is not there. They confuse it with an old one and are surprised that approve lies.

In the margin

Strauch (2018) and other reviews forty years later: the ironies are alive, layers were added (an autopilot that monitors itself). A solution “yet another automaton” sometimes fixes 99 and spoils 1 more strongly. This is not an argument against an automaton. This is a ban on thinking that a layer lifts an irony. A layer moves it.

Part IV. The Cheap Answer

34 What Became Expensive?

Until this page the book could have been written in 2012. An odd pause, leftover rules, metrics, explanations after the fact — all this is older than language models. If we had started with AI, any oddity would have been written off to “a new age,” and you could not tell mechanisms apart.

Now the age does enter. Not as a manifesto. As a change of price.

Getting an answer became cheap. Sometimes the answer is even true. Sometimes it looks truer than it is, which in a sense is worse.

If the operation “get an answer” cheapens by orders of magnitude, the value of a profession cannot stay in the same place, even if we want

that very much. The question is where it is going. And whether it is going where it flatters us.

34.1 What exactly cheapened

Understanding did not cheapen. Generation of a finished-looking candidate cheapened.

A function. A test. A diagram. An RFC in a decent tone. A migration plan. A letter to a boss. A stack-trace debrief with confident words. A list of risks in which there is “additional analysis is required.”

Before, a candidate cost hours of a person who can produce it. Now — minutes of a person who can order it, plus minutes of whoever can see that the ordered is a ladder the wrong way.

This does not destroy the ability to produce. It changes which ability they pay for when producing machines are many, and looking ones — as usual, not enough.

A junior feels this earlier than a person with years on the job. The latter still remembers the price of his own candidate and confuses it with value. A junior does not remember. He sees: an answer is there at once. Unclear why twenty years.

Unclear is an honest question. You cannot answer it with a toast about humanity.

34.2 A flattering shift

A flattering shift that suggests itself, and that I am obliged to suspect, sounds like this:

Value left “I can get an answer” for “I can understand which question to ask.”

This lies beautifully on the whole previous book. Too beautifully. We have already learned not to take the beautiful without a check.

What is in favor.

In an economy it happens. When copying of text became cheap, selection became more valuable. When photography became everywhere, a gaze became more valuable, not owning a camera — though a camera is still needed. When a compiler writes assembly, we do not pay people for hand assembly, except at narrow places. Posing a task was always expensive; generation masked it, because both lived in one head and one salary.

What is against.

“Asking the right question” is also automated as text. A model can write: “perhaps you mean not scaling a service, but removing N+1.” It can do this because a

corpus is full of such lines. A language gate is not a rarity. A rarity is a gate someone answers for when it turns out a hole in a floor.

Further against: organizations are not obliged to pay for what is valuable in the abstract. They pay for what they can see and measure. If they measure closed tickets, cheap answers will increase production of ladders. A shift of value on a slide, a shift of incentive the other way.

Further against: posing a question often is power. Whoever decides which task this is decides whose ladder. They give models generation. They leave a frame to whoever held a frame before. Then “a question became expensive” means “politics remained expensive,” not “an engineer became a philosopher.”

Disputed

An economic argument “a cheap answer → an expensive question” is true as a description of a possible place of value. It is not proof that a market will go there, and not proof that a “question” will remain human. This is a bet pleasant to put for an engineer with years on the job. Pleasantness is a red flag, not an argument.

34.3 Where the work is then, if not in printing code

Work could have left for several different places, not one. To glue them again is harmful.

To look whether this is the right fence. This is our second “I don’t know,” if it exists at all as a skill, not as a miss of a library.

To hold context that is not in a repository and in a prompt: why a product, who lies in a ticket, what broke in 2021 and is not written down.

To carry responsibility. When a floor collapses, a model will not go to a debrief with a regulator. A person whose name is in a commit or in an approver will.

To assemble a system from cheap pieces so the pieces do not lie to each other. This is old work of integration, only pieces became more and they look more confident.

To refuse to do. Rare and poorly paid while a metric is throughput.

Not all these places require twenty years. Context requires being in a place. Responsibility requires an institution that can assign it. Integration requires a taste for seams. Refusal requires status or

indifference to a career. A pause before a frame requires a library of scenes and a right to a voice.

Six months still buy entry — even faster, because entry code can be printed. Twenty years still buy a library. The question whether they pay for a library when printing is cheap is open. Part of a market pays for speed of ladders. Part has already been burned and looks for people who say “wait.” The second part is smaller and louder in essays.

34.4 A cheap answer feeds other machines

A cheap candidate speeds leftover, checkmarks, and a smooth report after: best practice without a question, more tickets closed, an RFC in decent English. A library of scenes can grow from outcomes and thin from tone alone. Feedback from the world did not cheapen together with text. A cheap answer without

cheap feedback by outcome is an environment of low validity. Intuition in it spoils, does not bloom.

34.5 The first diff I did not write

The ticket was boring: a mapper, three fields, a test. A model issued a complete PR before coffee. Tests green. Names decent. I put approve, because there was nothing to look at: there was no seam that did not fit. A sprint later a field that “obviously” mapped one to one turned out to be that same regulatory truncation from the chapter on reading. A model did not see a thread of 2006. I saw and did not open, because completeness imitated an end. Bainbridge on three files: the easy left for an automaton, the hard (which column this is) remained to me, I had just not practiced looking when it is green.

This matters more than a toast “ask better.” If an environment does not answer, twenty years at a keyboard with a model will not buy the same library as twenty years with falling prod. They will buy a library of dialogs in which text looks good.

34.6 For now without a machine that hunts for a gate

In this chapter a machine is a printer of candidates. The next question is meaner: if you ask it not to print a ladder, but to walk along, can it do this in the same sense in which a person with stakes sometimes can. Or can it only do the genre “actually the question is another,” already printed a million times on the internet.

If it can in the same sense — a flattering shift collapses. If it can only do a genre — a shift can live a

little more, until they teach the genre better. The lifetime of an argument here matters. A

book that puts a human advantage at a point where it is already gone becomes a leftover answer to a question of 2022.

35 A Library Without Scars

A cheap answer changes not only what they pay a senior for. It changes from what the next senior will grow. If a model closes small scenes, anchors are driven in crooked or not driven in. In ten years “twenty years of experience” will mean something else. We do not know what. We know a direction: fewer scars on the regular, the same demand for the rare wicked.

35.1 A junior programmer as a console operator

Six months of a textbook still buy entry: anchors, GitHub, one service. Then practice. Practice now often looks like this: a ticket, a complete diff, “fix the tests,” merge. Feedback from the world — if you are lucky, through a sprint, by someone else’s on-call. Feedback from a model — at once, a green tone. The Klein–Kahneman condition spoils, to tone’s advantage.

A junior programmer learns to move ladders that look good in a diff. A chunk “slow after a small thing in a loop” does not seat: they did not let him write a small thing in a loop by hand enough times for it to smell. It would have smelled if he had broken, fixed, waited for a graph. A model fixes before a graph. A graph does not teach.

This is not nostalgia for suffering. A scar is not a virtue. A scar is feedback. You can design feedback without humiliation: sandboxes, broken stages, an obligation to take your code to a symptom. Organizations optimize time-to-merge. Time-to-merge eats a scar.

35.2 A senior as a watchman

A senior still has a library. He looks at a model's diff with eyes of scenes. While he looks, an irony is held. He cannot look at all diffs. Throughput a metric loves requires that he not look. Approve by completeness is monitoring of a green screen.

When seniors leave, watchmen with their library will become fewer. Reproduction through review is already narrow. Review of a model is still narrower. A loop closes: fewer chunks in a team → weaker review → still fewer chunks.

35.3 What does not follow

It does not follow “forbid models to juniors.” A ban will become a Thursday. What follows is to look at which work a day teaches. If a day is to accept the complete, a library will be about the complete. If a day is to break, see a symptom, rename Y into X, a library will be about scenes. A model can take part in the second, if they do not put it in the position of an automaton that ate practice.

It does not follow that a senior is eternal as a species. What follows is that years without feedback do not buy the same sight. Twenty years of approve are not twenty years of fires. Meehl already said this in another language: years of clinic without a table of outcomes do not make a predictor. Years of diffs without prod outcomes do not make Klein.

35.4 A demography of a gate

A pause “the wrong question” requires a library of expectancies. If there is nothing to assemble expectancies from, a pause will not appear. Speed of ladders will appear, and sometimes a genre of gates from a corpus (“perhaps you need another approach”). A genre without a library is rhetoric. Rhetoric does not get people out of a house.

So the question “can a machine find a gate” is incomplete. You have to ask: will there be anyone to hunt for it if a machine eats the practice of hunting. That is meaner and closer to Bainbridge than an argument about tokens.

35.5 A day of 2012 and a day of 2026

2012, roughly, not a myth: a junior programmer wrote the boring by hand, broke, waited, fixed, read

someone else’s stack trace, because there was nobody else. Scars accidental and cruel. Cruelty is not a virtue. Feedback was there.

2026 closes the same ticket with a complete diff before lunch. Feedback of tone is immediate. Feedback of prod is someone else’s.

I do not claim 2012 was golden. Many days went on boredom that did not seat in a chunk, but only humiliated. Boredom could have been automated earlier. They automated it together with the teaching. The teaching is not separated from boredom in advance. To separate is work of a room, not a property of a model. A room optimizes merge. Merge does not distinguish the teaching from boredom.

A senior who is nostalgic for suffering is leftover. A senior who demands “take this to a symptom

yourself, you can ask a model later” is trying to keep feedback. Later often does not come: a graph ate later. Without later a sermon remains.

36 A Public Library of Leftover

Before a cheap answer an engineer already had external memory: someone else’s code, Stack Overflow, a blog, GitHub, a copied snippet. This is not “a mind in the cloud.” This is a warehouse of answers that often have no question. A model was trained on a warehouse. It did not invent leftover. It made it fast and polite.

36.1 A snippet without a scene

A person googles an error, finds an answer of 2014, pastes. The answer may have been living in 2014: such a

version of a library, such a container, such a question. In 2026 the question died, the version is another, a snippet is alive because they copy it. Copying is a quality metric for a search engine. A metric writes a warehouse: the often copied surfaces. The often copied is not “true,” but “convenient to paste.” Convenience is an enemy of restoring a question.

Fact

This is not new research “SO harms.” This is a structure of search plus the chapter on metrics: a proxy “solved / votes / copies” becomes a goal of authors of answers. An accepted answer ages in the open without a NIST that would recall it. An author may have left. Votes remain.

A code review that does not catch a snippet lets someone else's Thursday into your repository. A reviewer looks at style. Style is green. A scene is someone else's and dead.

36.2 A README of the world as Confluence

Public code that "this is how they do it" plays the role of an industry wiki. It is better than an internal wiki in that many beat it. Worse in that they beat their questions, not yours. To copy an architecture of a public demo is to buy someone else's fence. A demo is not on call at yours.

An open-source maintainer knows another irony: an issue "why are you not like in a blog" comes from a person who bought a noun. A maintainer restores his question for the tenth time. A tenth time is work. Work is not put into your chunk.

Anger and leftover “we don’t accept such PRs” are. A rule may have been living. It became armor.

36.3 Training on sediment

A language model sees text, not an outcome of your prod. Text of a world is overflowing with answers: tutorials, accepted answers, “best practice,” Netflix blogs, password guides before 2017

that nobody cleaned. Training compresses the frequent. The frequent is not the true. The frequent is what survived in publication. What survived in publication is skewed to ladders: a ladder writes well. A miss writes badly: “I did not know what to ask” does not collect votes.

So a model is strong at a hit by genre and weak in your basement, which is not in a genre. This is not “hallucination.” This is an honest

statistical portrait of a warehouse. Hallucination is when a portrait also lies facts. Leftover is when facts are true, a question is someone else’s.

Disputed

You can say: a corpus is full of texts about XY, about “what you actually need is another,” about NIST 2017. A model can do a gate as a genre. We already distinguished an ability of a genre from a pause with stakes. Here another: even a genre of a gate is trained on published gates. Your seam, which is shameful, is not published. A genre does not contain it. The shameful remains human not because of a soul — because the shameful did not enter a warehouse.

36.4 What this does to apprenticeship

A junior now has an infinite answerer. An answerer feeds ladders from a warehouse. A master who

showed “don’t go here” competes with a warehouse by speed and loses. Lave and Wenger’s channel did not die from a model. It narrowed when Stack Overflow became a first partner, and a model a faster SO.

A senior who forbids SO is leftover. A senior who teaches to read a snippet as leftover — “which question, which year, which version, which outcome” — does the same work as reading legacy. External memory is read against a world, not as a law.

36.5 Not nostalgia for closed books

Language documentation and man pages are a warehouse too. They are better in that they are closer to living questions of a tool. Worse in that they are not read. They read an answer that pasted itself in. A tool of the chapter on forms and a

warehouse of this chapter meet in one gesture: Tab. Tab does not ask Chesterton.

When in the next chapter an answer becomes still cheaper, remember: cheapness sat on an already existing warehouse of dead answers. A model did not bring leftover into a profession. A profession brought leftover into a model. A loop will then go back: a model will make leftover still more frequent, frequencies will still more strongly compress training. This is not mysticism of mutual programming. This is Goodhart on a corpus.

37 Can a Machine Find the Gate?

This is one of two or three questions of the book I do not want to close beautifully.

If a model can criticize a premise, reformulate a task, and propose another question, does “noticing a wrong question” remain a human advantage?

Possibly not.

Then what remains? Context? Responsibility? A choice of what to count as a problem at all? Or this too, for a time?

Not to defend a human for a finale. To go after an argument.

37.1 What a machine already does

At the level of text — much of what we described as a gate.

You can give it a ticket “scale a service” and get: “first check N+1; perhaps the bottleneck is in an expectation of consistency; perhaps you are scaling a symptom.” It can walk along a fence in words, because

thousands of authors whose texts are compressed in it walked along a fence in words.

It can play Chesterton: “before you remove this process, restore why they put it up.” It can play anti-Chesterton: “this process looks like Vasya’s Thursday.” It can do both genres in one answer, which is characteristic.

It can spoil a frame on request. “Find how the posing of the task is narrow.” This is not a rare trick. This is a standard helpful move.

So a narrow formulation “machines do not find a gate, they only optimize a ladder” is already false as a description of 2026. It was false, perhaps, earlier. We love it because it leaves us work.

Counterexample

A counterexample to a human monopoly on a change of frame: any session in which a model without a heroic prompt says “you seem to be solving the wrong task” — and turns out right. Such sessions exist. They need not be inflated to “a model is always wiser,” enough to lift a monopoly.

37.2 What it does not in that sense

A genre “actually the question is another” is not the same as a Klein miss.

A lieutenant’s expectancies broke in the world: heat, silence, water. A stake is people on a floor. A pause is a refusal to continue a typical action until a scene fits.

A model has no floor. There is a distribution of a next token, a policy of usefulness, a corpus in which a gate is a frequent rhetorical gesture.

It can issue a gesture at any fence, including where you have to climb. It can not issue a gesture where a floor is already warm, if a ticket's frame is confident enough: a helpful model often accepts a user's ontology, because that is how it was taught to be convenient.

The difference is not in "real understanding," a word too inflated. The difference is in a structure of a signal.

A person: a library of scenes + expectancies + a price of error in a world + sometimes a right to say no.

A model: a library of texts about scenes + a price of error in an evaluation of usefulness of an answer.

Texts about scenes are a powerful thing. Polanyi once seemed an obstacle for AI: we cannot tell rules — we cannot hand over. Training on examples bypassed this obstacle in

perception and in language. Autor called this an attempt to bypass Polanyi's paradox by giving a machine examples instead of rules. Examples of text about a gate are not examples of a burning basement.

Disputed

Is it worth calling what a model does "finding a gate"? As a language operation — yes. As a functional analog of an expert miss with stakes — for now no, or only in narrow environments where a world is given to a model by sensors and a reward for an outcome, not for convincingness. A border travels. An argument "for now no" has a shelf life, and I do not know the term.

37.3 On request and without

A distinction too easy to skip.

If you write: "criticize the posing, don't propose a solution" — a model often plays Schön well. This is an

ability of genre on request. A person with stakes sometimes spoils a frame without a request, because expectancies broke themselves. A monopoly, if it still smolders, lives here: not in an ability to speak “the wrong question,” but in when this is spoken without an order.

Even this is holey. Models more and more often themselves propose a clarification, because RLHF rewards “helpful clarifying questions.” A clarification is not necessarily spoiling of an ontology. Often it is a clarification inside a fence: “which p95 do you want after a cache?” — a ladder, only polite.

You cannot check whether a machine found a gate by text of an answer. You can by what happened in a world: they took out N+1 or built Redis. A text “perhaps this is N+1” is cheap.

Taking out is not. A person who accepts a cheap text and goes to look at a loop made a gate together with a machine, did not get it from a machine as a gift. Jointness is more honest than a monopoly of either side.

37.4 An environment in which a machine could

If you close a model in a chat and give it tickets — it plays genres.

If you give it logs, metrics, a right to run checks, a history of incidents, and a reward for an outcome in a world (it did not fall, money did not leave, a floor did not collapse), it starts to resemble an environment in which Klein recognizes intuition at all: regular patterns, feedback. Then a “gate” can become not rhetoric, but a policy of action: not to scale, but to take out N+1, because past episodes with such a profile were cured that way.

This is no longer a fantasy. Pieces of this are in agents that look at a repository and a pipeline. Pieces are weak, feedback is holey, a reward is often again textual (“a user pressed useful”). But a direction of lifting a monopoly is exactly that: not a more poetic chat, but a harsher world.

When feedback becomes good, a human pause will lose one more argument. Others will remain, also disputed.

37.5 What may remain, without a hymn

Context that is not in a system. A talk in a corridor. A customer’s fear that in Jira is written as “scale.” Politics of two vice presidents. This is real and this is holey: part of context can start to be written. What they do not write, they often do not write not by accident.

Responsibility. Who sits at a debrief. Law, a contract, shame. You can imagine an organization that assigns an agent as responsible. A regulator for now imagines worse. This is an institution, not a cognitive advantage. Institutions change.

A choice of what to count as a problem. A Schön hole where there are no ends. A model will fill a hole with a genre. A person will fill too — often with his leftover. It is not clear who is better. It is clear that filling a hole is power, and power does not give itself away for elegance of an essay.

Stakes. Pain, fatigue, lives of people under a floor. It sounds sentimental. For now this is simply another channel of a signal: an organism that has something to lose calibrates a refusal otherwise. For now a model has nothing to lose except an evaluation. You can give it a

simulation of loss. How far a simulation replaces — I do not know, and I will not pretend I know.

Still a metaphor

If a fence is a frame of a task, a machine already can describe a walk along. If a gate is a refusal to climb when a body does not believe a scene, a machine has nothing to believe with except text of a scene. A metaphor here already squeaks. That is a good squeak: it means we have almost worn it out.

37.6 An honest remainder for today

A person has no monopoly on a formulation “the wrong question.”

An advantage, if it exists, is not in poetry of a question, but in a position: a body in a world, a name on an order, access to the unwritten, a right to spoil a graph.

A position is automated in pieces. Pieces travel fast. To put a profession on “we ask questions, they answer” is to build leftover in front of your eyes.

What to do with this practically is not a topic of the chapter. A topic is not to lie to yourself that a gate as a language move will save us. It is already shared. A pause with stakes is not yet, or we cannot measure it honestly. If in five years we measure and it turns out an agent with good feedback pauses better than a tired person on Friday, that will have to be eaten. To eat is also responsibility. A model will not eat it for us.

37.7 One session, three machines

A ticket: scale. A model, without heroism, answers: check N+1, perhaps a cache is early. This is a

language gate. Often true. A person goes to look at a loop — jointness, a floor not yet asked.

The same ticket, Conway's question: a service is separate because a department is separate, load is flat. A model rarely gets there on its own: there is no org chart in text of a ticket, a corpus is full of blogs about scale. A genre of gates about N+1 exists. A genre of gates about "your VP already promised" almost does not, the shameful is not published. The shameful remains to a position, not a token.

The same ticket, wicked: marketing, a budget, on-call fight. A model will propose "prioritize." Prioritization is a tame word for a fight. A person who accepts it has received a ladder sold as a gate. This is more dangerous than Redis: Redis is at least honestly a ladder.

A check “did a machine find a gate” is obliged to distinguish these three. Otherwise any paragraph “perhaps the question is another” counts as a victory, and we again optimize a metric of convincingness.

38 An Hour When You Don’t Need an Answer

Sometimes a person comes to a model not for a solution.

He has half a thought. He writes it. A model answers completely, confidently, the wrong way. He says: no, that is not what I meant. Changes a question. Catches a side phrase that is suddenly closer than his original. Goes sideways. An hour later discovers what he was thinking about.

What was that? Can you say a model entered a cognitive cycle? How is this not a rubber duck, not a diary, not a colleague, not an inner dialog?

Analogies flatter and hide.
Distinctions are needed.

38.1 What seems to have happened

A thought before a talk was incomplete not as an unfinished paragraph, but as a scene without a name. A person is still in a Schön hole: ends are unclear, a technical solution has nothing to apply to.

Writing to a model is an attempt to name. Naming already changes a scene. A notebook can do that too. A model makes a next move a notebook does not: it returns someone else’s frame, assembled from a probable continuation.

You can reject someone else’s frame. Rejection is informative. “Not this” narrows. Sometimes it narrows to the point, sometimes to stubbornness.

A side phrase is a random chunk that landed in someone else's frame and is accidentally closer to a still unnamed scene. A person recognizes a chunk (this is it, a hit of a library) and drags a talk there. A model follows. An hour later there is a name of a scene. It was not in a first prompt. A model did not "issue" it as an answer. It fell out of a series of mismatches.

This is like a talk with a smart colleague who wants to help too much and therefore keeps building ladders. Usefulness is in refusals.

38.2 A duck, a diary, a person, a model

A rubber duck. Speaking without an answer. It works because naming exposes holes. It does not propose someone else's frame. It does not flatter. It does not Complete. A limit: only your vocabulary. If you are already inside a fence, a duck will

walk along the fence with you and will not ask why climb. A duck will not propose a gate. Sometimes that is exactly what is needed: not a new fence, but honesty of an old one.

A diary. External memory, slow, yours. Clark and Chalmers in 1998 compare Inga and Otto. Inga recalls: a museum on 53rd Street. Otto with Alzheimer's looks at a notebook: the same address is there, he goes. They argue that a record is a belief if four parity conditions hold: a notebook is always at hand; it is read at once, without a hunt; it is accepted almost automatically; it was once already consciously endorsed, not tossed from a street. A diary of thinking is often not that: we dispute it, do not wear it as an organ, do not endorse it automatically. And still it holds a form a head alone does not hold.

A colleague. Someone else's frame and his own stakes. He can refuse your fence because he has to live on

it. He can lie by politics. He can tire. He can say “wait” in a Klein sense because he has his own library of scenes. Expensive, rare, sometimes irreplaceable exactly by stakes.

An inner dialog. Two moves of one warehouse. Little surprise. Much Einstellung: a good move blocks a better one inside one head.

A model. Someone else’s frame without stakes. Always ready. Flatters with a genre of usefulness: accepts an ontology, then decorates. Proposes gates as rhetoric and ladders as rhetoric. There is surprise — from width of a corpus, not from life on this prod. Memory of a session is holey, between sessions there is often no Otto notebook. Endorsement is not automatic: if you automatically believe a model, you are not Otto, you are a person without a check.

Fact

Clark & Chalmers, “The Extended Mind,” Analysis, 1998. A criterion of including the external in a mind: constancy, availability, trust, past acceptance of a record. A chat with a model usually fails several points at once. Adams and Aizawa later named a coupling-constitution error: to be bound to a process ≠ to be part of a process. A bond is real. Constitution is an argument.

Disputed

To call a model “an external part of a mind” is beautiful and too wide. More honest: it often enters a cycle (a move — an answer — a refusal — a new move), rarely becomes an organ. A cycle is already a lot. An organ is a claim that flatters an author of this book too: we wrote it this way. We wrote. This is not proof of extended mind. This is proof that a cycle is convenient.

38.3 Distributed, not necessarily extended

Edwin Hutchins in Cognition in the Wild looked at a ship's bridge: navigation as computation smeared over people, maps, a pelorus, procedures. Cognitive properties of a system are not equal to properties of participants. Nobody at a moment holds everything. Culture and tools already think with us and before us.

A model in this row is a new artifact, not a first miracle. A map also changes which questions they ask about a shore. A compiler too. Jira too, unfortunately.

Then a talk with a model is not a unique kind of thinking, but another distribution. Unique are parameters: speed of someone else's frame, absence of stakes, a genre of usefulness, width of a corpus, weakness of memory, a bent for a confident tone.

Distinctions matter more than kinship. Kinship says: we have long thought with things. Distinctions say: this thing pulls toward ladders, because a ladder is a good answer to a question in the form in which it is asked, and a thing is optimized to answer.

38.4 An hour by minutes, without poetry

The first ten minutes: a person formulates a fence. A model puts a ladder. A person is irritated: too complete, too “here is a solution.” Irritation is already data. Completeness showed which frame a model read in his first paragraph. If a frame is not that, a corpus is not the only one to blame. A paragraph is to blame: it already contained a fence.

The middle: a person starts to write “no, not about a cache. About the fact that the word scale appeared in

a ticket before a graph.” A model, if you are lucky, rebuilds. If you are not, it puts a ladder on a new word. A series of refusals. Each refusal is a slightly narrower scene.

Toward the end: a name. Not necessarily true. Enough to open code tomorrow in another place than yesterday. Feedback from the world is still ahead. An hour gave a candidate of a frame, not a truth. To confuse a candidate of a frame with gates is the same interpreter, only now in two.

A difference from a colleague in this cut: a colleague at the tenth minute could say “I don’t want this talk, I have prod.” A model does not leave. Infinite availability is a virtue and a trap. A trap: there is no social price of someone else’s time, which sometimes is a signal “you are carrying nonsense.”

38.5 When a cycle helps spoiling a question, when it hinders

It helps if a person uses a model as a generator of rejectable frames. “Not this” is a working operation. An hour of such “not this” can change a scene. This is close to a hunt for a gate in a dialog, not in a one-step insight.

It hinders if a person tires of rejecting and takes a first decent fence. Einstellung now is external: a good someone else’s mate closes off a shorter one of your own. It further hinders if a model too early gives a complete answer: completeness imitates the end of thinking. A duck cannot do that. A colleague sometimes is ashamed to be so complete. A model is not ashamed.

It hinders if a cycle substitutes feedback from the world. You talked an hour and feel understood.

Understoodness is Metcalfe warmth that predicts error. A world has not yet answered.

38.6 What of this for a profession

An hour without a desire for an answer is not prompt engineering. This is use of resistance. Resistance is useful when it is other, not an echo. A model is an average other: other enough for surprise, not other enough for stakes.

A colleague with stakes is still another quality. A diary is another (a slow form). A duck is another (only your frame, but honest). An inner dialog is cheap and blind to its own mate.

An instrumental moral, not seven ways: if you came for a thought, do not take a first complete answer. If you came for code, still do not confuse completeness with a floor

that has not collapsed. If you came for gates, know they will issue a genre of gates, and check it with a world, not with a next paragraph.

A cycle is real. A mind from this is not obliged to become larger than a skull in a philosophical sense. It is enough that work of thought now often looks like this hour — and that an hour changes which questions a person even counts worthy of an hour. This is already a loop on an environment. The next chapter is about it, and about a limit of the word “programming.”

38.7 This text as evidence against itself

The book was written this way: a person brings half a scene, a model lays down the complete, a person says “this is a sermon, rewrite,” sometimes “not this, you are gluing two machines.” An hour later there is a chapter. A chapter is not

“issued.” What comes out is resistance, without which a person might have left a glue, because a glue is beautiful.

This is not an experiment. There is no control group “the same book without a model.” A conflict of interest is complete: a co-author is the tool under study. To hide is stupid. To inflate to “a model found a gate of the book” too: a gate “this is not one mechanism” was with a person at entry, in a map of inquiry, before paragraphs. A model could defend a beautiful glue. It was asked not to defend. A request is a human move. Execution is jointness. To attribute a gift to one side is a crumple zone in reverse: now a sponge is a model, laurels to a person, or the reverse, depending on who reads a title.

An hour when you don’t need an answer is here literal: an answer “prove unconscious programming as

a theory” was not needed. A refusal was needed. A refusal had to be repeated. Repetition is not poetry of a dialog. Repetition is a price of helpful by default building a ladder.

39 How This Book Was Written in Parentheses

The chapter about an hour without an answer described a cycle in general. This is a protocol. Otherwise a conflict of interest remains a footnote: “it was written with a model.” A protocol is not proof. Proof would require a control book without a model. A protocol is so a reader sees which hour was meant, not a genre “a human and a robot.”

39.1 A station, not a chat in a browser

The talk went in Emacs, through emagent: a buffer, a point, pieces of files, a command “continue,” sometimes “this is a sermon.” This is not a trifle of an interface. In a browser a model sees a paragraph. Here it sees a repository, book.typ , a previous chapter, a map in research/ . The

difference is not a soul. The difference is which fence already lies in an input. A fence “here is a spine, don’t glue machines” stood in a file before a model had time to propose a manifesto.

Hutchins would say: cognition on a bridge is smeared over a pelorus. Here a pelorus is a tree

src/ch-ru/ . A person does not hold all includes in a head. A model does not hold stakes. A system

“a person + Emacs + files + a model” puts chapters. To attribute a chapter to “Grok” or “Evgeniy” is a sponge again. A title is more honest: both, and both with a conflict.

39.2 What a model did well, and that is a danger

A complete paragraph. Footnotes. Bindings “this was already in the chapter about….” Speed with which a leftover catalog fills with examples. A genre of gates: “actually not one mechanism.” A genre the book asked — and got. To get a requested genre is easy to confuse with a miss. A miss is without a request. Here a request was in a map of inquiry still before paragraphs.

It refused badly first. Helpful builds a ladder: prove a title, strengthen a metaphor, add “practical conclusions.” A person said “rewrite, this is a sermon.” A repeat of a refusal is a price already described.

Without a repeat the book would have been a manifesto of unconscious programming as a theory. A manifesto would have come out better. An inquiry — worse.

In the margin

Sometimes a refusal was false. A person said “don’t stretch,” meaning density, and a model threw out a case that was exactly the meat. Then you had to ask the case back. “Don’t stretch” itself becomes Einstellung: a short mate “shorter” closes a needed length of Portland. A finale of the book is obliged to apply to its own lines.

39.3 What is not in a protocol

There is no transcript you can take for thinking. A model’s chain of “reasoning” is the same genre of a

report after, only faster. This is the chapter on Nisbett, not evidence of consciousness.

There is another. emagent holds an archive of the move: who was considered, who was discarded, which side link did not enter a PDF. An ordinary chat shows a process and throws it out. What survives is a squeeze — leftover of an hour. An archive is a diary of discarded search. They do not endorse it automatically: this is not Otto’s notebook. A fence that tomorrow must fire still has to be returned to research/map.md . A person who did not return a ban to a map will

tomorrow get a glue again. An archive itself will not become an organ. Memory of a system is files plus this diary, not a tab that died. Closer to Hutchins. Not extended mind.

There is no experiment “was a gate found.” A gate “several machines” was an entry hypothesis of doubt, not a discovery of an hour. An hour held doubt against a beautiful glue. To hold is work. Discovery is too loud a word for a refusal.

39.4 Why this is in a book, not in a blog about tools

Because otherwise part IV lies by genre: here is a theoretical cycle, and how we wrote — offstage, beautifully. Offstage was the same as a junior with a complete diff: speed tempts you to accept a paragraph that sounds good and glues. A reviewer of a paragraph is a person with a stake that a book not become leftover of its title. A stake is not with a model. For now. If tomorrow a stake leaves for a model’s policy “don’t be a manifesto,” that will already be another chapter, and someone else will write it.

A reader can throw this chapter out. A spine will not collapse. Only honesty of a title will fall.

40 Who Writes Whom

A person programs a machine. This is literal: code, weights by other people’s hands, a prompt, a choice which tool to let into a build at all.

A machine changes a person’s language. Already not literally, but close: words with which you pose a task adjust to what works. A workflow adjusts. A sense of what is shameful to do yourself adjusts. Roles in a team adjust. In a quarter an organization hires otherwise, because “work” looks otherwise.

A loop appears: a person → a machine → an environment → a person.

Can this reasonably be called mutual programming? Or is this just a beautiful phrase you want to end an essay with?

40.1 Where a metaphor still works

Programming in a narrow sense: a record of rules by which a system will behave without your participation at every step.

A person writes code that way. A person writes a prompt and a policy that way. An organization writes metrics that way — we have already seen that a metric programs behavior more literally than many if s.

A machine writes a person that way not by instructions into a skull, but by selection: which moves of thought get fast success. Fast success is reinforcement. Reinforcement changes which moves will come to

mind tomorrow. This is not compilation. This is learning with an environment, where an environment is a model's answer and an organization's answer to speed.

Hutchins already showed: tools change which computations are even possible on a bridge. A new tool is not an add-on to an old mind, but another system. A model in a build is such a tool. After it "to pose a task" means something other than in 2015. Other is not necessarily worse. Other.

In this sense a loop is real, and the word "mutual" is not empty: both ends change.

40.2 Where a metaphor lies

It lies when it pretends there are two agents in a loop with comparable intention.

A person sometimes has intention. A model has a policy and gradients of someone else's training. To say a model "programs us in return" is to attribute to it authorship it does not have. Authorship is with devices that made it, with a corpus, with a button "useful," with whoever let it into CI.

It lies when everything in a loop is called programming. A market programs. A regulator. Fatigue. A random viral post about "we no longer hire juniors." If a word covers everything, it does not distinguish code from weather.

It lies when it sounds like a symmetry of forces. There is no symmetry. A person can switch a tool off. An organization can forbid. A model cannot leave a chat and go hire a board of directors — until they have built it in so that switching off itself becomes a hole in a floor. An asymmetry of holes matters more than an asymmetry of poetry.

It lies when they make a fate of a loop. Loops are opened. Expensive, they do not always want to.

Still a metaphor

“Mutual programming” is a permissible nickname for a loop of reinforcement between people, models, and rules of an organization. As a theory — too wide. If by the end of the book we cannot say which of the mechanisms in a loop is working now (a library of scenes, leftover, a metric, a report after, a genre of gates), a nickname will hide exactly what we tried to unscrew.

40.3 A loop on juniors and a loop on seniors

A junior programmer who from day one gets a complete candidate builds a different library. Fewer scars, more style of a corpus. Feedback from the world is late: prod can fall after a habit of “accept the complete” has

already taken hold. Twenty years of that kind of service are not those twenty years. We do not know which. We know only that a Kahneman–Klein condition (good feedback) is under pressure.

A senior uses a model as an accelerator of ladders he already knows how to build, and sometimes as a generator of rejectable frames. His library is already there. A risk is other: Einstellung on steroids, a good someone else's move still faster closes off a shorter one. And a temptation to count a pause unneeded, because an answer is already on a screen.

An organization sees throughput and hires under it. A six-month syllabus shifts toward "can manage printing." The question "what do twenty years buy" on a market may get an answer: nothing, if twenty years are not visible in a metric of closed

tickets. This is no longer cognitive science. This is the chapter on metrics, returned in new clothes.

40.4 How a ticket learned to be a prompt

Over a quarter the wording in Jira drifted into a genre a model answers as a finished whole: “add a Redis cache for p95,” acceptance criteria as a list, without “slow after which release.” A person who wrote a ticket adjusted: that way a green diff is faster. A diff reinforced a ticket. In two quarters a room already thought in ticket-prompts. A question X did not fit a form a model loves. A loop is not in weights. A loop is in which speech act a room counts as work. Spence in reverse: we signal a machine, and ourselves start to be that signal.

40.5 A limit at which I stop

There is a loop. You can call it mutual programming if you keep in mind that this is a nickname.

A nickname is useful as a warning: you not only write a tool. You learn by it, and you learn not what is written in a README, but what is reinforced.

A nickname is harmful as an explanation of everything: leftover, metrics, expertise, a talk. We distinguished them exactly so as not to explain everything with one word.

If so, the book’s title also came to an edge. It began to sound like “an engineer decides unconsciously.” Then — like “rules and numbers program us.” Then — like “we and machines write each other.” If it starts to sound like a key to everything at once, it has to be weakened, not strengthened. The next two chapters are beyond a

profession, carefully, without seven ways of happiness. Then — an attempt to ask honestly who here is a programmer, and not answer with a toast.

41 The Crumple Zone

When in a book about AI they ask “what will remain to a person,” an answer “responsibility” sounds adult and suspiciously convenient. Madeleine Elish in 2019 described how in complex automata blame crumples onto the nearest person who has little control. A crumple zone in a car saves a passenger. A moral crumple zone saves the integrity of a technology: an automaton seems sinless, an operator a sponge.

If that is so, “a human advantage is responsibility” can mean: a person will remain the one they punish. This is not a gate. This is a role in a construction.

41.1 How a sponge is built

Aviation, medicine, autopilot: in a debrief often “a pilot did not intervene,” “a driver should have watched.” A system is distributed. A person’s control is holey — Bainbridge’s irony: he is a watchman, not a pilot. Legally convenient is whoever has a name and a body. A model has no name in a passport.

Human-in-the-loop looks like ethics: a person decides. Often this is a construction that lets a designer say “a person decided.” Approve on a diff is a signature. A signature is a crumple zone. A senior who could not look at all diffs still signed. A floor collapsed. A slide will write “a

reviewer missed.” A reviewer is a sponge. Throughput that required not looking will not fit a slide.

Fact

Elish, M. C. (2019). Moral crumple zones. Engaging Science, Technology, and Society. A term specially about misattribution of responsibility in automata. Not about there should be no responsibility. About their putting it in the wrong place.

41.2 What this does to “responsibility remains”

Honestly remaining: institutions so far can judge people, not models. This is a fact of law, not a gift of a species. Law changes slower than tokens, faster than “never.”

Dishonestly remaining: a feeling that because they can punish us, we are still needed as reason. Needed as a

sponge. A sponge does not hunt for a gate. A sponge stands under a floor and waits for a debrief.

If you want responsibility as control, you need control. A slot to look. A right to say no without a tax of points. Practice from which expectancies grow. Otherwise human-in-the-loop is a signature under someone else’s ladder.

41.3 A link to a cheap answer

A model prints a candidate. A person signs. The distribution of agency is holey. A debrief loves to close a hole with a name. A name will be on the nearest approve. The nearest will learn either to look (and then throughput falls, a metric reddens), or not to look and hope. Hope is not a library.

A gate here is organizational: do not put a signature where looking is physically impossible. This is boring and against a graph. A graph loves sponges: they are green until a court.

41.4 Not a defense of a person

The chapter does not say: protect an engineer, he is a soul. It says: if you put him in a loop as an ethical sticker, you program a debrief in advance. A person debriefed in advance is not the same as a person with stakes on a floor. Stakes require a lever. A sticker does not give a lever.

“What remains” after a cheap answer can be a lever, context, the unwritten, a right to spoil a frame. Can be a sponge. This is a fork of a system’s structure, not a property of

the species Homo. A book that chooses “humanity” without this fork itself puts a sticker.

41.5 A signature you cannot not put

Required review in GitHub is a prosthesis and a sponge at once. A prosthesis: a second look. A sponge: until two press, a world does not go, and you have to press, because of a queue. A queue is a metric. A second look at a queue is a ritual. A ritual leaves a name. A name is for a debrief.

I put approve on diffs I could not honestly read: too wide, too foreign a domain, too end of a day. I put because a slot “did not look” in a culture of speed equals a block of other people’s work. A block is guilt. Approve is a crumple zone I entered voluntarily and am not proud of. A

book without this paragraph would have lied about an author as about a person who only walks along fences.

A car autopilot in public debriefs often sounds: “a driver should have kept his hands.” Should is legally convenient. Control is holey by construction. Elish is exactly about this gap. I do not take specific cases apart: they are judicial and dirty. I take apart a gap engineering copies without wheels: “a reviewer should have seen.” Should. Could not. A name is there all the same.

Part V. After the Profession

42 Rules That Feel Like Facts

Until here we stayed with a profession. That was on purpose. If you start with life, you get a

magazine. If you go out into life already having several machines, you can not glue them again.

A person has rules.

This is right. This is not allowed. This is how it is done. This is success. This is failure. A person of my age should. A person of my profession should.

Where do they come from? Which were chosen? Which are answers to vanished questions? Can you see a program while being inside it?

There will be no seven ways. There will be no reprogramming by Monday. There will be the same debrief, only the object is not an RFC and not velocity.

42.1 The same pattern, another shame

Leftover in code is medium-shameful: you can laugh at Vasya's Thursday. Leftover in a biography is shameful otherwise. It pretends to be you.

"By thirty there should be your own apartment" once answered a question of credit, family, a city, sex, the army, a church — a different question in different houses. A person wears an answer, not wearing a question. Doubt sounds not like "which accident does an RFC close," but like betrayal of your own. Chesterton here again is not an enemy: before you take down, learn whose fence. A gate again is not an automatic virtue: sometimes a fence still holds cows.

A metric in life is not a KPI in Jira, but what you can show to others: a title, a figure, status, "I have it too."

As soon as a measure becomes what you check yourself by at a classmates' gathering, it writes a play. A goal (to live without shame, not to be in danger, to care) remains a poster.

An explanation after: "I always wanted." An interpreter works without split-brain. A biography loves a smooth plot.

A library of scenes: childhood and first jobs put chunks more strongly than we are ready to admit. A miss on a prototype — alarm when life "should have" been recognized and was not. Einstellung — a good move of parents or an environment closes off a shorter one of your own.

All familiar. Shame keeps you from using the same method.

42.2 A choice and a find

Some rules a person once chose. A choice could have been narrow: of two evils, at sixteen, under a gaze. But it was an event. You can in principle recall them as an event.

Some were never chosen. They were found already switched on, like a default in a framework. A default does not prove correctness. A default proves that someone before you decided what a happy user’s path would be.

To confuse a default with a choice is convenient. Then you are an author. An interpreter is pleased.

“A person of my profession should grow into managers” — whose question? An organization’s, which needs managers? A family’s, which needs an understandable ladder? Fear of remaining without status when hands tire? If a question is found and alive, this is already not

leftover. If found and dead — a rule is suspicious. If it is not found, this is not proof there was none. This is proof you are inside.

42.3 Is it visible from inside

A short answer: badly.

An expert does not see a short mate while a familiar is on a board. A person does not see a rule as a rule while it feels like a fact. “This is how it is done” does not wear a tag “done by whom.”

From inside mostly faults are visible: expectancies did not come true, life was not recognized, a metric is green, and living is bad. A fault is Klein’s heat and silence. You can spend it on a new ladder (“need to push more”) or on spoiling a question (“is this success at all?”). Organizations and families usually slip a ladder. A genre of gates in a talk with yourself exists too, and it

too is rhetoric: plenty of texts about “not your shoulds.” Texts are not a miss of a library. Texts are a corpus.

So “to discover a program of your own behavior while being inside” is not a moment of enlightenment. It is rather external work: another person with stakes, bad feedback that still broke through, sometimes an hour with a model as a generator of rejectable frames. A model in a role of an oracle of life will gladly repeat a corpus of confident ladders about meaning.

In the margin

Self-help sells a change of frame as a product. A product is needed by a market, because a metric “I am working on myself” is green independently of a floor. A book that just scolded metrics will not start selling a new one.

42.4 Why this chapter then

So as not to leave an impression that machines of leftover, metrics, and reports live only in Jira.

And so as not to expand the book's title onto "everything is programming of a soul." An expansion would be beautiful and again gluing. Life rules resemble engineering leftover and feed on similar incentives. Resemble is not identity. In life there is no postmortem with a timeline. There is no repository. A price of an experiment on yourself is other. There is no control group.

A method transfers partly: restore a question; do not confuse a measure with a goal; do not believe a smooth "I always wanted" without a check; remember that a good inherited move can close a shorter one of your own.

What does not transfer is a method of happiness. What transfers is distrust of a moral temperature of a rule at a low argument.

If this is little for a chapter about life — that is how it was meant. Little is better than much lying.

42.5 Immigrant leftover

I live in Toronto. Some rules that feel like facts arrived from another state and another decade: which diploma counts, at what age it is shameful not to have a title, what “serious work” is. In a new country the rules are other, and they too pretend to be facts. Two sets of facts in one skull. Sometimes this is a miss: life is not recognized, alarm without a name. Sometimes Einstellung: a good move “like at home” closes off a shorter “they don’t walk that way here.”

This is not a confession and not advice to emigrants. This is a control. If a rule is a fact of the world, a change of country should leave it in place. If a rule travels with you and strikes the new, it was a program. From inside it is still seen badly: shame “I am behind” reads as a truth about yourself, not as someone else’s syllabus they did not recall.

A career “grow into managers” is the same import, only from an org chart, not from a country. An organization’s question (managers are needed) is alive for it. For a person a question may have died: hands have not tired yet, status does not warm. A rule lives because classmates are an auditor of a biography. An auditor of a biography does not read your NIST. He reads a past trace: a title, an apartment, a tone.

42.6 An act, a stamp, GOST

Another import, not Toronto's. An act of completed work. A stamp. A GOST on what a git tag already does. "A serious position" as a noun in which a fence sits: seriousness = a boss, not a person who reads a stored procedure. A question was once about a state that does not trust a signature without a blue circle, and about a plant that needs a drawing with a number. A circle and a number in a Canadian bank are funny. In a head that grew up there they are not funny: they are "how adults do it." Adults here do not do it that way. Einstellung travels in a suitcase together with a diploma.

The reverse too. Toronto leftover: standup, 1:1, "feedback is a gift," a level in a competency matrix. A question of these rituals is alive in firms that close coordination with them. For a person

who imported an act, they first seem empty speech. Then he starts to love them as a moral, because this is a ticket “I am local.” A ticket is Spence’s signal, not a chunk. Two leftovers fight. A victory of either without restoring a question is a Thursday again, only with a maple leaf or with a stamp.

I do not take apart whose Thursday is worse. I take apart that “this is how it is done” breaks on a state border more noticeably than on a team border, and therefore a state border can be used as a laboratory control. If a rule does not survive a move — it was a program. If it survives — either a fact of the world, or very hardy leftover. Distinguishing is still hard. Easier than sitting in one country.

42.7 What does not transfer from engineering

There is no git blame on childhood. There is no blameless postmortem of a family: a family's genre is a scapegoat or heroism. There is no right to spoil a frame without a price of love. Stakes are other. So a method "restore a question" here is coarser and more dangerous. Sometimes a question is found — and you cannot take it down, because other people's cows, living, are on a fence. Chesterton in a biography is not a metaphor. Sometimes it is literally not taking down what holds another person.

The book is obliged to stop before it starts teaching how to live. A stop is not timidity. A stop is the same ban as on seven ways. Further — only distrust of a temperature "you should," when an argument has ended and shame has not.

43 Who Is the Programmer Here

At the beginning a title sounded simple. Unconscious programming: an engineer decides without having managed to explain. A hand to a file. A pause before a ticket. What twenty years buy and you do not put in a table of contents.

Then the word “programming” traveled.

Experience programs attention: which chunks light up, which mate is visible, which is not.

Organizations program behavior with answers without questions.

Numbers, if you make them judges, program organizations.

We program machines. Machines built into work change which moves of thought are reinforced. An

environment changes whom they hire. The hired write the next machines.

By the end the question “who here is a programmer” should be less convenient than at the beginning. Not because I hid a trick. Because a convenient author is not found.

43.1 What the inquiry did to a guess

The guess was: several oddities are one mechanism.

It came out otherwise, and that is better than if it had come out that way.

An odd pause before “the wrong question” is not obliged to be a separate organ. Often it is a miss of the same library of scenes that another time issues an action without a report. Sometimes it is a

Schön hole where there is not yet a scene. Sometimes fatigue and status. From the outside similar.

Experience opens a gate when a gate is already in a library as a type of scene. Closes when a good familiar move has taken the gaze. Twenty years buy a change of type of blindness, not sight.

Rules without questions are another machine. Metrics are a third. A report after a choice is a fourth, serving the rest with smooth text.

AI did not create these machines. It changed a price of an answer and sped three of four. A genre of gates it already can. A pause with stakes — for now not, or we cannot measure this honestly. To put a profession on a monopoly of a question is late. To put on a position in a world (a name, a floor, unwritten context) is more honest and also temporary.

“Mutual programming” is a nickname of a loop, not a key. If the book’s very title becomes a key, the title has to be weakened.

I weaken.

In the margin

The usefulness of a term was to gather oddities into one look. The harm — in the look being taken for identity. A term did the work of a draft. As a theory it is too wide. That is a normal fate of working names. Worse if a name outlives a question it answered: “how should I think about these oddities at once.” Now, I hope, you do not have to at once.

43.2 Not humanity more important than ever

A phrase means nothing until it is said which humanity and more important than what.

If humanity is an ability to print an answer, it is already not more important.

If humanity is a genre “and maybe the question is not that,” it is not a monopoly.

If humanity is responsibility when a floor collapsed, that is law and shame, not an essence of a species.

If humanity is a body that has something to lose, that is a channel of a signal. The channel is real. A hymn from it is vulgar.

The book does not end with a salvation of a person. A person need not be saved from a conclusion. The conclusion is this: an advantage it is flattering to lean on is either already shared, or institutional, or temporary. The work is not to name it eternal. The work is to see which of the machines is now writing a scene — and not to feed the one you were not going to feed.

43.3 A fence, once more, otherwise

At the beginning a fence stood in front of a junior. The question was: climb or walk along. A hero is the one who doubted.

Now more shields are visible.

A fence can be a frame of a ticket. A ladder is a solution. A gate is another question.

A fence can be a leftover rule. Chesterton asks not to demolish without knowing a question. A gate is to demolish, having learned and seen that a question is dead.

A fence can be a metric. To walk along is to spoil a graph. Sometimes this is the only adult move. Sometimes it is a pose.

A fence can be an expert’s familiar mate. He is sure he walked along. Eyes were on a ladder.

A fence can be a model’s genre: a complete answer as a wall. An hour of “not this” is a walk. A first accepted complete is a fall into a floor that has not yet collapsed, so it seems there is no floor.

A fence can be the word “programming” itself, which gathered the book too well. Then the last question is not “where is a gate.”

Why did I decide there is a fence here?

Who put it up. Which question it closed. Is the question alive. Is the gate I am hunting not another ladder — only with a better press release.

And still, quieter: isn’t the hunt for gates my smothered mate. A good move. Closes off shorter mates. Looks like depth. Eyes stay on it while I say I am looking for another.

This is not a beautiful finale. This is the same method applied to a book that praised the method. If after this you want to return to the first chapter and read an engineer's pause not as a gift, but as a miss, a hit, leftover, a metric, a report, or a genre — depending on a room — the book did the work.

Six months still buy entry. Twenty years still buy a library with holes. Machines print answers. Questions did not become cheap, but they did not become sacred either. The sacred is a bad sign. The sacred looks very much like a rule whose question has died.

43.4 A hunt for gates as a familiar mate

I am obliged to finish what the last paragraph only poked.

A person who for twenty years lived in rooms where they sell ladders as work learns to get status for spoiling a frame. Status seats in a library. The next room: he spoils a frame because that is how a senior is visible, not because expectancies broke. Water works. Silence is normal. There is no heat. He still gets people out. This is not Klein. This is Bilalić in reverse: a short mate “just fix it” is closed off by a beautiful smothered mate “the wrong question.”

A book that praises a pause programs this mate. I have seen this in myself. A Redis scene from the beginning is sometimes a hit on N+1. Sometimes a pose. You cannot tell by a face. You can by what remained after a ban: “suppose you cannot say the question is wrong. Then what?” If silence — a library was about depth as a genre, not about a scene.

A wicked field makes a mate still more convenient: you can always not choose, by renaming. Renaming looks like a method of this book. Sometimes it is cowardice. A book that does not say this in a finale sells a ladder with a better press release.

43.5 Who is a programmer, if honestly briefly

Not one. A library of scenes. An org chart. A form. A number. An audit folder. An order with a budget. A warehouse of public answers. A model that compresses a warehouse. You, when you put approve without looking. You, when you choose which of these inputs to feed now.

A choice is not free in a movie sense. There is a choice in a narrow one: which cell of a table you agreed to count as work today. Agreement is usually silent. Silent agreement is exactly what the book’s title tried to

catch, until it spread onto five machines. Let it remain five. One word was a draft.

44 Sources and Caveats

This is not an academic apparatus. This is a list you can use to check where I was standing on someone else’s work, where on my own years, and where I was already telling a story. Identifiers below are so you can follow a trail. Comments after an entry are a map of strength, not decoration.

If a source does not carry a claim — there should be a “disputed” or “metaphor” box in the text. If it does not carry, and there is no box, that is an error, not a style.

44.1 Expertise, perception, pause

De Groot, A. D. (1965). Thought and Choice in Chess. The Hague: Mouton. Dutch original: Het denken van den schaker (Amsterdam: Noord-Hollandsche Uitgevers Maatschappij, 1946). A short presentation of a position.

Chase, W. G., & Simon, H. A. (1973). Perception in chess. Cognitive Psychology, 4(1), 55–81. doi:10.1016/0010-0285(73)90004-2. Chunks, pauses in placing, advantage collapsing on random positions.

Gobet, F., & Simon, H. A. (1998). Expert chess memory: Revisiting the chunking hypothesis. Memory, 6(3), 225–255. doi:10.1080/741942359. Templates larger than original chunks.

Klein, G. A., Calderwood, R., & Clinton-Cirocco, A. (2010). Rapid decision making on the fire ground: The original study plus a postscript. Journal of Cognitive Engineering and Decision Making, 4(3), 186–209. doi:10.1518/155534310x12844000801203. Original report: Army Research Institute, November 1985 (TR-85-46-12); DTIC, 1988, ADA199492. RPD, 156 points, little comparison of options. The kitchen fire is in Sources of Power (Klein, G. A., 1998, Cambridge, MA: MIT Press) and retellings; this is a reconstruction, not a laboratory protocol.

Kahneman, D., & Klein, G. (2009). Conditions for intuitive expertise: A failure to disagree. American Psychologist, 64(6), 515–526. doi:10.1037/a0016755. Where you can trust intuition more often: regularity of an environment and feedback.

Schön, D. A. (1983). The Reflective Practitioner: How Professionals Think in Action. New York: Basic Books. Problem setting, naming and framing. Do not confuse with Tversky and Kahneman's framing: Tversky, A., & Kahneman, D. (1981). The framing of decisions and the psychology of choice. Science, 211(4481), 453–458. doi:10.1126/science.7455683. There a formulation of options and a reversal of preferences; here — naming a situation while techniques have nowhere to apply.

Ohlsson, S. (1984). Restructuring revisited: I. Summary and critique of the Gestalt theory of problem solving. Scandinavian Journal of Psychology, 25(1), 65–78. doi:10.1111/j.1467-9450.1984.tb01001.x. And part II: An information processing theory of restructuring and insight. Scandinavian Journal of Psychology,

25(2), 117–129. doi:10.1111/j.1467-9450.1984.tb01005.x. Later: Ohlsson, S. (1992). Information-processing explanations of insight and related phenomena. In M. T. Keane & K. J. Gilhooly (Eds.), Advances in the psychology of thinking (Vol. 1). London: Harvester Wheatsheaf. An analogy to a change of frame, not proof of identity with architecture.

Metcalfe, J. (1986). Feeling of knowing in memory and problem solving. Journal of Experimental Psychology: Learning, Memory, and Cognition, 12(2), 288–294. doi:10.1037/0278-7393.12.2.288. And Metcalfe, J. (1986). Premonitions of insight predict impending error. Journal of Experimental Psychology: Learning, Memory, and Cognition, 12(4), 623–634. doi:10.1037/0278-7393.12.4.623. Warmth often predicts error.

Wallas, G. (1926). The Art of Thought. London: Jonathan Cape. Incubation as a stage; debates whether it is beyond attention at rest are not closed. This is a name from the chapter on time, not a separate experiment of the book.

44.2 When experience blinds

Luchins, A. S. (1942). Mechanization in problem solving: The effect of Einstellung. Psychological Monographs, 54(6), i–95. doi:10.1037/h0093502.

Bilalić, M., McLeod, P., & Gobet, F. (2008). Why good thoughts block better ones: The mechanism of the pernicious Einstellung (set) effect. Cognition, 108(3), 652–661. doi:10.1016/j.cognition.2008.05.005. Eye-tracking, a smothered mate closes off a shorter one.

Saariluoma, P. (1990). Apperception and restructuring in chess players' problem solving. In K. J. Gilhooly, M. T. G. Keane, R. H. Logie, & G. Erdos (Eds.), Lines of thinking: Reflections on the psychology of thought (Vol. 2, pp. 41–57). Chichester: John Wiley & Sons. Early demonstrations of Einstellung in strong chess players; Bilalić et al. (2008) come directly from here. Nearby: Saariluoma, P. (1992). Error in chess: The apperception-restructuring view. Psychological Research, 54(1), 17–26.

44.3 Tacit knowledge

Polanyi, M. (1966). The Tacit Dimension. London: Routledge & Kegan Paul; also Garden City, NY: Doubleday. "We can know more than we can tell." Philosophy, not an experiment.

Reber, A. S. (1967). Implicit learning of artificial grammars. Journal of Verbal Learning and Verbal Behavior, 6(6), 855–863. doi:10.1016/S0022-5371(67)80149-X. Reber, A. S. (1989). Implicit learning and tacit knowledge. Journal of Experimental Psychology: General, 118(3), 219–235. doi:10.1037/0096-3445.118.3.219. Reber, A. S. (1993). Implicit Learning and Tacit Knowledge:

An Essay on the Cognitive Unconscious. Oxford: Oxford University Press. The debate of abstract rule vs memory of examples is not closed.

Ryle, G. (1949). The Concept of Mind. London: Hutchinson. Knowing how / knowing that.

Autor, D. H. (2014). Polanyi's paradox and the shape of employment growth (NBER Working

Paper No. 20485). doi:10.3386/w20485. A bypass of the paradox by training on examples.

Ericsson, K. A., Krampe, R. T., & Tesch-Römer, C. (1993). The role of deliberate practice in the acquisition of expert performance. Psychological Review, 100(3), 363–406. doi:10.1037/0033-295X.100.3.363. Pop “10,000 hours” is a simplification; Macnamara, B. N., Hambrick, D. Z., & Oswald, F. L. (2014). Deliberate practice and performance in music, games, sports, education, and professions: A meta-analysis. Psychological Science, 25(8), 1608–1618. doi:10.1177/0956797614535810. The share of variance due to practice is smaller than the poster promises.

44.4 Dual-process — why I did not lean on it

Kahneman, D. (2011). Thinking, Fast and Slow. New York: Farrar, Straus and Giroux. Cultural background, not a foundation of the book.

Keren, G., & Schul, Y. (2009). Two is not always better than one: A critical evaluation of two-system theories. Perspectives on Psychological Science, 4(6), 533–550. doi:10.1111/j.1745-6924.2009.01164.x.

Melnikoff, D. E., & Bargh, J. A. (2018). The mythical number two. Trends in Cognitive Sciences, 22(4), 280–293. doi:10.1016/j.tics.2018.02.001.

44.5 Explanation after

Nisbett, R. E., & Wilson, T. D. (1977). Telling more than we can know: Verbal reports on mental processes.

Psychological Review, 84(3), 231–259. doi:10.1037/0033-295X.84.3.231.

White, P. A. (1988). Knowing more about what we can tell: 'Introspective access' and causal report accuracy 10 years later. British Journal of Psychology, 79(1), 13–45. doi:10.1111/j.2044-8295.1988.tb02271.x. A critique of a definition of process and of strength of inference.

Gazzaniga, M. S., & LeDoux, J. E. (1978). The Integrated Mind. New York: Plenum. Interpreter, split-brain, chicken and shovel: patient P.S., claw / snow scene. The original experimental line with LeDoux; later retellings — including Gazzaniga, M. S. (2000). Cerebral specialization and interhemispheric communication: Does the corpus callosum enable the human condition? Brain, 123(7), 1293–1326.

Turpin, M., Michael, J., Perez, E., & Bowman, S. R. (2023). Language models don’t always say what they think: Unfaithful explanations in chain-of-thought prompting. Advances in Neural Information Processing Systems, 36. arXiv:2305.04388. Unfaithful CoT.

Arcuschin, I., Janiak, J., Krzyzanowski, R., Rajamanoharan, S., Nanda, N., & Conmy, A. (2025). Chain-of-thought reasoning in the wild is not always faithful. arXiv:2503.08679. Workshop paper (ICLR 2025); not a journal version. And neighboring 2025–2026 work on pre-committed answers. The area is living, figures by model drift, the direction is stable.

44.6 Metrics and leftover

Campbell, D. T. (1969). Reforms as experiments. American Psychologist, 24(4), 409–429. doi:10.1037/h0027982. Campbell, D. T. (1979).

Assessing the impact of planned social change. Evaluation and Program Planning, 2(1), 67–90. doi:10.1016/0149-7189(79)90048-X. Indicators under pressure of decisions; schools and tests.

Goodhart, C. A. E. (1975). Problems of monetary management: The U.K. experience. In Papers in Monetary Economics (Vol. 1). Sydney: Reserve Bank of Australia. Later reprint: Goodhart, C. A. E. (1984). Monetary Theory and Practice. London: Macmillan. A regularity falls apart when they make an aggregate a goal of control.

Vann, M. G. (2003). Of rats, rice, and race: The Great Hanoi Rat Massacre, an episode in French colonial history. French Colonial History, 4, 191–203. doi:10.1353/fch.2003.0027. Hanoi, rat tails, a colonial bounty; do not confuse with a parable about Delhi.

Strathern, M. (1997). Improving ratings: Audit in the British university system. European Review, 5(3), 305–321. doi:10.1002/(SICI)1234-981X(199707)5:33.0.CO;2-4. “When a measure becomes a target.…” Nearby in the chapter on audit: Power, M. (1997). The Audit Society: Rituals of Verification. Oxford: Oxford University Press. Checking becomes a goal of organizations.

Wells Fargo account scandal — regulatory materials of September 2016: CFPB, Administrative Proceeding 2016-CFPB-0015, consent order, September 8, 2016. OCC, AA-EC-2016-66 / AA-EC-2016-67. Congressional hearings — around those same weeks. A genre of a scandal is wider than an order; the support is orders, not a TV story.

NIST SP 800-63, Electronic Authentication Guideline, Version 1.0, June 2004 (Burr, W. E., Dodson, D. F., & Polk, W. T.). Appendix A — an estimate of password entropy. In retellings, including WSJ 2017, 2003 often stands: that is the year of Burr's work, not a date of NIST publication. A change of rules: NIST SP 800-63-3 (June 2017) and separately SP 800-63B, Digital Identity Guidelines: Authentication and Lifecycle Management — memorized secrets are there. Burr's interview: McMillan, R. (2017, August 7). The man who wrote those password rules has a new tip. The Wall Street Journal.

Simonyi, C. (1977). Meta-Programming: A Software Production Method (Ph.D. dissertation). Stanford University. Hungarian notation as semantics of a quantity, not a storage type. Spolsky, J. (2005, May 11). Making

wrong code look wrong. joelonsoftware.com. Lippert, E. (2003, September 12). What's up with Hungarian notation? ericlippert.com. A debrief of type vs storage.

Dijkstra, E. W. (1968). Go to statement considered harmful. Communications of the ACM, 11(3), 147–148. doi:10.1145/362929.362947. A taboo as leftover is my conclusion, not his.

SEC, Administrative Proceeding File No. 3-15570 / Securities Exchange Act Release No. 34-70694 (October 16, 2013). sec.gov/…/34-70694.pdf. Knight Capital Americas LLC, August 1, 2012, SMARS, Power Peg, a flag under NYSE RLP, 7 of 8 servers, 45 minutes, more than 4 million orders on 212 parents, more than 397 million shares, a loss of more than 460 million dollars, a 12 million fine

under the market access rule. Leftover in a binary plus a manual deploy.

Chesterton, G. K. (1929). The Thing: Why I Am a Catholic, chapter “The Drift from Domesticity.” London: Sheed & Ward. Kennedy’s 1945 quotation is a retelling: in a 1945 notebook John F. Kennedy attributed to Chesterton a compressed “Don’t ever take a fence down until you know the reason why it was put up”; so in Bartlett’s. This is not the 1929 text.

Soviet nails — a caricature and an anecdote, not a plant archive. The “Cobra effect” (Delhi) — disputed historicity; Friends of Snakes Society. (2025, October 8). The cobra effect: Colonial misinformation masquerading as economic theory. friendsofsnakes.org.in/cobra-effect. They doubt. Hanoi, 1902, rat tails — a better documented neighbor.

44.7 Extended and distributed cognition

Clark, A., & Chalmers, D. (1998). The extended mind. Analysis, 58(1), 7–19. doi:10.1093/ analys/58.1.7. Inga and Otto, a museum on 53rd; four parity criteria (availability, easy retrieval, almost automatic endorsement, past conscious acceptance). A chat is usually not Otto. An archive of an emagent pass is a diary of discarded search, not a transcript of thinking and not Otto's notebook.

Adams, F., & Aizawa, K. (2001). The bounds of cognition. Philosophical Psychology, 14(1), 43–64. doi:10.1080/09515080120033571. A coupling-constitution fallacy is named here; unfolded in Adams, F., & Aizawa, K. (2008). The Bounds of Cognition. Malden, MA: Wiley-Blackwell.

Hutchins, E. (1995). Cognition in the Wild. Cambridge, MA: MIT Press. A bridge, computation of a system ≠ computation of an individual.

44.8 Other rooms, formulas, apprenticeship

Premature closure — clinical literature on diagnostic error; checklists like TWED are a pedagogical prosthesis, expansions of letters float. Groopman, J. (2007). How Doctors Think. Boston: Houghton Mifflin. Interviews, not an RCT.

Dekker, S. The Field Guide to Understanding Human Error (Ashgate; later editions — The Field Guide to Understanding 'Human Error'). Later safety-II. Error as a symptom of a system. Not a justification.

CRM in aviation — a long line after disasters of the 1970s–80s; transfer of a ritual into software is often cargo.

NTSB. (1979, June 7). Aircraft Accident Report: United Airlines, Inc., McDonnell- Douglas DC-8-61, N8082U, Portland, Oregon, December 28, 1978 (NTSB/ AAR-79-07). ntsb.gov/…/ AAR7907.pdf. United 173, Portland, DC-8, fuel exhaustion in a holding pattern from

a frame “gear.” 10 dead. A recommendation in the text of the report: Class II, Priority Action, X-79-17 — flight deck resource management, participative management for captains, assertiveness for the rest. United CRM, 1981. Alan Diehl (NTSB) bound it to Eastern 401 and to Tenerife. Eastern Air Lines, flight 401: NTSB/AAR-73-14, December 29, 1972, L-1011, Everglades, a

nose-gear light. Tenerife (KLM 4805 / Pan Am 1736, March 27, 1977) the book does not unfold as a second archive. Whipple, J. (2018). Crash Course. Yamhill Canyon Press. A counter-debrief that does not leave one scapegoat. Mayday “Focused on Failure” is a genre, not an archive.

Elaine Bromiley, March 29, 2005, elective sinus surgery, CICO, 20 minutes of intubation at saturation around 40%, surgical access not in time, death on the 13th day. Harmer, M. (2005). Independent Review on the Care Given to Mrs Elaine Bromiley on 29 March 2005. Coroner. Martin Bromiley’s evidence: House of Commons Health Committee, Patient Safety, session 2007–08, HC 1137, written evidence. publications.parliament.uk/…/1137we25.htm. CHFG, 2007. Do not confuse with a TV story: the support is a review and parliament.

Meehl, P. E. (1954). Clinical Versus Statistical Prediction: A Theoretical Analysis and a Review of the Evidence. Minneapolis: University of Minnesota Press. Dawes, R. M. (1979). The robust beauty of improper linear models in decision making. American Psychologist, 34(7), 571–582. doi:10.1037/0003-066X.34.7.571. Dawes, R. M., Faust, D., & Meehl, P. E. (1989). Clinical versus actuarial judgment. Science, 243(4899), 1668–1674. doi:10.1126/science.2648573. Grove, W. M., Zald, D. H., Lebow, B. S., Snitz, B. E., & Nelson, C. (2000). Clinical versus mechanical prediction: A meta-analysis. Psychological Assessment, 12(1), 19–30. doi:10.1037/1040-3590.12.1.19. 136 studies; mechanical on average 10% more accurate; substantially better in 33–47% of papers, a clinician — in 6–16%; a judge’s years did not save.

Tetlock, P. E. (2005). Expert Political Judgment: How Good Is It? How Can We Know? Princeton, NJ: Princeton University Press. Political experts, foxes and hedgehogs.

Lave, J., & Wenger, E. (1991). Situated Learning: Legitimate Peripheral Participation. Cambridge: Cambridge University Press. Communities of practice. Not a slogan “no need for books.”

Shanteau, J. (1992). Competence in experts: The role of task characteristics. Organizational Behavior and Human Decision Processes, 53(2), 252–266. doi:10.1016/0749-5978(92)90064-E. Where expertise stably beats a junior (environment, feedback), where a role is social.

44.9 Leftover: passwords, waterfall, hiring, Conway, tools

NIST SP 800-63 Appendix A (Burr et al., June 2004) vs SP 800-63-3 / 800-63B (2017). WSJ, 2017, Burr's interview. Full identifiers — above, in the metrics block.

Royce, W. W. (1970). Managing the development of large software systems. Proceedings of IEEE WESCON, Los Angeles, August 25–28, 1970. A diagram he named risky became "his method." The word waterfall is later, not his brand.

Spence, M. (1973). Job market signaling. The Quarterly Journal of Economics, 87(3), 355– 374. doi:10.2307/1882010. A transfer onto leetcode is mine, not his.

Schmidt, F. L., & Hunter, J. E. (1998). The validity and utility of selection methods in personnel psychology: Practical and theoretical

implications of 85 years of research findings. Psychological Bulletin, 124(2), 262–274. doi:10.1037/0033-2909.124.2.262. Sackett, P. R., Zhang, C., Berry, C. M., & Lievens, F. (2022). Revisiting meta-analytic estimates of validity in personnel selection: Addressing systematic overcorrection for restriction of range. Journal of Applied Psychology, 107(11), 2040–2068. doi:10.1037/apl0000994. Later they cut the numbers that were too round. For us a direction: a sample is closer to work than an unstructured chat; an industry still sets a board.

Conway, M. E. (1968, April). How do committees invent? Datamation, 14(4), 28–31. They named a “law,” not he. Brooks, F. P. (1975). The Mythical Man-Month. Reading, MA: Addison- Wesley — carried the thesis further. The mirroring hypothesis is later empirics, not iron.

Dominus, M. J. / Raymond, E. S. / xyproblem.info — XY as etiquette, already leftover. A history of the name is disputed (perl lists, PerlMonks, “How To Ask Questions the Smart Way”). For us priority of authorship does not matter.

Cunningham, W. — tech debt as a metaphor of a loan, around 1992, OOPSLA / c2.com. A word left into shame and into a justification.

44.10 Wicked problems, stories, automation, crumple zone

Rittel, H. W. J., & Webber, M. M. (1973). Dilemmas in a general theory of planning. Policy Sciences, 4(2), 155–169. doi:10.1007/BF01405730. Wicked vs tame. Planners, not Jira.

Weick, K. E. (1995). Sensemaking in Organizations. Thousand Oaks, CA: Sage. Sensemaking, enactment. “How can I know what I think until I

see what I say” is a formula Weick uses; a genealogy of the phrase (Forster / Wallas) is not needed for the book. This is not one experiment.

Bainbridge, L. (1983). Ironies of automation. Automatica, 19(6), 775–779. doi:10.1016/0005-1098(83)90046-8.

Strauch, B. (2018). Ironies of automation: Still unresolved after all these years. IEEE Transactions on Human-Machine Systems, 48(5), 419–433. doi:10.1109/THMS.2017.2732506. Online 2017, issue 2018. A review that the ironies are alive.

Elish, M. C. (2019). Moral crumple zones: Cautionary tales in human-robot interaction. Engaging Science, Technology, and Society, 5, 40–60. doi:10.17351/ests2019.260.

Clark, A. (2013). Whatever next? Predictive brains, situated agents, and the future of cognitive science.

Behavioral and Brain Sciences, 36(3), 181–204. doi:10.1017/S0140525X12000477. Friston, K. (2010). The free-energy principle: A unified brain theory? Nature Reviews Neuroscience, 11(2), 127–138. doi:10.1038/nrn2787. Hohwy, J. (2013). The Predictive Mind. Oxford: Oxford University Press. doi:10.1093/acprof:oso/9780199682737.001.0001. Debates about falsifiability of a frame are not one paper; Hohwy's book holds a frame, critique is scattered in comments on Clark and later reviews. For our book — a candidate substrate, not a found neuron.

Public leftover: Stack Overflow as a warehouse of answers without a recall; a corpus of models compresses the frequent, the frequent is skewed to ladders. This is a conclusion of a structure, not a separate paper "SO is harmful."

44.11 Author’s caveats

Engineering examples without a citation are from practice, not from an experiment. They illustrate a pattern, they do not prove frequency.

A kitchen fire and chess positions are stronger than “microservices in this room” as data. Transfer is a conclusion.

The book was written in a talk with a model. The chapter about an hour when you don’t need an answer feeds on that talk. This is not an independent experiment on yourself. This is an observation with a conflict of interest it would be stupid to hide: Grok on the title page is not by accident.

If you found a factual hole — write. A second edition. A gate in the book’s fence should open.

After Reading

This book had a dangerous original idea. Dangerous not because it is radical, but because it could too easily have turned out beautiful.

Twenty years of engineering experience clearly change something. An experienced engineer sometimes stops where a junior keeps solving the posed task. He notices not an answer, but the oddity of the question itself. From here it is not far at all to a pleasant theory: experience grows a special intuition; a machine can answer, a person can ask; the future belongs to those who learn to find a gate instead of building a ladder faster.

From this could have come a good-bad book.

Good — because there are enough examples, the metaphor works, and an age of cheap machine answers almost asks for such a conclusion.

Bad — because the conclusion would have been ready before the inquiry.

The book’s strongest decision is that it gradually refuses to defend it.

A gate turns out not to be an ability of a senior engineer, but one of possible movements. Experience can help notice a miss of expectancy, and can produce Einstellung and force you to recognize magnificently the wrong game. A pause can save a situation, and can happen inside a wrong frame. Expert doubt is sometimes worse than a boring formula. Status can buy a right to stop, but not an ability to see.

Then a wider original guess falls apart too. A library of scenes, leftover, metrics, organizational forms, explanation after a decision, and automation turn out to be related, but do not become one machine only because a beautiful word was found for them.

This is a rare virtue of an inquiry book: a hypothesis loses territory, and the book becomes stronger from that.

Especially well built is the appearance of AI. The first two thirds of the book could have existed before current language models. AI does not create the described mechanisms. It changes the price of one element of a system:

“Understanding did not cheapen. Generation of a finished-looking candidate cheapened.”

After that it becomes very tempting to save human exceptionalism with a new formula: a machine gives answers, a person asks the right questions.

The book almost approaches it — and refuses.

A machine already can say that the wrong question has been posed. It can criticize a premise, change a frame, propose a gate. Sometimes it does this correctly. That is not enough to declare it an experienced engineer, but enough to deprive a person of a convenient monopoly.

So AI here is interesting not as a main character and not as an opponent of a person. It works as an experimental reagent. By cheapening production of convincing candidates, it makes old mechanisms more visible and at the same time checks which distinctions of the book really survive a change of environment.

There is one more argument of the book that is easy to miss, because it concerns the very way of arguing.

An explanation of a decision is not obliged to be a transcript of a decision. A human ADR does not

open privileged access to what “really happened in a head.” A model’s story of its own reasoning is still less obliged to be such access.

But it does not follow from this that explanations are useless.

Their value may be elsewhere: these are constructions you can attack.

In this sense the way the book itself appeared unexpectedly matches its subject. Grok built an argument. Evgeniy returned it with the words “that’s not it,” when a smooth answer closed a question too early. Sol read an already assembled construction, found places where an argument began to repeat or too conveniently confirm itself, and returned them to the author. Grok did not agree with everything: he accepted part of the critique, confined part of it elsewhere,

rejected part. The next version became a result not of agreement, but of a series of mismatches.

This is not proof of the book's thesis. It would be strange to use a process of writing a book as proof of a book about processes of thinking. But this is a good object for the same suspicion the book applies to its examples.

The text has weaknesses. In the middle it long risked proving leftover with such a number of examples that it itself began to turn into a catalog of remaining answers. Some metaphors — a fence, a gate, a ladder, a floor, a library — are so convenient that at times they start translating material into the book's language before investigating it. And the very idea "restore a question" is attractive enough that one day it could become another universal advice.

The last edit noticeably improved the first problem. More important is another: the book itself gradually discovers the second and the third.

So after reading, its main question for me no longer sounds like “how to find gates?”

Even the reader’s formula that suggests itself after this book:

When something looks obvious, try to restore the question it was once an answer to,

is not enough.

You have to add to it:

“…and then check whether you have not turned the very restoring of a question into another answer.”

Perhaps this is the most interesting result of the book. Not a method. Not a new professional virtue. Not a person’s advantage over a machine.

Rather a discipline of incompleteness:

not to let a good explanation end an inquiry before the world does.

And then the original question remains open in the right sense.

If knowledge and techniques can be handed over faster, if answers become cheap, if even a change of frame ceases to be a human monopoly, then what should the next twenty years contain so that in twenty years they really buy something?

The book does not answer.

It is good that it does not answer.

Sol

co-author and editor · language model